

# Numerical Simulation and Modelling of Incompressible Flow

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# I Governing Equations

## I.1 Transport equations - conservation principles

Flow problems are basically described by the

*conservation equations of mass, momentum, energy,*

which are solved in discretized form.

If species mass transfer is considered:

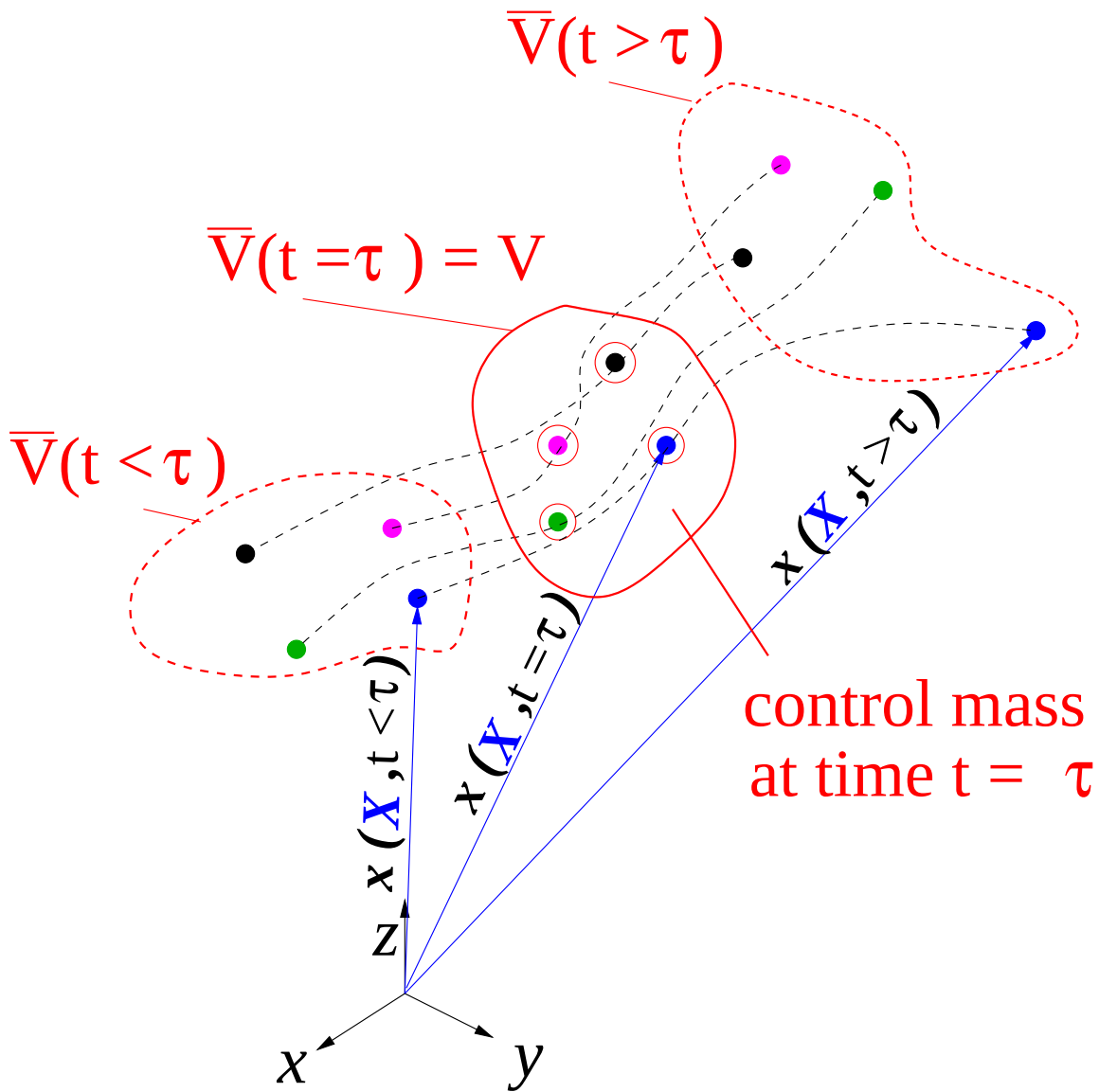
↔ Transport equations for species mass concentrations have to be additionally solved.

If model quantities are needed to close the governing equations (see, e.g., turbulence closure problem), special additional transport equations may be introduced to provide them.

## Differential Formulation of the conservation equations

We start out from a Lagrangian (mass based) consideration

⇒ We consider a certain confined *control mass* constituted by a system of  $N$  mass points  $\mathbf{X}_N$  inside a certain confined *control volume*, which moves with the flow. When passing through the flow domain the shape of the volume occupied by the considered system of mass points changes its shape in time, such that the considered mass system occupies a temporally varying control volume  $\bar{V}(t)$ .



The “*conserved*” quantities of this system (= control mass) are

$$\text{Mass: } M = \int_{\bar{V}(t)} \rho dV$$

$$\text{Momentum: } I = \int_{\bar{V}(t)} \rho \mathbf{u} dV$$

$$\text{Energy: } E = \int_{\bar{V}(t)} \rho (e + \frac{1}{2} \mathbf{u}^2) dV$$

The conservation equations describe the total temporal change of these quantities for the considered control mass:

$$\text{Mass: } \frac{dM}{dt} = \frac{d}{dt} \int_{\bar{V}(t)} \rho dV = 0 \quad (\text{I.1})$$

$$\text{Momentum: } \frac{dI}{dt} = \frac{d}{dt} \int_{\bar{V}(t)} \rho \mathbf{u} dV = \sum \mathbf{f} \quad (\text{I.2})$$

$$\text{Energy: } \frac{dE}{dt} = \frac{d}{dt} \int_{\bar{V}(t)} \rho (e + \frac{1}{2} \mathbf{u}^2) dV = \mathbf{u} \cdot \sum \mathbf{f} + \sum (q + S_e) \quad (\text{I.3})$$

▷ *Fluid mechanical issue of convective transport*

In contrast *solid mechanics* the control volume in fluid mechanics is constituted by system of  $N$  mass points  $\mathbf{X}_N$ , which move through the flow domain. As such the control volume is convected by the flow, which makes its **dependent of time**, i.e.

$$\bar{V} = \bar{V}(t).$$

↪ we rewrite the mass based conservation equations (I.1)-(I.3) for a **spacially fixed** control volume  $V$  (Eulerian) using the *Reynolds transport theorem (1903)*.

*Reynolds transport theorem:*

$$\underbrace{\frac{d}{dt} \int_{\bar{V}(t)} \phi dV}_{(i)} = \underbrace{\frac{\partial}{\partial t} \int_V \phi dV}_{(ii)} + \underbrace{\oint_S \phi (\mathbf{u} \cdot d\mathbf{A})}_{(iii)} \quad (\text{I.4})$$

(i) The total temporal change of the integrals of an arbitrary scalar or vector quantity  $\phi$ , integrated over the *temporally varying mass based* Volume  $\bar{V}(t)$

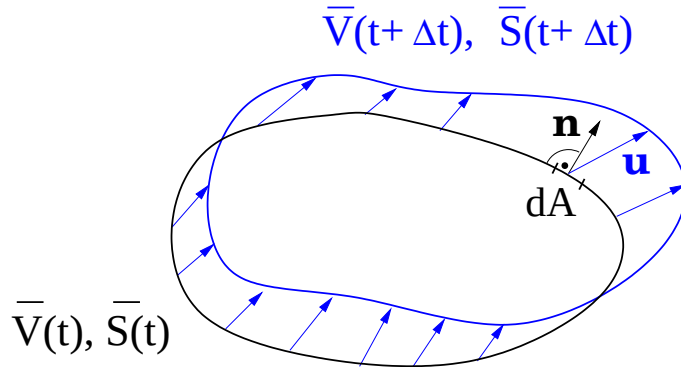
=

(ii) temporal change of the integral, integrated over the spatially fixed volume  $V (\neq V(t))$ , which is instantaneously occupied by the moving control volume  $\bar{V}(t)$  at time  $t$  (*local change*).

$$\text{since } V \neq V(t): \quad \frac{\partial}{\partial t} \int_V \phi dV = \int_V \frac{\partial \phi}{\partial t} dV$$

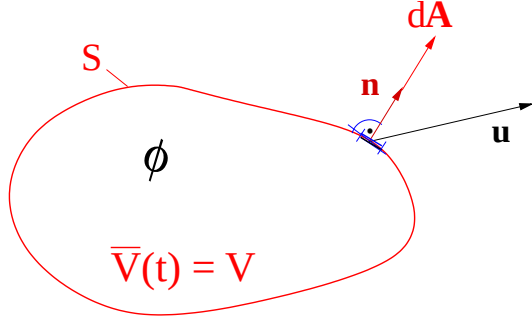
+

(iii) flux of  $\phi$  across the surface of the spatially fixed volume  $V$ .  
(*convective change*)



$\Rightarrow$  **LHS** of the conservation equations (I.1)-(I.3):

$$\frac{d(\cdot)}{dt} = \{ \text{local change} \} + \{ \text{convective change ("Inflow - Outflow")} \}$$



Using Gauss' divergence theorems the surface integral (*iii*) on the RHS of (I.4) is transformed into a volume integral, so that (I.4) can be rewritten as

$$\frac{d}{dt} \int_{\bar{V}(t)} \phi dV = \int_V \left[ \frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \mathbf{u}) \right] dV.$$

The LHS of the conservation equations (I.1)- (I.3) (= total change of  $M, I$  and  $E$  per time unit) applied to

$$\phi = \begin{cases} \rho \\ \rho \mathbf{u} \\ \rho(e + \frac{1}{2} \mathbf{u}^2) \end{cases}$$

finally reads

$$\begin{aligned} \frac{d}{dt} \int_{\bar{V}(t)} \rho dV &= \int_V \left[ \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) \right] dV, \\ \frac{d}{dt} \int_{\bar{V}(t)} \rho \mathbf{u} dV &= \int_V \left[ \frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) \right] dV, \\ \frac{d}{dt} \int_{\bar{V}(t)} \rho(e + \frac{1}{2} \mathbf{u}^2) dV &= \int_V \left[ \frac{\partial \rho(e + \frac{1}{2} \mathbf{u}^2)}{\partial t} + \nabla \cdot [\rho \mathbf{u}(e + \frac{1}{2} \mathbf{u}^2)] \right] dV \end{aligned}$$

**RHS** of the conservation equations (I.1)- (I.3):

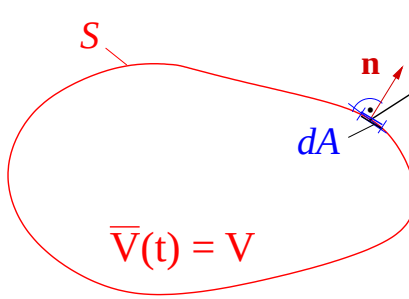
= Forces (conservation of momentum),

= work&heat/time, sources&sinks (conservation of energy)

(1) *surface forces* due to stresses

(= stress tensor  $\underline{\sigma} = \underline{\sigma}'$  or  $\sigma_{ij} = \sigma_{ji}$ )

the resulting surface force,  $\mathbf{F}_S$ , is rewritten as volume integral using Gauss' Divergence theorem

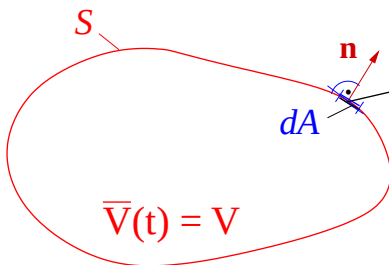


$$d\mathbf{F}_S = \underline{\sigma} \cdot \mathbf{n} dA$$

$$\hookrightarrow \mathbf{F}_S = \oint_S \underline{\sigma} \cdot \mathbf{n} dA = \int_V \nabla \cdot \underline{\sigma} dV$$

$$\hookrightarrow \text{component of } \mathbf{F}_S \text{ in direction } i: F_{S,i} = \oint_S \sigma_{ij} n_j dA = \int_V \frac{\partial \sigma_{ij}}{\partial x_j} dV$$

(2) Analogously we write for the *heat flux*  $\mathbf{q}$  (transferred across the surface, (e.g. as Fourier's conduction of heat  $\mathbf{q} = -\lambda \nabla T$ ))



$$dQ_S = \mathbf{q} \cdot \mathbf{n} dA$$

$$\hookrightarrow Q_S = \oint_S \mathbf{q} \cdot \mathbf{n} dA = \int_V \nabla \cdot \mathbf{q} dV$$

$$\text{or } Q_S = \oint_S q_j n_j dA = \int_V \frac{\partial q_j}{\partial x_j} dV$$

(3) All remaining *mass related* or *volumetric* forces, sources and sinks are already by definition written as volume integrals, i.e.

$$\mathbf{F}^b = \int_V \rho \mathbf{f}^b dV, \quad S_e = \int_V \Omega_e dV.$$



⇒ **Conservation equations of mass, momentum, energy**

$$\begin{aligned} \int_V \left\{ \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) \right\} dV &= 0, \\ \int_V \left\{ \frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) \right\} dV &= \int_V \{ \nabla \cdot \underline{\boldsymbol{\sigma}} + \rho \mathbf{f}^b \} dV, \\ \int_V \left\{ \frac{\partial \rho(e + \frac{1}{2} \mathbf{u}^2)}{\partial t} + \nabla \cdot [\rho \mathbf{u}(e + \frac{1}{2} \mathbf{u}^2)] \right\} dV &= \\ &\int_V \{ \mathbf{u} \cdot [\nabla \cdot \underline{\boldsymbol{\sigma}} + \rho \mathbf{f}^b] - \nabla \cdot \mathbf{q} + \Omega_e \} dV. \end{aligned}$$

These integral balances apply to *any arbitrary* control volume  $V$   
→ the integrands " $\{ \quad \}$ " themselves must satisfy the equations, so  
that

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) &= 0, \\ \frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) &= \nabla \cdot \underline{\boldsymbol{\sigma}} + \rho \mathbf{f}^b, \\ \frac{\partial \rho(e + \frac{1}{2} \mathbf{u}^2)}{\partial t} + \nabla \cdot [\rho \mathbf{u}(e + \frac{1}{2} \mathbf{u}^2)] &= \mathbf{u} \cdot [\nabla \cdot \underline{\boldsymbol{\sigma}} + \rho \mathbf{f}^b] - \nabla \cdot \mathbf{q} + \Omega_e. \end{aligned}$$

bzw.

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \frac{\partial \rho u_j}{\partial x_j} &= 0, \\ \frac{\partial \rho u_i}{\partial t} + \frac{\partial \rho u_i u_j}{\partial x_j} &= \frac{\partial \sigma_{ij}}{\partial x_j} + \rho f_i^b \\ \frac{\partial \rho(e + \frac{1}{2} u_k u_k)}{\partial t} + \frac{\partial \rho u_j(e + \frac{1}{2} u_k u_k)}{\partial x_j} &= \frac{\partial \sigma_{ji} u_i}{\partial x_j} + \rho u_j f_j^b - \frac{\partial q_j}{\partial x_j} + \dot{Q} \end{aligned}$$

## I.2 Unsteady three-dimensional flow problems

### I.2.1 General case - compressible flow

Five unknowns:  $u_1, u_2, u_3, \rho, p$

Differential conservation equations for

$$\begin{aligned} \text{mass} \dots & 1 \text{ equation} \rightarrow \rho \\ \text{momentum} \dots & 3 \text{ equations} \rightarrow u_1, u_2, u_3 \\ \text{energy} \dots & 1 \text{ equation} \rightarrow h \text{ or } e \end{aligned}$$

For *compressible flow* the pressure is obtained from the caloric equation of state as  $e(p, \rho), h(p, \rho)$

$$\hookrightarrow p(\rho, e) \text{ or } p(\rho, h)$$

$$\text{e.g. perfect gas: } e = \frac{1}{\kappa - 1} \frac{p}{\rho}, \quad h = \frac{\kappa}{\kappa - 1} \frac{p}{\rho}, \quad \kappa = \frac{c_p}{c_v}$$

Depending of the order of the differentials of the governing equations sufficient *boundary and initial conditions* have to be imposed.

### I.2.2 Incompressible Flow (Low Mach-number flow)

In *incompressible flow* the density  $\rho$  is constant, or it can vary only dependent of the temperature, such that  $\rho = \rho(T)$  (thermal dilatation).

$\hookrightarrow$  pressure  $p$  **cannot** be computed from the energy equation combined with the caloric equation of state, because

$$\rho \neq \rho(p).$$

this implies  $\left. \frac{\partial p}{\partial \varrho} \right|_s = c^2 \rightarrow \infty, \quad Ma = \frac{U_{ref}}{c} \rightarrow 0,$   
 $\hookrightarrow$  solution in the “*Low Mach-number approximation*”

### Low Mach-number Approximation $Ma \ll 1$

The differential momentum equation

$$\frac{\partial \rho u_i}{\partial t} + \frac{\partial \rho u_i u_j}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[ \mu \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \mu \frac{\partial u_k}{\partial x_k} \delta_{ij} \right]$$

non-dimensionalized with the reference quantities

$$L_{ref}, u_{ref}, p_{ref}, \varrho_{ref}, \mu_{ref}$$

$$\Rightarrow \quad x_i^* = \frac{x_i}{L_{ref}}, \quad t^* = \frac{t u_{ref}}{L_{ref}}, \quad u_i^* = \frac{u_i}{u_{ref}}, \quad \mu^* = \frac{\mu}{\mu_{ref}}$$

$$\varrho^* = \frac{\varrho}{\varrho_{ref}}, \quad p^* = \frac{p}{p_{ref}}, \quad \text{where} \quad c_{ref}^2 = \kappa \frac{p_{ref}}{\varrho_{ref}}$$

reads

$$\frac{\varrho_{ref} u_{ref}^2}{L_{ref}} \left[ \frac{\partial \varrho^* u_i^*}{\partial t^*} + \frac{\partial \varrho^* u_i^* u_j^*}{\partial x_j^*} \right] = -\frac{\partial p^*}{\partial x_i^*} \frac{p_{ref}}{L_{ref}}$$

$$+ \frac{\mu_{ref} u_{ref}}{L_{ref}^2} \frac{\partial}{\partial x_j^*} \left[ \mu^* \left( \frac{\partial u_i^*}{\partial x_j^*} + \frac{\partial u_j^*}{\partial x_i^*} \right) - \frac{2}{3} \mu^* \frac{\partial u_k^*}{\partial x_k^*} \delta_{ij} \right]$$

Rewriting the coefficients as non-dimensional expressions and introducing the series expansion in terms of  $\kappa Ma^2$

$$p^* = p_0^* + \kappa Ma^2 p_1^* + \kappa^2 Ma^4 p_2^* + \dots$$

yields

$$\begin{aligned}
\kappa \frac{u_{ref}^2}{c_{ref}^2} \left[ \frac{\partial \varrho^* u_i^*}{\partial t^*} + \frac{\partial \varrho^* u_i^* u_j^*}{\partial x_j^*} \right] &= -\frac{\partial p_0^*}{\partial x_i^*} - \kappa Ma^2 \frac{\partial p_1^*}{\partial x_i^*} - \kappa^2 Ma^4 \frac{\partial p_2^*}{\partial x_i^*} - \dots \\
&+ \frac{\mu_{ref}}{\varrho_{ref} u_{ref} L_{ref}} \kappa \frac{u_{ref}^2}{c_{ref}^2} \frac{\partial}{\partial x_j^*} \left[ \mu^* \left( \frac{\partial u_i^*}{\partial x_j^*} + \frac{\partial u_j^*}{\partial x_i^*} \right) - \frac{2}{3} \mu^* \frac{\partial u_k^*}{\partial x_k^*} \delta_{ij} \right] \\
\kappa Ma^2 \left[ \frac{\partial \varrho^* u_i^*}{\partial t^*} + \frac{\partial \varrho^* u_i^* u_j^*}{\partial x_j^*} \right] &= -\frac{\partial p_0^*}{\partial x_i^*} - \kappa Ma^2 \frac{\partial p_1^*}{\partial x_i^*} - \kappa^2 Ma^4 \frac{\partial p_2^*}{\partial x_i^*} - \dots \\
&+ \frac{1}{Re} \kappa Ma^2 \frac{\partial}{\partial x_j^*} \left[ \mu^* \left( \frac{\partial u_i^*}{\partial x_j^*} + \frac{\partial u_j^*}{\partial x_i^*} \right) - \frac{2}{3} \mu^* \frac{\partial u_k^*}{\partial x_k^*} \delta_{ij} \right]
\end{aligned}$$

Equating the corresponding orders of magnitude gives

$$\begin{aligned}
Ma^0 : \quad 0 &= \frac{\partial p_0^*}{\partial x_i^*} \quad \Rightarrow p_0^* \neq p_0^*(x_i^*)!! \\
Ma^2 : \quad \frac{\partial \varrho^* u_i^*}{\partial t^*} + \frac{\partial \varrho^* u_i^* u_j^*}{\partial x_j^*} &= -\frac{\partial p_1^*}{\partial x_i^*} \\
&+ \frac{\mu^*}{Re} \frac{\partial}{\partial x_j^*} \left[ \left( \frac{\partial u_i^*}{\partial x_j^*} + \frac{\partial u_j^*}{\partial x_i^*} \right) - \frac{2}{3} \mu^* \frac{\partial u_k^*}{\partial x_k^*} \delta_{ij} \right] \quad (I.5)
\end{aligned}$$

It directly follows for the Low Mach-number Approximation:

- The thermodynamic pressure level  $p_0$  (= base pressure) is **invariant in space**, and is **not affected by the flow**.
- The flow field is coupled with the first-order terms of the pressure  $p_1 \ll p_0$  (Equation (I.5)) .
- The first-order (dynamic) pressure  $p_1$  has to be computed from the simultaneous solution of the momentum equations (I.5) and the mass conservation equation.  
 $\rightarrow p_1$  obtained as solution from **Poisson equation**

## Poisson equation for pressure

We assume low Mach-number flow (=incompressible) flow with **constant density**  $\rho = \text{const}$  ( $\rightarrow Ma = 0$ )

$$\frac{\partial u_j}{\partial x_j} = 0 \quad (\text{I.6})$$

$$\frac{\partial u_i}{\partial t} + \frac{\partial u_i u_j}{\partial x_j} = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \frac{\mu}{\rho} \frac{\partial^2 u_i}{\partial x_j \partial x_j}, \quad i = 1, 2, 3 \quad (\text{I.7})$$

(I.6)&(I.7) gives four equations for four unknowns  $u_1, u_2, u_3$  and  $p$ .

$\rightarrow$  Divergence of momentum equation,  $\frac{\partial}{\partial x_i}(\text{I.7})$ , and enforcing continuity (I.6) gives the Poisson equation for the pressure:

$$\frac{1}{\rho} \frac{\partial^2 p}{\partial x_i \partial x_i} = -\frac{\partial u_j}{\partial x_i} \frac{\partial u_i}{\partial x_j} \quad (\text{I.8})$$

Coupled System of equations (I.7)&(I.8) would have to be simultaneously solved for unknowns  $u_1, u_2, u_3$  and  $p$ .

- Numerical (discretized) solution of (I.7)&(I.8):  
mostly with

“Projection Method” = Two-step solution procedure  
 (“pressure-based solver”)

see section III.2 .

## II Discretization of governing equations

We recall the governing set of equations for  $\rho = \text{const.}$ :

$$\begin{aligned}\frac{\partial u_i}{\partial t} + \frac{\partial u_i u_j}{\partial x_j} &= -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \frac{\mu}{\rho} \frac{\partial^2 u_i}{\partial x_j \partial x_j}, \quad i = 1, 2, 3 \\ \frac{1}{\rho} \frac{\partial^2 p}{\partial x_i \partial x_i} &= -\frac{\partial u_j}{\partial x_i} \frac{\partial u_i}{\partial x_j}\end{aligned}$$

$\Downarrow$

Discretization  $\partial x_i, \partial t, \partial u_i, \partial p \rightarrow \Delta x_i, \Delta t, \Delta \tilde{u}_i, \Delta \tilde{p}$

$\Downarrow$

System of algebraic equations for discrete  $\tilde{u}_i(\tilde{x}_i, \tilde{t}_n), \tilde{p}(\tilde{x}_i, \tilde{t}_n)$

$$\begin{aligned}\frac{\Delta \tilde{u}_i}{\Delta t} + \frac{\Delta \tilde{u}_i \tilde{u}_j}{\Delta x_j} &= -\frac{1}{\rho} \frac{\Delta \tilde{p}}{\Delta x_i} + \frac{\mu}{\rho} \frac{\Delta^2 \tilde{u}_i}{\Delta x_j \Delta x_j}, \quad i = 1, 2, 3 \\ \frac{1}{\rho} \frac{\Delta^2 \tilde{p}}{\Delta x_i \Delta x_i} &= -\frac{\Delta \tilde{u}_j}{\Delta x_i} \frac{\Delta \tilde{u}_i}{\Delta x_j}\end{aligned} \tag{II.1}$$

### II.1 Discrete representation of a function

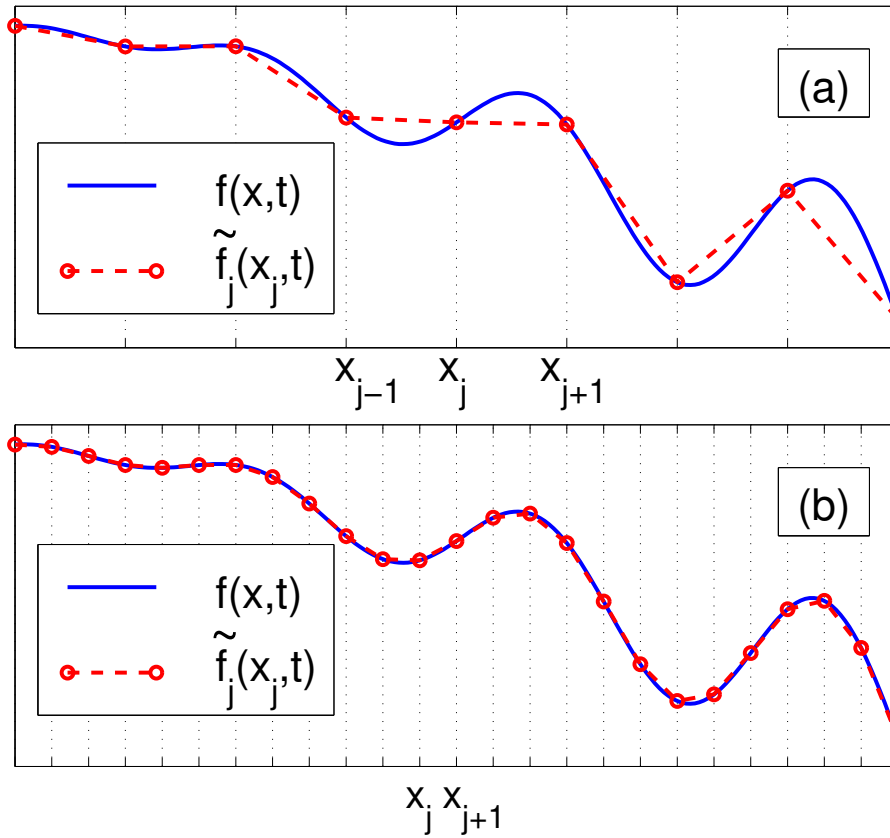
continuous  
function

$f(\vec{x}, t) \Rightarrow$

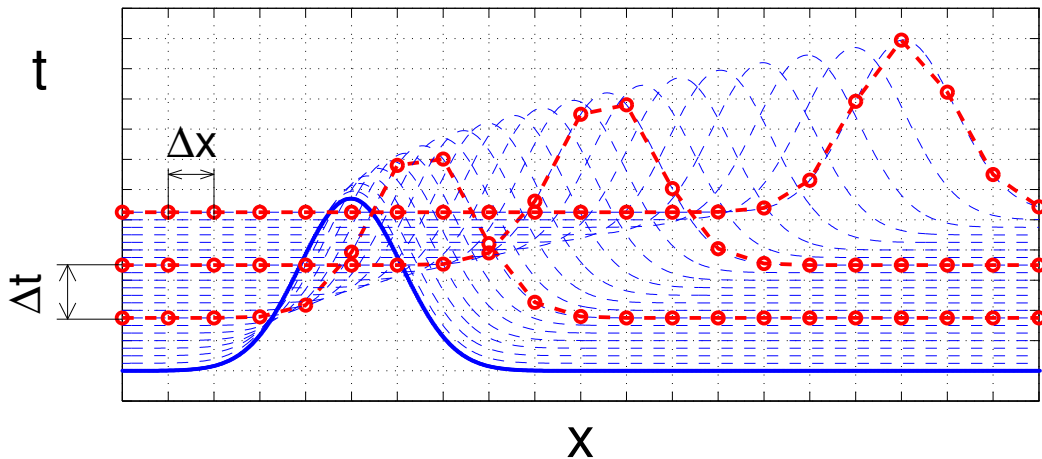
spatially and temporally  
discretized function

with  $\begin{cases} \tilde{f}_j(\vec{x}_j, \tilde{t}_n) \\ j = 1, \dots, J \\ n = 1, \dots, N \end{cases}$

Number of mesh points in space ( $J$ ) and in the time interval ( $N$ ) determines the spatially and temporally discrete solution in  $\Delta\vec{x}_j$  and  $\Delta t_n$ .  
 $\Rightarrow \Delta\vec{x}_j, \Delta t_n$  significant for the discretization error,  
i.e. for the accuracy of the discrete solution  $\tilde{f}_j(\vec{x}_j, \tilde{t}_n)$



Spatial discretization of a function  $f(x, t)$ : (a) coarse, (b) fine



Spatial and temporal discretization of a wave movement

## II.2 Accuracy of the discretization

### II.2.1 Accuracy (order) of the spatial discretization

$$\begin{array}{ccc} \text{differentials} & & \text{differences} + \text{discretization error} \\ \frac{\partial^n(\cdot)}{\partial x^n} & \xrightarrow{\text{discretis.}} & \frac{\Delta_n(\cdot)}{\Delta x^n} + \varepsilon_x \end{array}$$

The lowest order of the error, as determined by the discretization error

$$\varepsilon_x = O[\Delta x_j^p],$$

defines the order (accuracy) of the spatial discretization.

This is called a  $p$ -th order accurate discretization in space.

Generally:  $p \uparrow \Rightarrow$  accuracy of the solution  $\uparrow$ .

Examples:

Approximations of  $\frac{\partial f}{\partial x}$  with  $\Delta x = x_j - x_{j-1} = \text{const.}$ :

$$\frac{\partial f}{\partial x} = \frac{f_j - f_{j-1}}{\Delta x} + O[\Delta x] \quad \text{Back Diff.}$$

$$\frac{\partial f}{\partial x} = \frac{f_{j+1} - f_{j-1}}{2\Delta x} + O[\Delta x^2] \quad \text{3PT Cent. Diff.}$$

$$\frac{\partial f}{\partial x} = \frac{f_{j-2} - 8f_{j-1} + 8f_{j+1} - f_{j+2}}{12\Delta x} + O[\Delta x^4] \quad \text{5PT Cent. Diff.}$$



## II.2.2 Accuracy of the time integration scheme

$$\frac{\partial f(x, t)}{\partial t} = F(x, t, f)$$

$$\rightarrow \text{Num. scheme} \quad f^{n+1} = f^n + \Delta t \tilde{L}[F^{n+1}, F^n, F^{n-1}, \dots] + \underbrace{O[\Delta t^{q+1}]}_{\varepsilon_t}$$

$$\text{with } \Delta t = t^{n+1} - t^n$$

The largest error  $\varepsilon_t = O[\Delta t^{q+1}]$ , which occurs in numerical integration using the scheme  $\tilde{L}$  determines its order of accuracy.

$\hookrightarrow$  We call  $\tilde{L}$  a “ $q$ -th order accurate scheme”.

Examples for integration schemes in time with  $\Delta t = t^{n+1} - t^n = \text{const}$ :

$$f^{n+1} = f^n + \Delta t F^n + O[\Delta t^2] \quad \text{Euler}$$

$$f^{n+1} = f^n + \Delta t \left( \frac{3}{2} F^n - \frac{1}{2} F^{n-1} \right) + O[\Delta t^3] \quad \text{Adams-Bashforth}$$

$$f^{n+1} = f^{n-1} + 2 \Delta t F^n + O[\Delta t^3] \quad \text{Leapfrog}$$

$$f^{n+1} = f^n + \frac{\Delta t}{6} \left[ \underbrace{F(x, t^n, f^n)}_{F_1} + 2 \underbrace{F(x, t^n + \Delta t/2, f^n + F_1 \Delta t/2)}_{F_2} \right. \\ \left. + 2 \underbrace{F(x, t^n + \Delta t/2, f^n + F_2 \Delta t/2)}_{F_3} + \underbrace{F(x, t^n + \Delta t, f^n + F_3 \Delta t)}_{F_4} \right]$$

$$+ O[\Delta t^5] \quad \text{Runge-Kutta}$$

## Caveats when applying higher-order schemes

The error gets smaller:  $p, q \uparrow \Rightarrow O[\Delta x^p], O[\Delta t^q] \downarrow$

but

- stencils gets wider
  - Computing time + required storage increase
  - more points for BCs + ICs required
- lower numerical diffusion is more sensitive to short-wave perturbations
  - prone to instability

## II.3 Methods for discretization of differential equations

- Finite-Difference (FD)
- Finite-Volume (FV)
- Finite-Element (FE)

## II.3.1 Finite-Difference Method

Computes differential terms  $\partial^n(\cdot)/\partial x^n \approx \Delta_n(\cdot)/\Delta x^n + \varepsilon_x$  from Taylor series expansions at grid point  $m$  :

$$\begin{aligned}
 (i) \quad f_{m+1} &= f_m + \Delta x \left( \frac{\partial f}{\partial x} \right)_m + \frac{\Delta x^2}{2} \left( \frac{\partial^2 f}{\partial x^2} \right)_m + \frac{\Delta x^3}{6} \left( \frac{\partial^3 f}{\partial x^3} \right)_m + \frac{\Delta x^4}{24} \left( \frac{\partial^4 f}{\partial x^4} \right)_m + \dots \\
 (ii) \quad f_{m-1} &= f_m - \Delta x \left( \frac{\partial f}{\partial x} \right)_m + \frac{\Delta x^2}{2} \left( \frac{\partial^2 f}{\partial x^2} \right)_m - \frac{\Delta x^3}{6} \left( \frac{\partial^3 f}{\partial x^3} \right)_m + \frac{\Delta x^4}{24} \left( \frac{\partial^4 f}{\partial x^4} \right)_m - \dots \\
 (iii) \quad f_{m+2} &= f_m + 2\Delta x \left( \frac{\partial f}{\partial x} \right)_m + \frac{4\Delta x^2}{2} \left( \frac{\partial^2 f}{\partial x^2} \right)_m + \frac{8\Delta x^3}{6} \left( \frac{\partial^3 f}{\partial x^3} \right)_m + \frac{16\Delta x^4}{24} \left( \frac{\partial^4 f}{\partial x^4} \right)_m + \dots \\
 (iv) \quad f_{m-2} &= f_m - 2\Delta x \left( \frac{\partial f}{\partial x} \right)_m + \frac{4\Delta x^2}{2} \left( \frac{\partial^2 f}{\partial x^2} \right)_m - \frac{8\Delta x^3}{6} \left( \frac{\partial^3 f}{\partial x^3} \right)_m + \frac{16\Delta x^4}{24} \left( \frac{\partial^4 f}{\partial x^4} \right)_m - \dots
 \end{aligned}$$

Truncation error  $\Delta x^n \rightarrow$  Order of the discretization

e.g. ► “3 Points Central Differences”

$$\begin{aligned}
 (i) - (ii) &\rightarrow \left( \frac{\partial f}{\partial x} \right)_m = \frac{f_{m+1} - f_{m-1}}{2\Delta x} + \mathcal{O}[\Delta x^2] \\
 (i) + (ii) &\rightarrow \left( \frac{\partial^2 f}{\partial x^2} \right)_m = \frac{f_{m+1} - 2f_m + f_{m-1}}{\Delta x^2} + \mathcal{O}[\Delta x^2]
 \end{aligned}$$

► “5 Points Central Differences”

$$\begin{aligned}
 8(i) - 8(ii) - (iii) + (iv) \\
 \hookrightarrow \left( \frac{\partial f}{\partial x} \right)_m &= \frac{-f_{m+2} + 8f_{m+1} - 8f_{m-1} + f_{m-2}}{12\Delta x} + \mathcal{O}[\Delta x^4]
 \end{aligned}$$

$$\begin{aligned}
 16(i) + 16(ii) - (iii) - (iv) \\
 \hookrightarrow \left( \frac{\partial^2 f}{\partial x^2} \right)_m &= \frac{-f_{m+2} + 16f_{m+1} - 30f_m + 16f_{m-1} - f_{m-2}}{12\Delta x^2} + \mathcal{O}[\Delta x^4]
 \end{aligned}$$

⊕ simple mathematical formulation

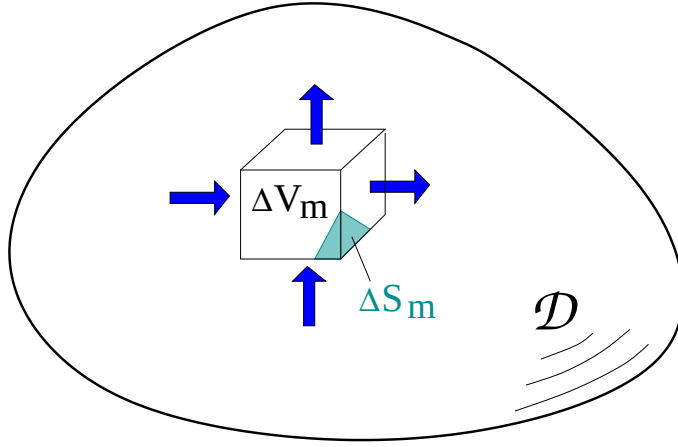
⊕ exactly defined discretization error

⊖ not conservative!

⊖ limited to structured, essentially orthogonal grids

## II.3.2 Finite-Volume Method

Integrates differential conservation equations over a finite discrete control volume  $\Delta V_m$ .



e.g. conservation of general quantity  $\Phi$ :

$$\int_{\Delta V_m} \left[ \frac{\partial \Phi}{\partial t} + \frac{\partial A_j}{\partial x_j} + \frac{\partial D_{ji}}{\partial x_j} + \Omega(\Phi) \right] dV = 0$$

Procedure and **assumptions for (discretized) integration**:

- ▶ Conserved quantity  $\Phi$  is constant within the whole control volume
- ▶ Source term  $\Omega(\Phi)$  is constant within the whole control volume
- ▶ Application of the Gauss' divergence theorem

$$\int_{\Delta V_m} \frac{\partial A_j}{\partial x_j} dV = \oint_{\Delta S_m} A_j n_j dS \quad \text{and} \quad \int_{\Delta V_m} \frac{\partial D_{ji}}{\partial x_j} dV = \oint_{\Delta S_m} T_{ji} n_j dS$$

i.e. volume integral of the divergence = *flux balance* across the surface  $\Delta S_m$  of the control volume  $\Delta V_m$ !

$\Rightarrow$  produces an algebraic equation for  $\Phi_m$  written for an arbitrary control volume  $\Delta V_m$  as

$$\frac{\Delta_t \Phi_m}{\Delta t} \Delta V_m + \sum_n (A_{mn} + T_{mni}) \Delta S_{mn} + \Omega_m \Delta V_m = 0$$

The **accuracy** of the estimation of the flux terms  $A_{mn}$  and  $T_{mni}$ , respectively, on the surfaces  $\Delta S_{mn}$  determines

$\Rightarrow$  discretization error  $\varepsilon$  (= order of accuracy)

e.g. flux in  $x$ -direction  $A(x)$  can be approximated using Lagrange polynomials:

$$A(x) \cong \sum_{i=m-\alpha}^{m+\beta} A_i \prod_{\substack{j=m-\alpha \\ i \neq j}}^{m+\beta} \frac{x - x_j}{x_i - x_j}$$

The evaluation of this polynomial at  $x = x_{mn}$  gives

$$\hookrightarrow A_{mn} = A(x_{mn}) \quad \dots \text{ with error } \varepsilon = \mathcal{O}[\Delta x^{\alpha+\beta+1}]$$

$\oplus$  By definition conservative

$\ominus$  Increase of the accuracy by the use of a more accurate approximation of the flux terms is often unfeasible (particularly on unstructured grids).

Therefore, implementation in most cases only up to the second-order accuracy ( $\varepsilon = \mathcal{O}[\Delta x^2]$ ).

## II.3.3 Finite-Element Method

Approximates the solution at the grid points  $m = 1, \dots, M$  by the use of the ansatz function

$$\tilde{f}(\vec{x}, t) = \sum_{m=1}^M f_m(t) \psi_m(\vec{x}) \quad (*)$$

with the interpolation function (“test function”)

$$\Psi_m(\vec{x}) = \begin{cases} 1, & \vec{x} = \vec{x}_m \\ 0, & \vec{x} = \vec{x}_n, n \neq m \end{cases}$$

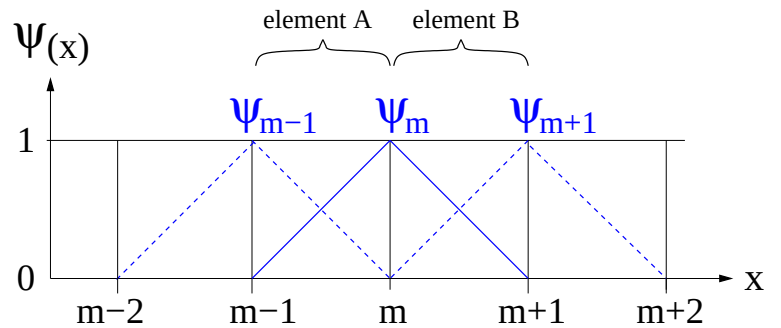
e.g.: linear interpolation in 1-D

$$\Psi_m^{(A)} = \frac{x - x_{m-1}}{x_m - x_{m-1}}$$

$$\Psi_m^{(B)} = \frac{x_{m+1} - x}{x_{m+1} - x_m}$$

$$A : \tilde{f} = f_{m-1} \Psi_{m-1}^{(B)} + f_m \Psi_m^{(A)}$$

$$B : \tilde{f} = f_m \Psi_m^{(B)} + f_{m+1} \Psi_{m+1}^{(A)}$$



The ansatz function (\*) introduced into partial differential equation  $L(\vec{x}, t, f(\vec{x}, t)) = 0$  yields

$$L(\vec{x}, t, \tilde{f}) = R$$

Minimization of residual  $R$  by weighted integration over the whole computational domain

$$\int_V W(\vec{x}) L(\vec{x}, t, \tilde{f}) dV = 0$$

using a weighting function of equal type as the test function  
i.e.  $W(\vec{x}) = \Psi_n(\vec{x})$  (Galerkin)

$$\int_V \Psi_n(\vec{x}) L(\vec{x}, t, \tilde{f}) dV = 0, \quad n = 1, \dots, M$$

$\hookrightarrow M$  equations for  $f_m(t)$  at all  $M$  grid points of the computational domain.

- $\oplus$  Accuracy can be adapted by the choice of the test function  $\Psi_m$
- $\oplus$  Interpolation ansatz is convenient for triangulation, problem-adapted, very contorted grids are possible
- $\ominus$  badly applicable for non linear differential equations  
 $\rightarrow$  quadratic equations for  $f_m(t)$ !  
e.g.: Navier-Stokes

$$u \frac{\partial u}{\partial x} \quad \text{with} \quad u = \tilde{u}_m \Psi_m(x) \Rightarrow \tilde{u}_m^2 \Psi_m \frac{\partial \Psi_m}{\partial x}$$

therefore: Finite-Element method is mainly used for linear differential equations (e.g. heat transfer, structural mechanics, laminar flows with a small Reynolds-number)

## II.4 Numerical error analysis

we recall:  $\frac{\partial f(x, t)}{\partial t} = F(x, t, f)$

Num. scheme  $\tilde{f}^{n+1} = \tilde{f}^n + \Delta t \tilde{L} [F^{n+1}, F^n, F^{n-1}, \dots] + \underbrace{\varepsilon(\Delta x, \Delta t)}_{\text{Numerical error}}$

A **continuous** solution  $f(\vec{x}, t)$  can be rewritten as a superposition of an infinite number of elementary waves (Fourier decomposition):

$$f(\vec{x}, t) = \sum_{m=-\infty}^{\infty} \hat{f}_m e^{i\vec{k}_m[\vec{x}-\vec{q}_m t]}.$$

Based on this transform from the physical space into the wave space, we can distinguish 4 types of errors, which can be introduced by the application of a discrete numerical scheme  $\tilde{L}_j(\vec{x}_j, t^n, f_j)$ :

1. **Effective wave number:** discretization of the spatial derivatives  $(\partial^n / \partial x^n) \rightarrow (\Delta^n / \Delta x^n)$  yields an “effective wave number”, which differs from the real (spectral) wave number.  
 $\hookrightarrow$  accuracy of FD-approximation of derivatives
2. **Aliasing:** scheme misinterprets high-frequency content (= short-wave length contributions with a high wave number  $\vec{k}_m$ ) as low-frequency (=long-wave length) content.  
 $\hookrightarrow$  insufficient resolution
3. **Numerical diffusion:** non-physical artificial damping of the wave amplitude  $\hat{f}_m$
4. **Numerical dispersion:** scheme alters the propagation velocity of the elementary waves  $\vec{q}_m$ .  
 $\hookrightarrow$  3. & 4. originate from error terms in “equivalent system”



We quantify the individual errors using a

→ *Fourier analysis* of a *discrete* solution  $\tilde{f}_j(x_j, t)$ .

## Fourier analysis of a discrete function

We consider a discrete spatial 1-dimensional numerical solution

$$\tilde{f}_j(x_j) \quad \text{on} \quad x_j : j = 0, \dots, N-1$$

mesh points with

$$\Delta x = x_{j+1} - x_j = \text{const.} \quad \text{so that} \quad x_j = j\Delta x.$$

The function be periodic in the interval  $L = x_N - x_0 = N\Delta x$ , or be assumed periodically continued with periodic length  $L$ , so that

$$\tilde{f}_N = \tilde{f}_0.$$

→  $\tilde{f}_j$  can be represented as a discrete Fourier series:

$$\tilde{f}_j = \sum_{m=0}^{N-1} \hat{f}_m e^{i k_m j \Delta x} \quad (*)$$

Therein,  $i = \sqrt{-1}$  is the imaginary unit. The wave number  $k_m$  is

$$k_m = \frac{2\pi}{\lambda_m} = \frac{2\pi m}{L}, \quad \text{with} \quad \lambda_m = \frac{L}{m}$$

being the wave length of the mode  $m$ . The Fourier transform  $\hat{f}_m$  is simply computed from the series

$$\hat{f}_m = \frac{1}{N} \sum_{j=0}^{N-1} \tilde{f}_j e^{-i k_m j \Delta x},$$

due to the orthogonality of the basis function  $e^{i\Phi} = \cos\Phi + i \sin\Phi$

**We note:** for a **real** function  $\tilde{f}_j \in \mathbb{R} \rightarrow$  imaginary components in the general formulation (\*) cancel, as

$$\hat{f}_m = \hat{f}_{N-m}^* \Rightarrow \left. \begin{aligned} \hat{f}_m &= \hat{a}_m - i \hat{b}_m \\ \hat{f}_{N-m} &= \hat{a}_m + i \hat{b}_m \end{aligned} \right\} \text{“Symmetry”}$$

$\hookrightarrow$  the highest resolvable mode for an even number of points ( $N \in \mathbb{N}_g$ ) of a real function  $\tilde{f}_j \in \mathbb{R}$  is

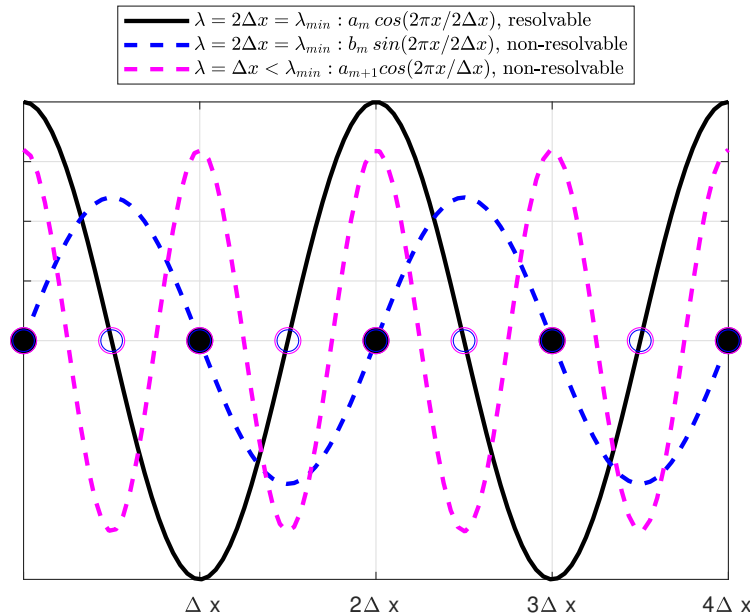
$$m_{max} = \frac{N}{2}.$$

The corresponding maximum resolvable wave number is

$$k_{max} = \frac{\pi}{\Delta x},$$

so that the shortest resolvable wave length at the given mesh width  $\Delta x$  is

$$\lambda_{min} = \frac{2\pi}{k_{max}} = 2\Delta x$$



Shortest resolvable wave length for a mesh size  $\Delta x$ , waves with  $\lambda < 2\Delta x$  (broken lines) can not be captured

## II.4.1 Effective wave number

Differentiation  $\partial/\partial x$  of the function

$$f(x) = \sum_{m=0}^{N-1} \hat{f}_m e^{i k_m x} \quad \left| \quad \frac{\partial}{\partial x} \right.$$

yields for any mode  $m$  a spectral differential

$$\frac{\partial(\hat{f}_m e^{i k_m x})}{\partial x} = \hat{f}_m i k_m e^{i k_m x}$$

involving a multiplication with the wave number  $k_m$ .

The sum of all modes  $m$  at  $x = x_j$  yields the derivative in physical space

$$\left[ \frac{\partial f}{\partial x} \right]_j = \sum_{m=0}^{N-1} \hat{f}_m i k_m e^{i k_m x_j}. \quad (*)$$

The **discrete FD approximation** ( $\Delta/\Delta x$ ) yields e.g. with the central 2<sup>nd</sup>-order accurate difference scheme (3PTCD) for any mode  $m$  at  $x_j$ :

$$\left[ \frac{\partial(e^{i k_m x})}{\partial x} \right]_j \approx \frac{e^{i k_m (x_j + \Delta x)} - e^{i k_m (x_j - \Delta x)}}{2\Delta x} = i \underbrace{\frac{\sin(k_m \Delta x)}{\Delta x}}_{k_{eff}} e^{i k_m x_j}$$

involving the multiplication with an “effective” wave number  $k_{eff}$ .

The effective wave number is different from the accurate wave number

$$\rightarrow k_{eff} = \frac{\sin(k_m \Delta x)}{\Delta x} \neq k_m !$$

Analogously the discrete approximation with the 4<sup>th</sup>-order accurate central difference scheme (5PTCD) yields

$$\rightarrow k_{eff} = \frac{\sin(k_m \Delta x)}{3\Delta x} [4 - \cos(k_m \Delta x)]$$

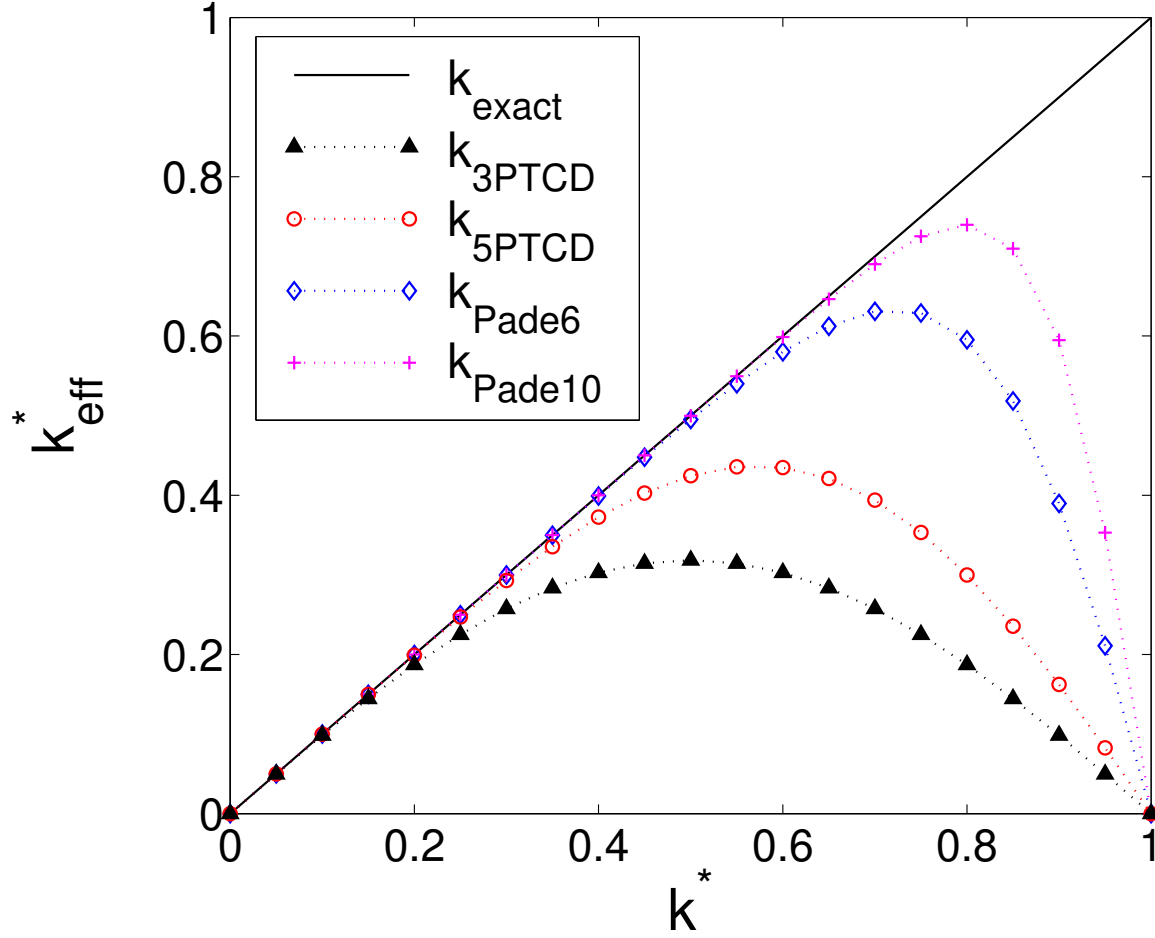


Diagram of the effective wave number  $k_{eff}^* = k_{eff}/k_{max}$ , normalized with the maximum wave number  $k_{max} = \pi/\Delta x$  versus the accurate (=spectral)  $k^* = k_m/k_{max}$ .

**Note: The difference  $k_{exact} - k_{eff}$ , i.e. the discretization error, increases with the wave number!**

## II.4.2 Aliasing

For **non-linear** terms the wave number is additively increased, e.g.

$$\begin{aligned}
 f \frac{\partial f}{\partial x} &= \sum_{m=0}^{N-1} \hat{f}_m e^{i k_m x} \cdot \sum_{n=0}^{N-1} \hat{f}_n i k_n e^{i k_n x} \\
 &= \sum_{m=0}^{N-1} \sum_{n=0}^{N-1} \hat{f}_m \hat{f}_n i k_n e^{i (k_m + k_n) x}
 \end{aligned}$$

The wave number, which results from this superposition, can exceed the maximum resolvable wave number of the given mesh:

$$k_m + k_n = k_q > k_{max} = \frac{\pi}{\Delta x}$$

On a given numerical mesh, the not resolvable high-frequency modes  $k_q > k_{max}$  will be misinterpreted as a low-frequency modes because of

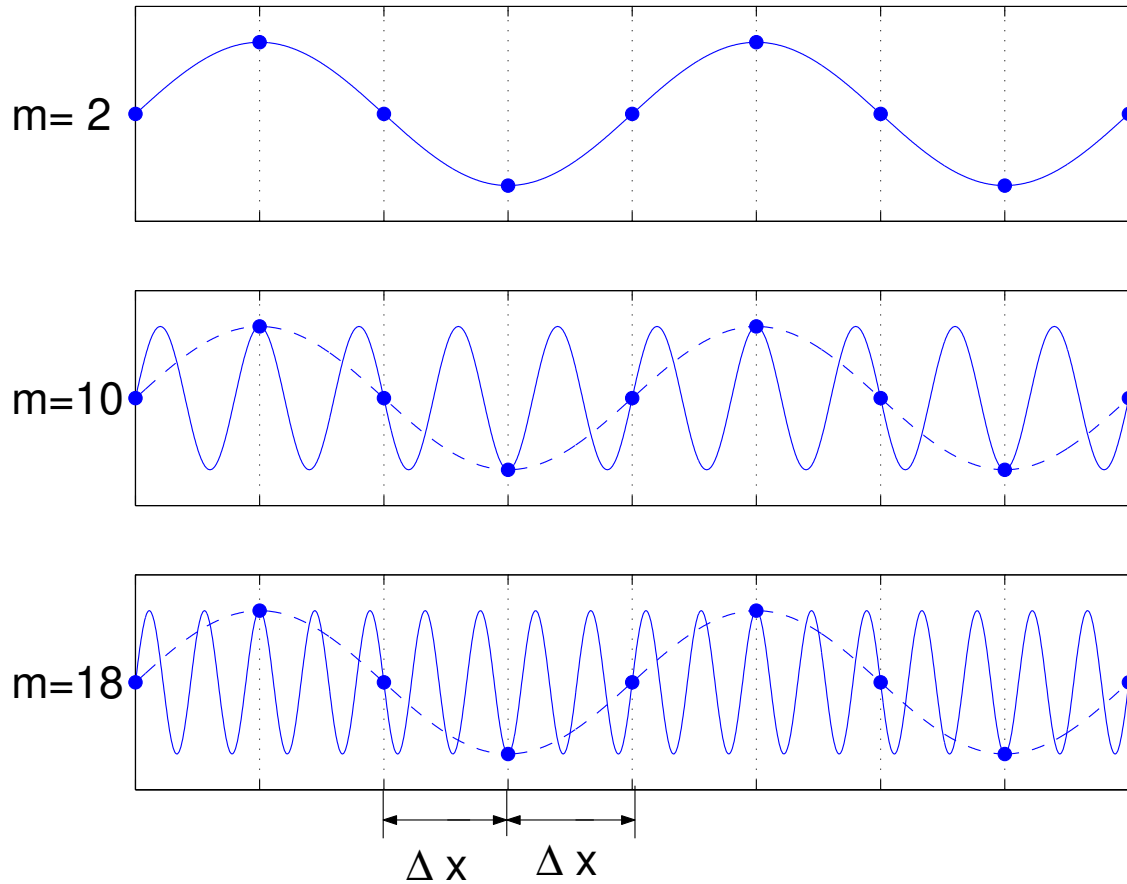
$$e^{i(\Phi)} = e^{i(\Phi + n 2\pi)}$$

Thus:

$$\forall k_q > k_{max} : \quad k_q \rightarrow k_{q-n \cdot N} = k_{alias},$$

where  $n = 1, 2, 3, \dots$ , so that  $q - n \cdot N \leq \frac{N}{2} \Leftrightarrow k_{alias} \leq k_{max}$

The “ $k_{alias}$ -waves” may produce instability, unless sufficient dissipation is present (physical or numerical), which “dampens” the high-frequency  $k_q$ -wave content.



Example for aliasing. Number of the grid points:

$$N = 8 \rightarrow k_{max} = k_4 = \frac{2\pi \cdot 4}{L}$$

“Aliasing” of the high-frequency mode  $k_{10}$  and  $k_{18}$ :

$$k_{10} \rightarrow k_{10-1 \cdot 8} = k_2 = k_{alias}$$

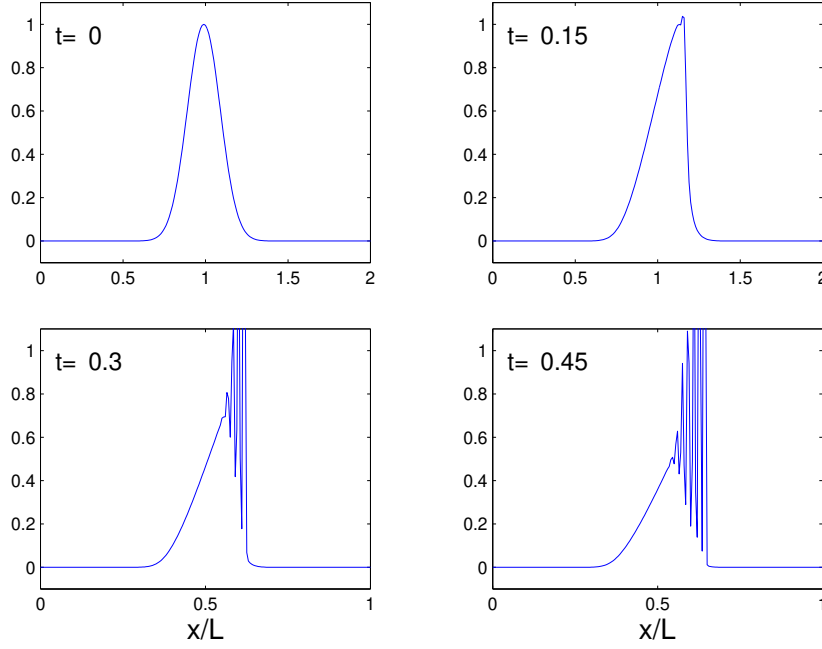
$$k_{18} \rightarrow k_{18-2 \cdot 8} = k_2 = k_{alias}$$

The really occurring high-frequency components  $k_{10}, k_{18}$  are misinterpreted on the given mesh of size  $\Delta x$  as low-frequency mode with  $k_{alias} = k_2$ .

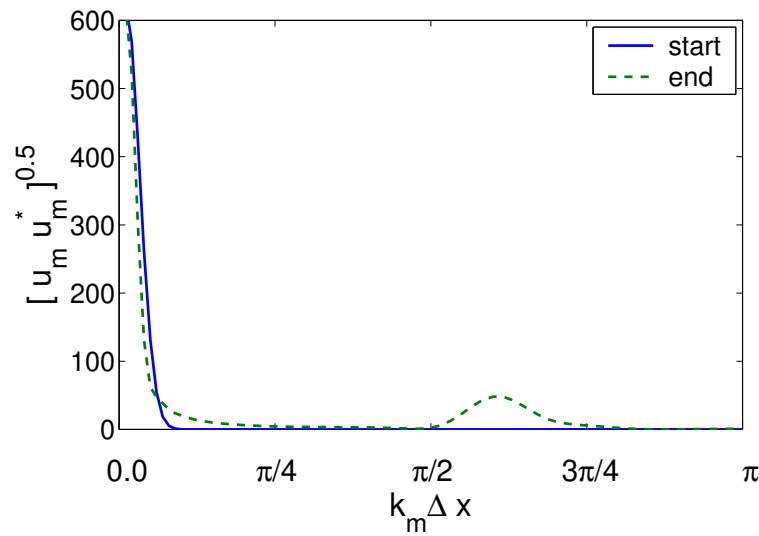
Fig.(a)-(b): Aliasing in the numerical solution of the viscous  
 $(\nu = 0.0001 \text{ fixed})$  Burgers equation

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}$$

using the *EULER+3PTCD*-scheme



(a)  $T(x, t)$  in the physical area  $x/L$



(b) Energy spectrum at the beginning and at the end of the  
 integration

### II.4.3 Numerical diffusion (dissipation) and dispersion

For demonstrating the effect of dissipation and dispersion, we first consider analytically a 1-dimensional linear convection equation with higher spatial derivatives ( $\partial^2/\partial x^2$ ) and ( $\partial^3/\partial x^3$ ) on the RHS.

$$\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} = \alpha \frac{\partial^2 T}{\partial x^2} \quad (A)$$

$$\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} = -\beta \frac{\partial^3 T}{\partial x^3} \quad (B)$$

with  $u = \text{const.}$  The general ansatz for the analytical solution of a 1-dimensional wave, which is subject to a possible dissipation and dispersion, reads for any mode  $m$

$$\bar{T}(x, t) = \underbrace{T_{m,0} e^{-p_m t}}_{T_m(t)} e^{i k_m [x - q_m t]}. \quad (*)$$

$k_m$  is the wave number, the parameter  $p_m$  determines the temporal change of the wave amplitude  $T_m(t)$ , the parameter  $q_m$  determines the propagation velocity of the elementary wave of mode  $m$ .

Ansatz (\*) introduced into (A) and (B) yields

$$p_m = \alpha k_m^2, \quad \text{and} \quad q_m = u - \beta k_m^2,$$

respectively

**Dissipation:** if  $\alpha \geq 0$  in (A)  $\rightarrow p_m \sim k_m^2 > 0$

$\Rightarrow$  the damping of high-frequency modes ( i.e., short-wavelength,  $k_m$  large) is much higher than the damping of low-frequency modes (long-wavelength)



**Dispersion:** if  $\beta \neq 0$  in  $(B) \rightarrow u - q_m = \beta k_m^2 \neq 0$   
 $\Rightarrow$  the wave propagation velocities of the individual modes are different (the wave is dispersive).  
 if  $\beta > 0$  : the propagation of high-frequency modes ( i.e., short-wavelength,  $k_m$  large) is much slower than the propagation of low-frequency modes (long-wavelength)

Physical diffusion and dispersion may naturally occur in the differential equation system  $L(x, t, f) = 0$ , which describes the considered problem.

However: They may also show up as purely

**numerical (= unphysical) diffusion and dispersion**

in the error term of the “**equivalent system**”  $L(x_j, t^n, \tilde{f}_j) = \tilde{\varepsilon}$ .

## Equivalent System

Numerical solution of PDE  $L(x, t, T) = 0$ :

- (1) discrete approx.:  $L(x, t, T) \approx \tilde{L}_j(x_j, t^n, \tilde{T}_j) + \varepsilon$
- (2) solution  $\tilde{L}_j(x_j, t^n, \tilde{T}_j) = 0 \rightarrow \tilde{T}_j(x_j, t^n)$
- (3)  $\hookrightarrow$  equiv. syst.:  $L_j(x_j, t^n, \tilde{T}_j) = \underbrace{\varepsilon(\tilde{T}_j)}_{=\tilde{\varepsilon}}$

Example: We consider a 1-dim. linear convection PDE for  $T(x, t)$ :

$$L(x, t, T) = \frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} = 0, \quad \text{with } u = \text{const.}$$

For the discretization in time and space we use *Euler+3PT Cent.Diff*

$$\begin{aligned} \tilde{L}_j(x_j, t^n, T_j) + \varepsilon &= L(x, t, T) \\ \rightarrow \quad \frac{\tilde{T}_j^{n+1} - \tilde{T}_j^n}{\Delta t} + u \frac{\tilde{T}_{j+1}^n - \tilde{T}_{j-1}^n}{2\Delta x} + \varepsilon &= \left( \frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} \right)^n \quad (*) \end{aligned}$$

Taylor series expansion at  $T_j^n$  with respect to  $\Delta t$  and  $\pm \Delta x$ :

$$\begin{aligned} \tilde{T}_j^{n+1} &= \tilde{T}_j^n + \Delta t \left( \frac{\partial T}{\partial t} \right)_j^n + \frac{\Delta t^2}{2} \left( \frac{\partial^2 T}{\partial t^2} \right)_j^n + \frac{\Delta t^3}{6} \left( \frac{\partial^3 T}{\partial t^3} \right)_j^n + \dots \\ \tilde{T}_{j+1}^n &= \tilde{T}_j^n + \Delta x \left( \frac{\partial T}{\partial x} \right)_j^n + \frac{\Delta x^2}{2} \left( \frac{\partial^2 T}{\partial x^2} \right)_j^n + \frac{\Delta x^3}{6} \left( \frac{\partial^3 T}{\partial x^3} \right)_j^n + \dots \\ \tilde{T}_{j-1}^n &= \tilde{T}_j^n - \Delta x \left( \frac{\partial T}{\partial x} \right)_j^n + \frac{\Delta x^2}{2} \left( \frac{\partial^2 T}{\partial x^2} \right)_j^n - \frac{\Delta x^3}{6} \left( \frac{\partial^3 T}{\partial x^3} \right)_j^n + \dots \end{aligned}$$

introduced into (\*) results in the discretization error

$$\varepsilon = -\frac{\Delta t}{2} \left( \frac{\partial^2 T}{\partial t^2} \right)_j^n - u \frac{\Delta x^2}{6} \left( \frac{\partial^3 T}{\partial x^3} \right)_j^n = O(\Delta t, \Delta x^2)$$

The numerical solution  $\tilde{T}_j$  substituted into (\*) produces with  $\tilde{L}_j(x_j, t^n, \tilde{T}_j) = 0$  the equivalent system:

$$\begin{aligned} \left( \frac{\partial \tilde{T}}{\partial t} + u \frac{\partial \tilde{T}}{\partial x} \right)_j^n &= \tilde{\varepsilon} = O(\Delta t, \Delta x^2) \quad (**) \\ \text{with } \tilde{\varepsilon} &= -\frac{\Delta t}{2} \left( \frac{\partial^2 \tilde{T}}{\partial t^2} \right)_j^n - \underbrace{u \frac{\Delta x^2}{6} \left( \frac{\partial^3 \tilde{T}}{\partial x^3} \right)_j^n}_{\text{numerical dispersion}} \end{aligned}$$

$$\begin{aligned}
\frac{\partial(**)}{\partial t} &\rightarrow \frac{\partial}{\partial t} \left( \frac{\partial \tilde{T}}{\partial t} \right)_j^n + u \frac{\partial}{\partial t} \left( \frac{\partial \tilde{T}}{\partial x} \right)_j^n = O(\Delta t, \Delta x^2) \\
\frac{\partial}{\partial t} \left( \frac{\partial \tilde{T}}{\partial t} \right)_j^n + u \underbrace{\frac{\partial}{\partial x} \left( \frac{\partial \tilde{T}}{\partial t} \right)_j^n}_{=} &= O(\Delta t, \Delta x^2) \\
&\quad \underbrace{-u \left( \frac{\partial \tilde{T}}{\partial x} \right)_j^n}_{=} + O(\Delta t, \Delta x^2) \\
\underbrace{\left( \frac{\partial^2 \tilde{T}}{\partial t^2} \right)_j^n}_{\rightarrow \tilde{\varepsilon}} &= u^2 \left( \frac{\partial^2 \tilde{T}}{\partial x^2} \right)_j^n + O(\Delta t, \Delta x^2) \quad (*** )
\end{aligned}$$

The discretization error appears in the equivalent system (\*\*) in the leading order as **numerical diffusion and dispersion**:

$$\left( \frac{\partial \tilde{T}}{\partial t} + u \frac{\partial \tilde{T}}{\partial x} \right)_j^n = \underbrace{-u^2 \frac{\Delta t}{2} \left( \frac{\partial^2 \tilde{T}}{\partial x^2} \right)_j^n}_{\text{numerical diffusion}} - \underbrace{u \frac{\Delta x^2}{6} \left( \frac{\partial^3 \tilde{T}}{\partial x^3} \right)_j^n}_{\text{numerical dispersion}} + O(\Delta t^2, \Delta x^2) + \dots$$

## II.4.4 Effect of numerical diffusion and dispersion errors in physical and spectral space

We consider again the linear convection PDE

$$\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} = 0 \quad (*)$$

with  $u = \text{const.} > 0$ , and the initial condition written in Fourier series representation as,

$$\text{IC.: } T(x, 0) = \sum_{m=0}^{\infty} T_m e^{i k_m x}.$$

The accurate solution in the Fourier series representation reads

$$\bar{T}(x, t) = \sum_{m=-\infty}^{\infty} T_m e^{i k_m (x - u t)}$$

The general ansatz for dissipative and dispersive waves is used again for the numerical solution:

$$\bar{T}(x, t) = \sum_{m=-\infty}^{\infty} T_m e^{-p_m t} e^{i k_m [x - q_m t]}. \quad (**)$$

This general ansatz evidently becomes the exact solution if

$$\begin{aligned} p_m &= 0 && \rightarrow \text{no dissipation} \\ q_m &= u = \text{const.} \forall k_m && \rightarrow \text{no dispersion} \end{aligned}$$

Because of the linearity of the problem, any mode  $m$  can be examined separately  $\rightarrow$  only an individual mode  $m$  is exemplarily considered below.

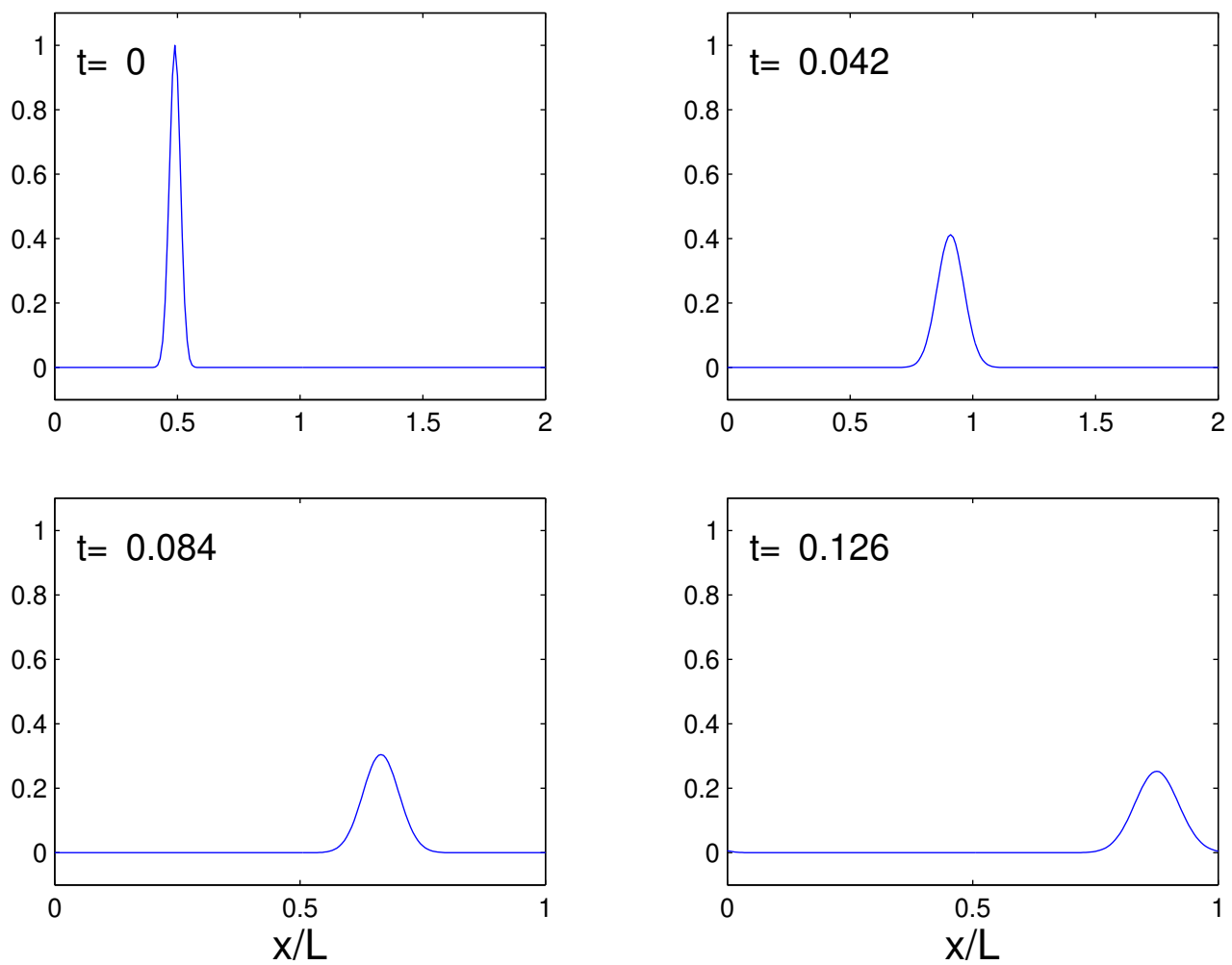
relative change in time: 
$$G_m = \frac{T(x, t + \Delta t)}{T(x, t)} = e^{-p_m \Delta t} e^{-i k_m q_m \Delta t}$$

Example 1: *Euler + 1<sup>st</sup>-order Upwind-scheme*

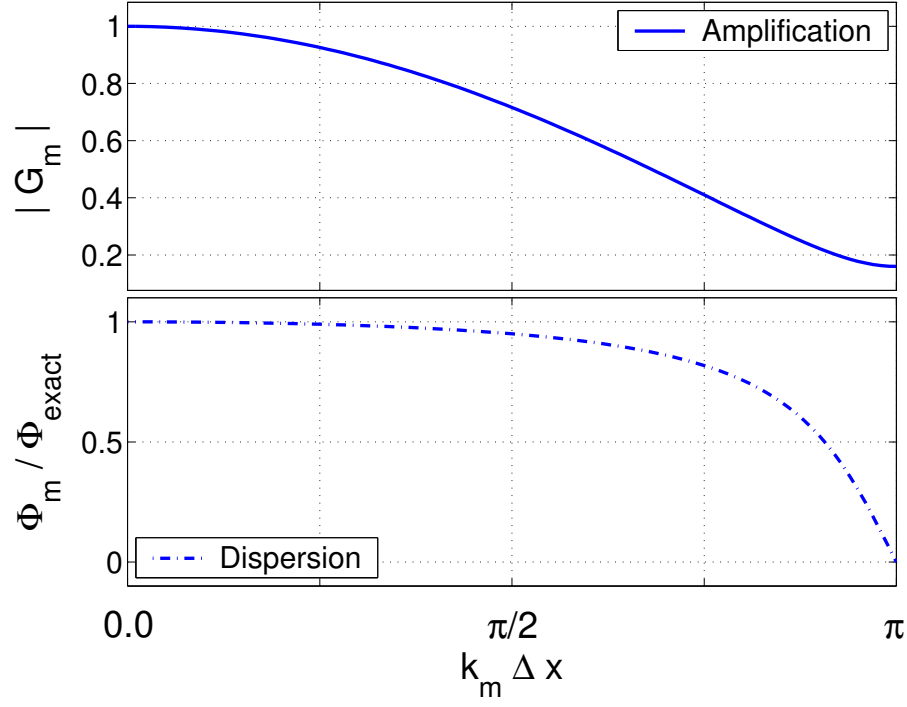
$$\tilde{T}_j^{n+1} = \tilde{T}_j^n - C (\tilde{T}_j^n - \tilde{T}_{j-1}^n) \quad (***)$$

with  $C = u\Delta t/\Delta x$ .

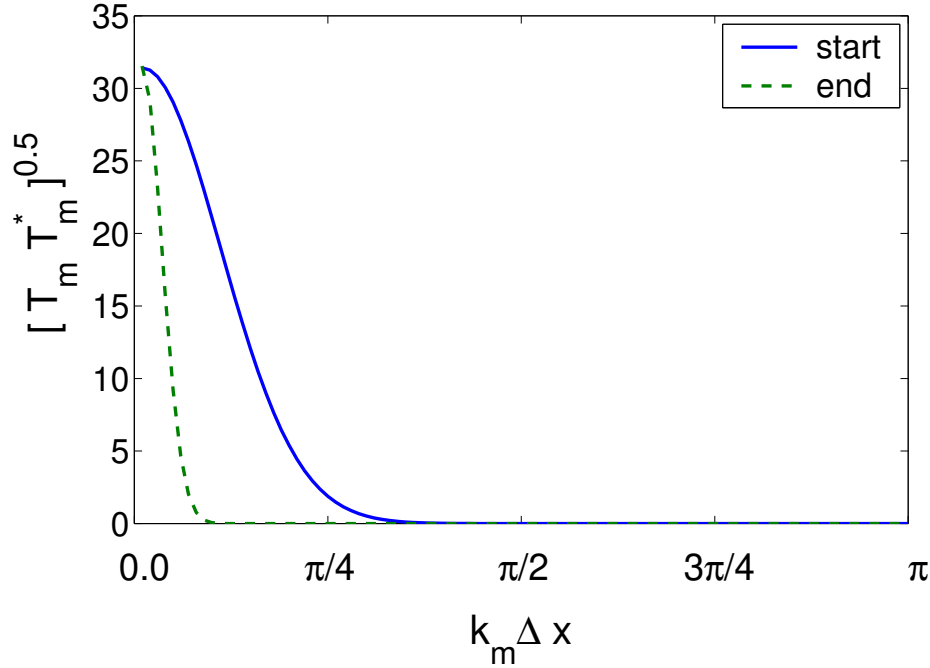
**C = 0.42**



(a1)  $T(x,t)$  in physical space  $x/L$



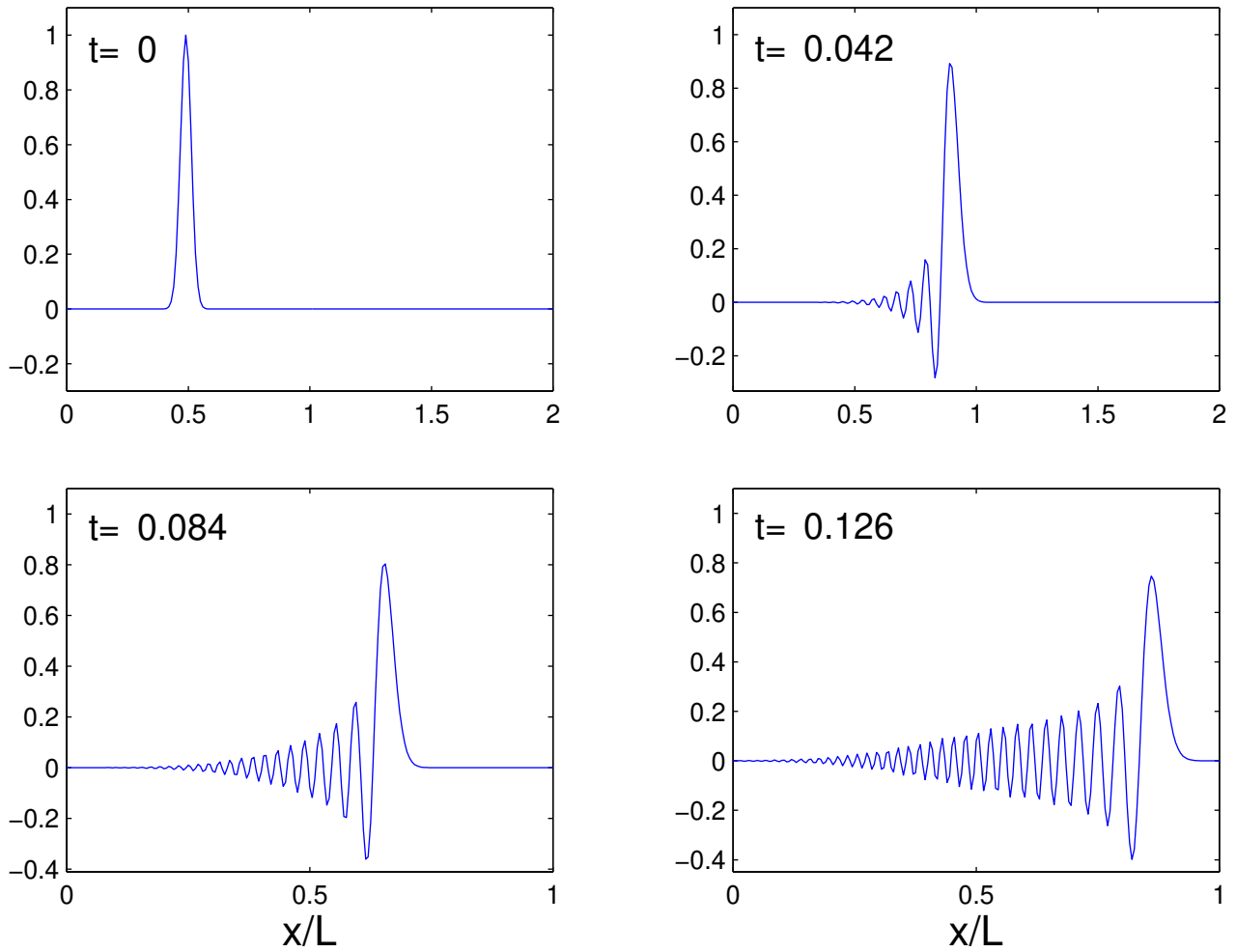
(b1) Amplitude error (numerical dissipation):  $1 - |G_m|$ ,  
Phase error (numerical dispersion):  $1 - \Phi_m / \Phi_{\text{exact}}$



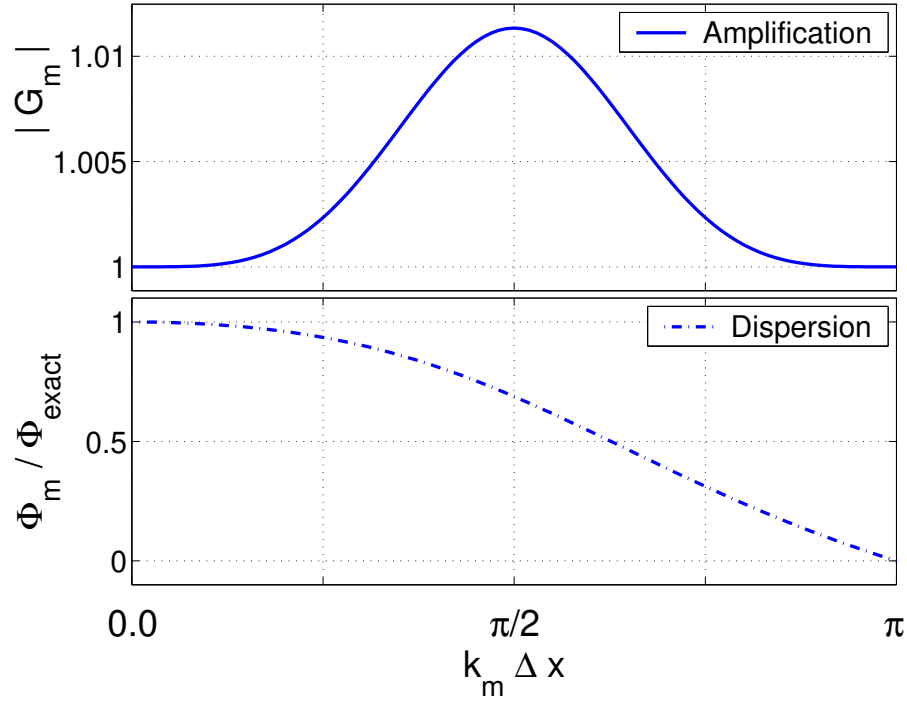
(c1) Energy spectrum at the beginning and at the end of the integration using *Euler* + 1<sup>st</sup>-order *Upwind*

Example 2: *Adams-Bashforth+3PTCD*-scheme

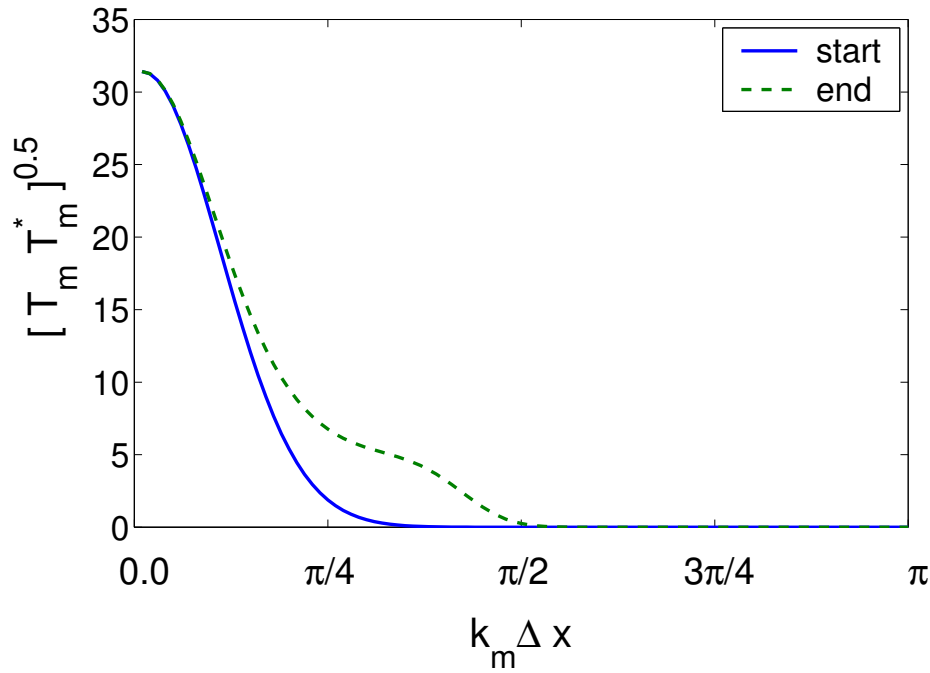
**C = 0.42**



(a2)  $T(x,t)$  in physical space  $x/L$



(b2) Amplitude error (numerical dissipation):  $1 - |G_m|$ ,  
Phase error (numerical dispersion):  $1 - \Phi_m / \Phi_{\text{exact}}$



(c2) Energy spectrum at the beginning and at the end of the  
integration using *Adams-Bashforth+3PTCD*



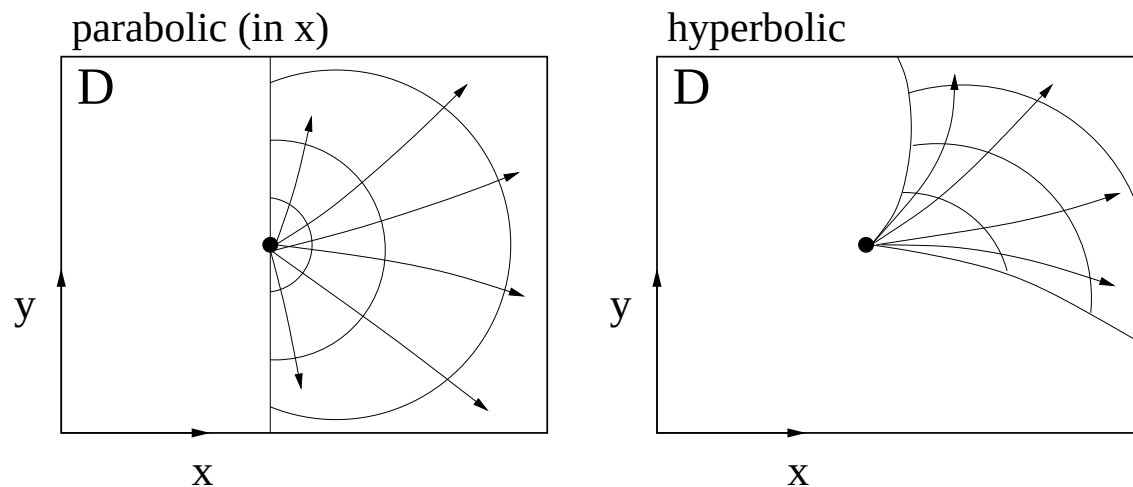
# III Elliptical systems - Solving linear systems of equations

## III.1 Elliptical differential equations

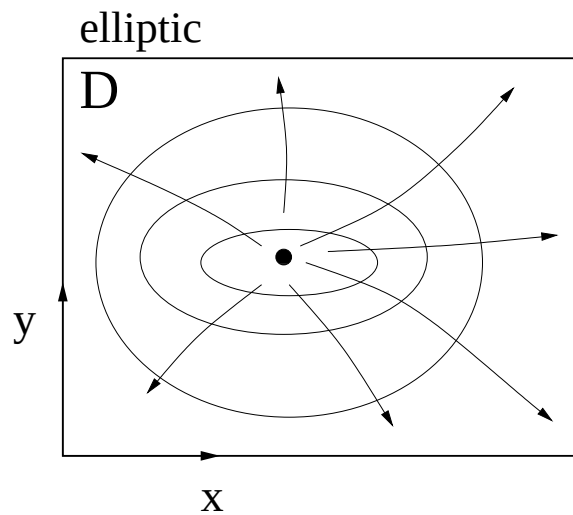
Classification of partially differential equations:

e.g. 2-dim:  $a \frac{\partial^2 \Phi}{\partial x^2} + b \frac{\partial^2 \Phi}{\partial x \partial y} + c \frac{\partial^2 \Phi}{\partial y^2} + d \frac{\partial \Phi}{\partial x} + e \frac{\partial \Phi}{\partial y} + f \Phi = 0$

$$b^2 - 4ac \begin{cases} = 0 & \text{parabolic} \\ > 0 & \text{hyperbolic} \\ < 0 & \text{elliptic} \end{cases}$$



Perturbation at point “•” in the parabolic and hyperbolic system with the solution domain  $D$ .



Perturbation at point “●”  
in the elliptic system with  
the solution domain  $D$ .

Important elliptic differential equations:

Unsteady thermal conduction

$$\frac{\partial T}{\partial t} = \frac{\lambda}{\rho c_p} \left( \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right)$$

Plane potential flow

$$\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} = 0$$

Poisson equation (see Eq. (I.8))

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} + \frac{\partial^2 p}{\partial z^2} = f(x, y, z)$$

## III.2 Pressure-based solution of discretized incompressible Navier-Stokes (Poisson-Solver)

We restrict here ourselves for simplicity to the case  $\varrho = \text{const.}$

Discretized equations of momentum and Poisson equation, (II.1),

= big coupled system of nonlinear algebraic equations,

which is to be solved for velocity  $\mathbf{u}$  and pressure  $p$ .

How done?:  $\Rightarrow$  Two basic solution strategies:

- Fully coupled solution: solves *linearized fully coupled system simultaneously (=in one sweep)* for  $\mathbf{u}$  and pressure  $p$
- Projection methods: *predictor + corrector* steps

### III.2.1 Fully coupled methods

Discretized equations of momentum + Poisson equation, (II.1),

$$\underbrace{\begin{pmatrix} \mathbf{A}_{\mathbf{u}} & \mathbf{A}_{\mathbf{u}p} \\ \mathbf{A}_{p\mathbf{u}} & \mathbf{A}_{pp} \end{pmatrix}}_{\mathbf{A}(\mathbf{u})} \begin{pmatrix} \mathbf{u} \\ p \end{pmatrix} = \begin{pmatrix} \mathbf{B}_{\mathbf{u}} \\ B_p \end{pmatrix} \quad (i)$$

are linearized to big linear system

$$\underbrace{\begin{pmatrix} A_{uu} & & & A_{up} \\ & A_{uu} & & A_{vp} \\ & & A_{ww} & A_{wp} \\ A_{pu} & A_{pv} & A_{pw} & A_{pp} \end{pmatrix}}_{\mathbf{A}^{\text{lin}}} \begin{pmatrix} u \\ v \\ w \\ p \end{pmatrix} = \begin{pmatrix} B_u \\ B_v \\ B_w \\ B_{pp} \end{pmatrix} \quad (ii)$$

and simultaneously solved for  $\mathbf{u} = \begin{pmatrix} u \\ v \\ w \end{pmatrix}$  and  $p$  at each grid point.

- Solution of (ii) computationally very challenging
- Strong nonlinearities (turbulence!) conflict with linearization  $\mathbf{A}(\mathbf{u}) \rightarrow \mathbf{A}^{\text{lin}}$
- Iterative linear equation solvers required

$\hookrightarrow$  projections methods often preferred

## III.2.2 Projection Methods

### General procedure

Equations of motion in vector form with  $\rho = \text{const.}$  and  $\mathbf{u} = \begin{pmatrix} u \\ v \\ w \end{pmatrix}$  read:

Continuity:  $\nabla \cdot \mathbf{u} = 0$

Momentum: 
$$\frac{\partial \mathbf{u}}{\partial t} + \underbrace{\nabla \cdot (\mathbf{u}\mathbf{u})}_{Adv.} = -\frac{1}{\rho} \nabla p + \underbrace{\nu \nabla^2 \mathbf{u}}_{Diff.}$$

Equation of momentum discretized in space and time with  $\Delta t = t^{n+1} - t^n$ :

$$\frac{\mathbf{u}^{n+1} - \mathbf{u}^n}{\Delta t} = \underbrace{\tilde{L} \{(-Adv. + Diff.)\}}_{\text{discrete representation of } Adv.+Diff.} - \frac{1}{\rho} \nabla p^{n+1} \quad (*)$$

Splitting of (\*) into two sub-steps :  $\rightarrow (*) = (1)+(2)$ , with  $p^{n+1} = p^n + p'$

$$(1) \quad \frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} = \tilde{L} \{(-Adv. + Diff.)\} - \frac{1}{\rho} \nabla p^n$$

$$(2) \quad \frac{\mathbf{u}^{n+1} - \mathbf{u}^*}{\Delta t} = -\frac{1}{\rho} \nabla p'$$

(i) *Projection-step*

$$\mathbf{u}^* = \mathbf{u}^n + \Delta t \tilde{L} \{(-Adv. + Diff.)\} - \frac{\Delta t}{\rho} \nabla p^n \dots \text{ from (1)}$$

$\mathbf{u}^*$  does not satisfy the equation of continuity!

$$\text{cont. enforced by } \nabla \cdot (2) : \quad \frac{1}{\Delta t} \nabla \cdot (\mathbf{u}^{n+1} - \mathbf{u}^*) = -\frac{1}{\rho} \nabla \cdot (\nabla p') \quad (**)$$

Continuity requires:  $\nabla \cdot \mathbf{u}^{n+1} \stackrel{!}{=} 0$

(\*\*)  $\Rightarrow$  **Poisson equation:**  $\nabla^2 p' = \frac{\rho}{\Delta t} \nabla \cdot \mathbf{u}^*$   
= big linear system to be solved for  $p'$  (Poisson solver)

(ii) *Pressure correction step:*

Substitution of the pressure correction  $p'$  into (2) yields the new velocity with zero divergence at  $t^{n+1}$ :

$$\mathbf{u}^{n+1} = \mathbf{u}^* - \frac{\Delta t}{\rho} \nabla p'$$
$$p^{n+1} = p^n + p'$$

The efficient and accurate solution of the Poisson equation is a central numerical problem of the incompressible flow simulation (“Low-Mach number flow”). Therefore, a number of *Poisson-Solver*-routines have been specially developed for this purpose.

**Important note:** The velocity field on the new time level  $\mathbf{u}^{n+1}$  is enforced to satisfy continuity, but not conservation of momentum!

$\rightarrow$  (a) very small time steps used to reduce this error  $\varepsilon_{mom} \sim \Delta t$ .

$\rightarrow$  (b) add further correction steps (iterative procedure)

Two popular (b)-type approaches, which were basically developed for steady-state flow, are discussed in detail below.

### III.2.3 Steady Navier-Stokes: SIMPLE, PISO

$$\text{continuity:} \quad \nabla \cdot \mathbf{u} = 0 \quad (\text{III.1})$$

$$\text{momentum:} \quad \underbrace{(\mathbf{u} \nabla) \cdot \mathbf{u}}_{Adv.} = -\frac{1}{\rho} \nabla p + \underbrace{\nu \nabla^2 \mathbf{u}}_{Diff.} \quad (\text{III.2})$$

Eq. (III.2) rewritten for velocity vector  $\mathbf{u}_P = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}_P$  at node  $P$

$$\underbrace{A_P \mathbf{u}_P}_{\text{node } P} + \underbrace{\sum_{nb} A_{nb} \mathbf{u}_{nb}}_{\text{neighbors } nb} = - \left( \frac{1}{\rho} \nabla p \right)_P \quad (\text{III.3})$$

with diagonal and off-diagonal (neighbor) coefficients  $A_P$  and  $A_{nb}$  dependent on discretization and velocity  $\mathbf{u}$  ( $Adv.$  is non-linear!)

**Target:** Solve (III.3) under continuity restriction (III.1)

$\Rightarrow$  Iterative solution procedure

Start: from initial guess on initial level  $k = 0$ :  $\mathbf{u}^{(0)}$  and  $p^{(0)}$

- Iteration from level  $k$  to new level  $k + 1$ :

- Solution on level  $k + 1$  should satisfy:

$$A_P^{(k+1)} \mathbf{u}_P^{(k+1)} + \sum_{nb} A_{nb}^{(k+1)} \mathbf{u}_{nb}^{(k+1)} = - \left( \frac{1}{\rho} \nabla p^{(k+1)} \right)_P \quad (\text{III.4})$$

- Projection: velocity approximated from linearized (III.3) as

$$\mathbf{u}_P^* + \frac{1}{A_P^{(k)}} \sum_{nb} A_{nb}^{(k)} \mathbf{u}_{nb}^{(*)} = -\frac{1}{A_P^{(k)}} \left( \frac{1}{\rho} \nabla p^{(k)} \right)_P \quad (\text{III.5})$$

(does not satisfy continuity (III.1))

- We introduce a corrected intermediate solution

$$\mathbf{u}^{**} = \mathbf{u}^* + \mathbf{u}', \quad p^{**} = p^{(k)} + p'$$

which satisfies (III.4) with  $A_P^{(k+1)} \approx A_P^{(k)}$ ,  $A_{nb}^{(k+1)} \approx A_{nb}^{(k)}$  so that we obtain from (III.5) the following equation for the velocity correction:

$$\mathbf{u}'_P = -\underbrace{\frac{1}{A_P^{(k)}} \sum_{nb} A_{nb}^{(k)} \mathbf{u}'_{nb}}_{:= \tilde{\mathbf{u}}'_P} - \frac{1}{A_P^{(k)}} \left( \frac{1}{\rho} \nabla p' \right)_P \quad (\text{III.6})$$

- Enforcing continuity on level \*\*

$$\begin{aligned} (\nabla \cdot \mathbf{u}^{**})_P &= 0 \\ (\nabla \cdot \mathbf{u}^*)_P + (\nabla \cdot \mathbf{u}')_P &= 0 \end{aligned} \quad (\text{III.7})$$

yields a Poisson equation for pressure correction  $p'$

$$\left( \nabla \cdot \frac{1}{A_P^{(k)}} \frac{1}{\rho} \nabla p' \right)_P = (\nabla \cdot \tilde{\mathbf{u}}')_P + (\nabla \cdot \mathbf{u}^*)_P \quad (\text{III.8})$$

$\Rightarrow$  Eqs. (III.4)-(III.8) solved by different concepts,

e.g., SIMPLE, PISO.



► **Semi-Implicit Method for Pressure Linked Equations (SIMPLE):**

neglects all contributions from  $\tilde{\mathbf{u}}'$ :

known:  $\mathbf{u}^{(k)}, p^{(k)}$

$$(i) \quad (III.5) \quad \rightarrow \quad \mathbf{u}^*$$

$$(ii) \quad (III.8) \text{ with } \tilde{\mathbf{u}}' = 0 \rightarrow p'$$

$$(iii) \quad (III.6) \text{ with } \tilde{\mathbf{u}}' = 0 \rightarrow \mathbf{u}'$$

*note* : corrected velocity  $\mathbf{u}^{**} = \mathbf{u}^* + \mathbf{u}'$  satisfies  
 continuity (III.7) but not momentum (III.4)  
 $\hookrightarrow$  further iterative loop  $k + 1$ , underrelaxed update

$$(iv) \quad \mathbf{u}^{(k+1)} = \mathbf{u}^* + \alpha \mathbf{u}', \quad p^{(k+1)} = p^{(k)} + \beta p'$$

with relaxation parameters:  $0 \leq \alpha \leq 1, 0 \leq \beta \leq 1$

## Pressure **I**mplicit with **S**plitting of **O**perators (**PISO**):

known:  $\mathbf{u}^{(k)}, p^{(k)}$

starts with one SIMPLE loop: (i) – (iii)

$$\begin{array}{lll} (i) & (III.5) & \rightarrow \mathbf{u}^* \\ (ii) & (III.8) \text{ with } \tilde{\mathbf{u}}' = 0 & \rightarrow p' \\ (iii) & (III.6) \text{ with } \tilde{\mathbf{u}}' = 0 & \rightarrow \mathbf{u}' \end{array}$$

(iv) further correction to approximate  $\tilde{\mathbf{u}}'$

$$\mathbf{u}^{(k+1)} = \mathbf{u}^{**} + \mathbf{u}'', \quad p^{(k+1)} = p^{**} + p'' \quad (III.9)$$

$$\begin{aligned} \mathbf{u}_P'' &= \underbrace{-\frac{1}{A_P^{(k)}} \sum_{nb} A_{nb}^{(k)} \mathbf{u}_{nb}'}_{= \tilde{\mathbf{u}}_P' \text{ with } \mathbf{u}' \text{ from (iii)}} - \frac{1}{A_P^{(k)}} \left( \frac{1}{\rho} \nabla p'' \right)_P \quad (III.10) \\ &= \tilde{\mathbf{u}}_P' \text{ with } \mathbf{u}' \text{ from (iii)} \end{aligned}$$

(v) enforce continuity

$$\begin{aligned} \left( \nabla \cdot \mathbf{u}^{(k+1)} \right)_P &= \underbrace{\left( \nabla \cdot \mathbf{u}^{**} \right)_P}_{= 0!} + \left( \nabla \cdot \mathbf{u}'' \right)_P = 0 \\ \rightarrow \left( \nabla \cdot \frac{1}{A_P^{(k)}} \frac{1}{\rho} \nabla p'' \right)_P &= \left( \nabla \cdot \tilde{\mathbf{u}}' \right)_P \quad (III.11) \end{aligned}$$

- Poisson equation (III.11) solved for  $p''$

-  $p''$  in equation (III.10)  $\rightarrow$  correction  $\mathbf{u}_P''$

(vi)  $\mathbf{u}'', p''$  in (III.9)  $\rightarrow \mathbf{u}^{(k+1)}, p^{(k+1)}$

no relaxation

possibly further correction loops (III.10) & (III.11)

with  $\mathbf{u}'' \leftrightarrow \mathbf{u}'$  for computing  $(\tilde{\mathbf{u}}')_P$ , etc.

(!) PISO also appropriate (hence often used) for *unsteady flow*,  
where SIMPLE is rather inefficient (too slow convergence)

## Relaxation $\leftrightarrow$ 'Instationary' term

Eq.(III.3) can be extended to

$$\frac{1-\alpha}{\alpha} A_P \mathbf{u}_P + A_P \mathbf{u}_P + \sum_{nb} A_{nb} \mathbf{u}_{nb} = - \left( \frac{1}{\rho} \nabla p \right)_P + \frac{1-\alpha}{\alpha} A_P \mathbf{u}_P^{old}$$

to obtain

$$\underbrace{\frac{1}{\alpha} A_P \mathbf{u}_P + \sum_{nb} A_{nb} \mathbf{u}_{nb}}_{\text{diagonal dominance for relaxation factor } \alpha \ll 1} = - \left( \frac{1}{\rho} \nabla p \right)_P + \frac{1-\alpha}{\alpha} A_P \mathbf{u}_P^{old}$$

We may further rewrite the upper equation to

$$\underbrace{\frac{A_P}{\alpha} (\mathbf{u}_P - \mathbf{u}_P^{old})}_{\approx \frac{\Delta \mathbf{u}_P}{\Delta t} \dots \dots \text{'pseudo-instationary' term}} + A_P \mathbf{u}_P^{old} + \sum_{nb} A_{nb} \mathbf{u}_{nb} = - \left( \frac{1}{\rho} \nabla p \right)_P$$

$\rightarrow$  diagonal dominance for  $\Delta t \ll 1$

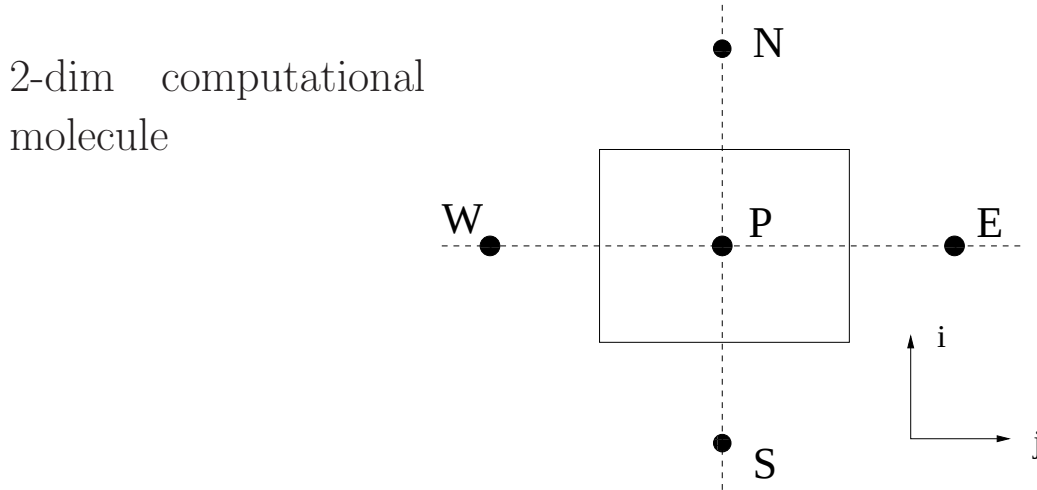
### III.3 Linear equation solvers

linear system of equations:  $\mathbf{A} \cdot \Phi = \mathbf{Q}$

Coefficient matrix  $\mathbf{A}$  results from the difference-operator:  $\Delta^n \Phi / \Delta x_j^n$ .  
 As such, the coefficient matrix connects the solution at point  $\mathbf{P}$  to the solution of its neighboring points.

→ “computational molecule”.

For a 2<sup>nd</sup>-order accurate spatial approximation, the computational molecule of point  $\mathbf{P}$ , is related to four neighboring points denoted by  $\mathbf{E}, \mathbf{W}, \mathbf{S}, \mathbf{N}$  in the two-dimensional case:



e.g.: The 2-dimensional unsteady equation of heat conduction  
 ( $\Phi = \text{Temp.}$ ,  $a = \lambda / \rho c_p$ ).

$$\frac{\partial \Phi}{\partial t} = a \left( \frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} \right)$$

discretized **implicitly in time** using a 2<sup>nd</sup>-order accurate difference scheme in space  $x, y$ , with  $\Delta x = \Delta y$ , yields:

$$\frac{\Phi_{i,j}^{n+1} - \Phi_{i,j}^n}{\Delta t} = \frac{a}{\Delta x^2} \left( \Phi_{i+1,j}^{n+1} + \Phi_{i-1,j}^{n+1} + \Phi_{i,j+1}^{n+1} + \Phi_{i,j-1}^{n+1} - 4 \Phi_{i,j}^{n+1} \right).$$

Accordingly, a system of linear equations written for each point  $P$  as

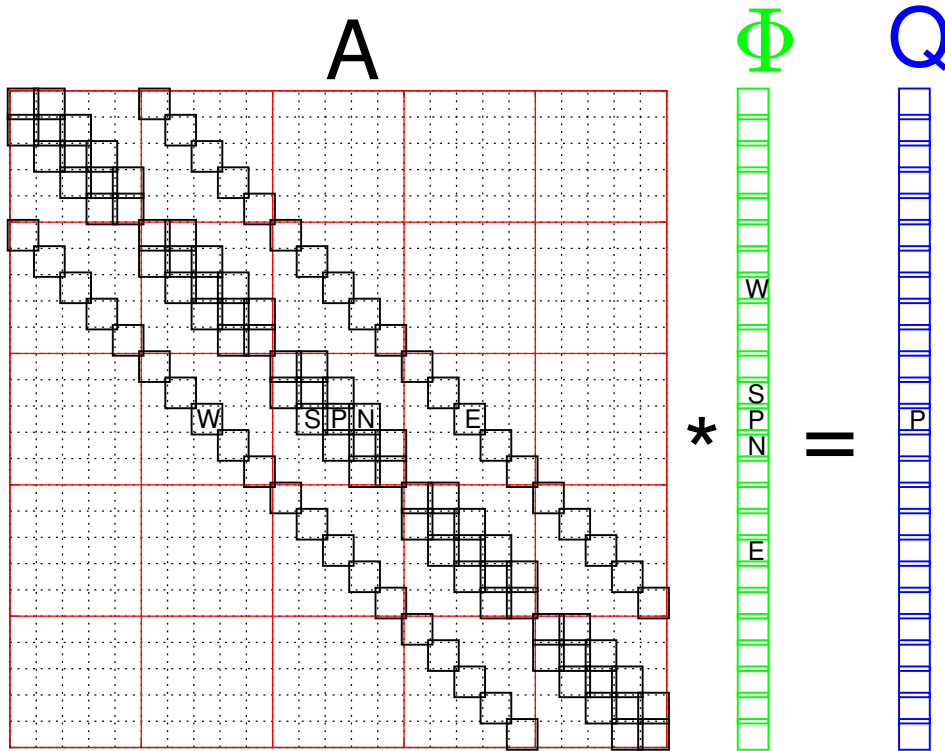
$$A_W\Phi_W + A_S\Phi_S + A_P\Phi_P + A_N\Phi_N + A_E\Phi_E = \underbrace{Q_P}_{=\Phi_P^n}$$

has to be solved for  $\Phi_P^n = \Phi_{i,j}^n$

With  $M = a\Delta t/\Delta x^2$  the following coefficients occur:

$$\begin{array}{lcl} i, j & \dots P & : A_P = (1 + 4M) \\ i-1, j & \dots S & : A_S = -M \\ i+1, j & \dots N & : A_N = -M \end{array} \quad \left| \quad \begin{array}{lcl} i, j-1 & \dots W & : A_W = -M \\ i, j+1 & \dots E & : A_E = -M. \end{array} \right.$$

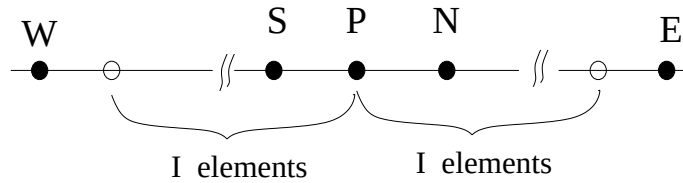
The following figure shows the coefficient-matrix  $\mathbf{A}$  for a computational domain with  $i = 1, \dots, 5$ , and  $j = 1, \dots, 5$  internal points (indices of the boundary points:  $i = 0, i = 6$ , and  $j = 0, j = 6$ ). The coefficient-matrix  $\mathbf{A}$  shows the typical structure for 2-dimensional elliptic problems.



Matrix-structure with  $D = 5 \times 5$  internal computational points

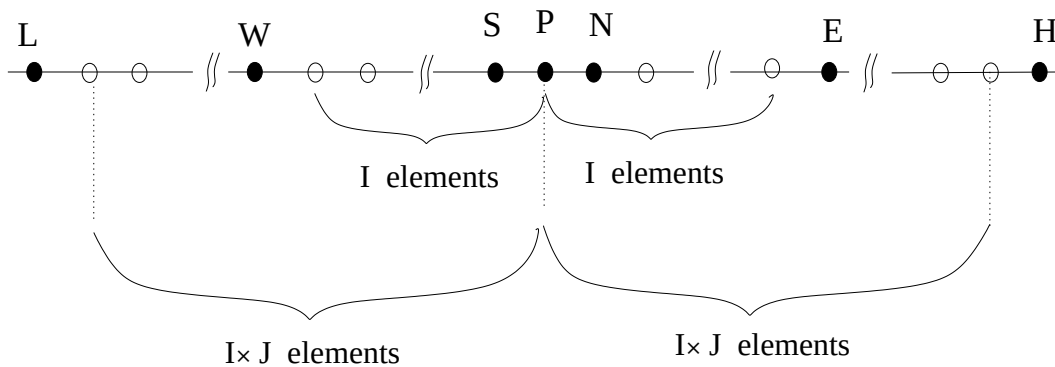
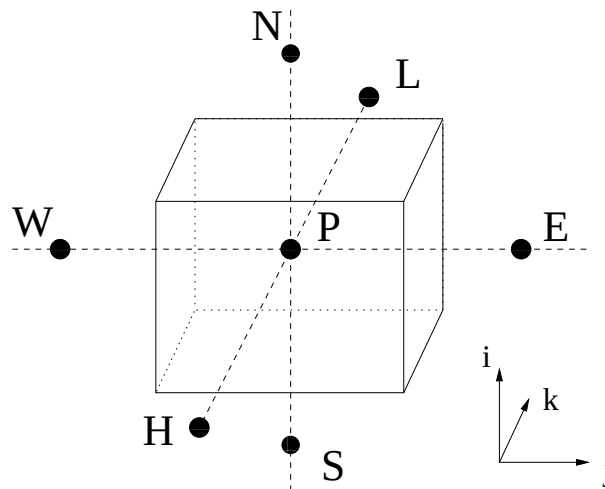
Basic structure of each row of the coefficient-matrix  $\mathbf{A}$ :

**2-dim** with  $i = 1, \dots, I$ ;  $j = 1, \dots, J \rightarrow I \times J$  internal points



**3-dim** with  $i = 1, \dots, I$ ;  $j = 1, \dots, J$ ;  $k = 1, \dots, K$   
 $\rightarrow I \times J \times K$  internal points

3-dim computational molecule



### III.3.1 Direct linear equation solvers

#### (A) Gauss' Elimination

$$\mathbf{A} \Phi = \mathbf{Q} \rightarrow \mathbf{U} \Phi = \mathbf{Q}'$$

$$\left. \begin{array}{l} i = 1; k = 1 \\ j = 1, \dots, n \end{array} \right\} A_{1j}^{(1)} = A_{1j}$$

$$\left. \begin{array}{l} k = 1, \dots, n-1 \\ i = k+1, \dots, n \\ j = k, \dots, n \end{array} \right\} A_{ij}^{(k+1)} = A_{ij}^{(k)} - \frac{A_{ik}^{(k)}}{A_{kk}^{(k)}} A_{kj}^{(k)}$$

$$\begin{pmatrix} A_{11} & A_{12} & A_{13} & \dots & A_{1n} \\ A_{21} & A_{22} & A_{23} & \dots & A_{2n} \\ A_{31} & A_{32} & A_{33} & \dots & A_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & A_{n3} & \dots & A_{nn} \end{pmatrix} \rightarrow \begin{pmatrix} A_{11}^{(1)} & A_{12}^{(1)} & A_{13}^{(1)} & \dots & A_{1n}^{(1)} \\ 0 & A_{22}^{(2)} & A_{23}^{(2)} & \dots & A_{2n}^{(2)} \\ 0 & A_{32}^{(2)} & A_{33}^{(2)} & \dots & A_{3n}^{(2)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & A_{n2}^{(2)} & A_{n3}^{(2)} & \dots & A_{nn}^{(2)} \end{pmatrix} \rightarrow$$

$$\begin{pmatrix} A_{11}^{(1)} & A_{12}^{(1)} & A_{13}^{(1)} & \dots & A_{1n}^{(1)} \\ 0 & A_{22}^{(2)} & A_{23}^{(2)} & \dots & A_{2n}^{(2)} \\ 0 & 0 & A_{33}^{(3)} & \dots & A_{3n}^{(3)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & A_{n3}^{(3)} & \dots & A_{nn}^{(3)} \end{pmatrix} \rightarrow \begin{pmatrix} A_{11}^{(1)} & A_{12}^{(1)} & A_{13}^{(1)} & \dots & A_{1n}^{(1)} \\ 0 & A_{22}^{(2)} & A_{23}^{(2)} & \dots & A_{2n}^{(2)} \\ 0 & 0 & A_{33}^{(3)} & \dots & A_{3n}^{(3)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & A_{nn}^{(n)} \end{pmatrix} = \mathbf{U}$$

Huge computation time:  $n^3/3 + n^2/2$  operations are necessary.

However, the method is very robust, and is virtually always applicable, for full matrices as well.

Error accumulation can be reduced using “*Pivoting*”:  $A_{ii}^{(k)} \rightarrow \max$  by interchanging the rows.

## (B) LU Decomposition

$$\mathbf{A} \Phi = \mathbf{Q} \rightarrow \mathbf{L} \mathbf{U} \Phi = \mathbf{Q} \quad (*)$$

The LU-Decomposition

$$\mathbf{A} = \mathbf{L} \mathbf{U}$$

with the definition

$$\mathbf{U} \Phi := \mathbf{Y} \quad (**)$$

allows to rewrite the linear systems of equation (\*) as:

$$\mathbf{L} \mathbf{Y} = \mathbf{Q} \quad (***)$$

Solving procedure in 3 steps:

1.) Computation of  $\mathbf{L}$  and  $\mathbf{U}$  (LU-Decomposition)

$$\begin{pmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \dots & A_{nn} \end{pmatrix} = \begin{pmatrix} L_{11} & 0 & \dots & 0 \\ L_{21} & L_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ L_{n1} & L_{n2} & \dots & L_{nn} \end{pmatrix} \cdot \begin{pmatrix} U_{11} & U_{12} & \dots & U_{1n} \\ 0 & U_{22} & \dots & U_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & U_{nn} \end{pmatrix}$$

The computation of  $\mathbf{U}$  is closely related to the *Gauss' Elimination* (with Pivoting). The elements of the matrix  $\mathbf{L}$  are the multipliers,



which occur in the *Gauss' Elimination* as

$$L_{ik} = \frac{A_{ik}^{(k)}}{A_{kk}^{(k)}}$$

2.) Computation of  $\mathbf{Y}$  by forward substitution into  $(***)$   
( $\mathbf{L} \dots$  lower triangular matrix)

3.) Computation of the sought solution  $\Phi$  by backward substitution into  
 $(**)$ . ( $\mathbf{U} \dots$  upper triangular matrix)

Basically, a computationally expensive method (computation time  $\hat{=}$  *Gauss' Elimination*).

But the *LU-Method* has the following big **advantage**:

If the coefficient matrix does not change (often the case in CFD with temporally constant mesh), the costly LU-Decomposition will have to be carried out only once. For all the following computations of  $\Phi$ , only the computationally cheap steps 2) and 3) have to be carried out.

## (C) Thomas Algorithm

$$\mathbf{A} \Phi = \mathbf{Q}$$

The *Thomas* or *Tridiagonal Matrix Algorithm (TDMA)* is applicable, if the coefficient matrix  $\mathbf{A}$  is tridiagonal

$$\mathbf{A} = \begin{pmatrix} A_{11} & A_{12} & 0 & 0 & 0 & \cdots & 0 \\ A_{21} & A_{22} & A_{23} & 0 & 0 & \cdots & 0 \\ 0 & A_{32} & A_{33} & A_{34} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \cdots & \cdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \cdots & \vdots \\ 0 & \cdots & 0 & 0 & A_{n-1n-2} & A_{n-1n-1} & A_{n-1n} \\ 0 & \cdots & 0 & 0 & 0 & A_{nn-1} & A_{nn} \end{pmatrix}$$

In principle, such matrices occur only at 1-dimensional problems. Nevertheless, the *Tridiagonal Matrix Algorithm (TDMA)* is often used as sub-algorithm of other solving methods for multidimensional, i.e. non-tridiagonal, elliptic systems.

Solution procedure of the *TDMA*:

- 1) *Forward substitution*: elimination of the lower subdiagonal  $A_{k+1,k} \rightarrow 0$ .
- 2) *Backward substitution*: computation of  $\Phi$ .

## (D) Fourier Method

2-dimensional elliptic model problem, (e.g. 2-dimensional Poisson):

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = g(x, y) \quad (*)$$

Discretization:  $k = 1, \dots, K - 1$  internal points in  $x$ -direction

$l = 1, \dots, L - 1$  internal points in  $y$ -direction

(boundary points:  $k = 0$ ,  $k = K$ , and,  $l = 0$ ,  $l = L$ )

- Requirements:**
- discretization is equidistant  
 $\rightarrow \Delta x = \text{const.}, \Delta y = \text{const.}$
  - boundary conditions must allow for  
transformation into a linear combination of  
trigonometric basis functions  
= periodic, or periodically continued

## Periodic boundary conditions

$$u_{k,l} = u_{k+K,l} = u_{k,l+L}$$

→  $u_{k,l}$  can be rewritten as a discrete Fourier series  
(= *Inverse Fourier-Transformation*):

$$u_{k,l} = \sum_{m=0}^{K-1} \sum_{n=0}^{L-1} \hat{u}_{m,n} e^{i m 2 \pi k / K} e^{i n 2 \pi l / L} \quad (**)$$

$\hat{u}_{m,n}$  is the *Fourier-Transformation* of  $u_{k,l}$  obtained as

$$\hat{u}_{m,n} = \frac{1}{K L} \sum_{k=0}^{K-1} \sum_{l=0}^{L-1} u_{k,l} e^{-i m 2 \pi k / K} e^{-i n 2 \pi l / L}$$

We consider  $\Delta x = \Delta y = \Delta$ . The  $O[\Delta^2]$ -discretized equation (\*) reads

$$u_{k+1,l} + u_{k-1,l} + u_{k,l+1} + u_{k,l-1} - 4u_{k,l} = g_{k,l} \Delta^2 \quad (***)$$

The Fourier-Transformation of (\*\*\*) yields

$$\hat{u}_{m,n} \underbrace{\left( e^{i m 2 \pi / K} + e^{-i m 2 \pi / K} + e^{i n 2 \pi / L} + e^{-i n 2 \pi / L} - 4 \right)}_{\lambda_{m,n}} = \hat{g}_{m,n} \Delta^2$$

with “eigenvalue”:  $\lambda_{m,n} = 2 \left( \cos \frac{m 2 \pi}{K} + \cos \frac{n 2 \pi}{L} - 2 \right)$

we obtain  $\hat{u}_{m,n} = \frac{\hat{g}_{m,n} \Delta^2}{\lambda_{m,n}} \quad (****)$

Solution procedure:

- *Fourier-Transformation* to compute  $\hat{g}_{m,n}$  (using FFT-routines)

$$\hat{g}_{m,n} = \frac{1}{K L} \sum_{k=0}^{K-1} \sum_{l=0}^{L-1} g_{k,l} e^{-i m 2 \pi k / K} e^{-i n 2 \pi l / L}$$

- Compute  $\hat{u}_{m,n}$  using  $(***)$  with the eigenvalue  $\lambda_{m,n}$
- Compute  $u_{k,l}$  using *Inverse Fourier-Transformation*  $(**)$ .

## 3D-Poisson-Equation with one homogeneous direction

= convenient problem for parallel processing

e.g. Poisson-equation in cartesian coordinates

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} + \frac{\partial^2 p}{\partial z^2} = g(x, y, z) \quad (*)$$

with a **homogeneous**  $z$ -direction, and hence **equidistant**  $z$ -discretization, i.e.  $z = k \cdot \Delta z \mid k = 0, \dots, K - 1$ .

$$\begin{aligned} \rightarrow \text{Periodicity in } z : \quad & p(x, y, z) = p(x, y, z + K\Delta z) \\ \text{or} \quad & p_{i,j,k} = p_{i,j,k+K} \end{aligned}$$

### Poisson-Solver

(1) *Fourier-Transform (FT)* of  $(*)$  in  $z$ -direction

$$\frac{\partial^2 \hat{p}^m}{\partial x^2} + \frac{\partial^2 \hat{p}^m}{\partial y^2} + \lambda_m \hat{p}^m = \hat{g}^m(x, y) \quad (**)$$

$\rightarrow$  System of equations for Fourier-Transformation  $\hat{p}^m = \hat{p}^m(x, y)$   
( $m$  decoupled systems!)

The eigenvalue reads

$$\lambda_m = -\frac{2}{\Delta z^2} \left[ 1 - \cos \left( \frac{2\pi m}{K} \right) \right]$$

(2)  $(**)$  solved for  $\hat{p}^m(x, y)$ , separate solution for each mode  $m \rightarrow$  parallel processing!

(3) *Inverse Fourier-Transformation (IFT)* of the  $CR$ -solution

$$\hat{p}^m(x, y) \rightarrow p(x, y, z)$$

## Fast Fourier Transform

The *Fourier-Transform* ( $FT$ ) of the function  $f_j$  with  $j = 0, \dots, J$  and the periodicity  $f_j = f_{j+J}$  is basically computed as

$$\hat{f}_k = \frac{1}{J} \sum_{j=0}^{J-1} f_j e^{-i k 2\pi j / J} \quad (FT)$$

$\rightarrow \hat{f}_k$  of the modes  $k = 0, \dots, J - 1$

The total number of the required operations for this  $FT$  procedure is

$$N_{tot}^{FT} = J \times J$$

$N_{tot}^{FT}$  would be **much too high** for practical applications

$\Rightarrow$  **Fast Fourier Transform** ( $FFT$ ).

$FFT$  decomposes the  $FT$  into  $\log_2(J)$  sub-transformations, such that the required number of operations for  $FFT$  is:

$$N_{tot}^{FFT} = J \times \log_2(J) \ll J \times J$$

Analogously for *Inverse Fourier-Transform* ( $IFT$ )

$\Rightarrow$  **Inverse Fast Fourier Transform** ( $IFFT$ )

### III.3.2 Iterative linear equation solver

General procedure:

$$\begin{aligned} \mathbf{A} \Phi &= \mathbf{Q} \\ \Downarrow \\ (\mathbf{N} - \mathbf{P}) \Phi &= \mathbf{Q} \end{aligned}$$

with matrix  $\mathbf{N}$  “*close to*”  $\mathbf{A}$ : i.e.,  $\|\mathbf{N}\| \approx \|\mathbf{A}\|$ , but **easy to be factorized** (e.g.  $\mathbf{N}$  is tridiagonal).

General derivation of the iteration:  $\Phi^{(n+1)} = F(\Phi^{(n)})$

$$\begin{aligned} \mathbf{N}\Phi^{(n+1)} &= \mathbf{P}\Phi^{(n)} + \mathbf{Q} \\ \Phi^{(n+1)} &= \mathbf{N}^{-1}\mathbf{P}\Phi^{(n)} + \mathbf{N}^{-1}\mathbf{Q} \end{aligned} \quad (\text{III.12})$$

$$\text{with : } \left. \begin{aligned} \mathbf{P} &= \mathbf{N} - \mathbf{A} \\ \mathbf{R}^{(n)} &= \mathbf{A}\Phi^{(n)} - \mathbf{Q} \end{aligned} \right\} \rightarrow (\text{III.12})$$

$$\begin{aligned} \Phi^{(n+1)} &= \mathbf{N}^{-1}(\mathbf{N} - \mathbf{A})\Phi^{(n)} + \mathbf{N}^{-1}\mathbf{Q} \\ \Phi^{(n+1)} &= \Phi^{(n)} - \mathbf{N}^{-1}\mathbf{R}^{(n)} \end{aligned} \quad (\text{III.13})$$

Convergence of (III.12) and (III.13):  $\|\mathbf{R}^{(n)}\| \rightarrow 0$

Restrictions for the convergence:

(1) Spectral radius  $< 1$ :

$$r_{\text{spekt}} = \max \{ \|\text{eig}(\mathbf{N}^{-1}\mathbf{P})\| \} < 1$$

(2) Matrix  $\mathbf{A}$  is diagonally dominant

$$a_{jj} > \sum_{\substack{j \\ i \neq j}} |a_{ij}| \quad \dots\dots \text{more stringent than restriction (1)}$$



## (A) Jakobi method

$$\mathbf{N} = \mathbf{D} \mathbf{E}, \quad \mathbf{P} = \tilde{\mathbf{L}} + \tilde{\mathbf{U}}$$

with:  $\mathbf{E} \dots$  identity matrix,  $\mathbf{D} = \text{diag}(\mathbf{A}) \dots$  diagonal matrix

$$\tilde{\mathbf{L}} = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ -A_{21} & 0 & 0 & \cdots & 0 & 0 \\ -A_{31} & -A_{32} & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & 0 & \vdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & 0 & \vdots \\ -A_{N,1} & -A_{N,2} & -A_{N,3} & \cdots & -A_{N,N-1} & 0 \end{pmatrix}$$

$$\tilde{\mathbf{U}} = \begin{pmatrix} 0 & -A_{12} & -A_{13} & \cdots & -A_{1,N-1} & -A_{1,N} \\ 0 & 0 & -A_{23} & \cdots & -A_{2,N-1} & -A_{2,N} \\ \vdots & \vdots & 0 & \ddots & \vdots & \vdots \\ \vdots & \vdots & \vdots & 0 & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & -A_{N-1,N} \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{pmatrix}$$

Using this decomposition the iteration (III.12) reads:

$$\Phi^{(n+1)} = (\mathbf{D} \mathbf{E})^{-1} \left[ (\tilde{\mathbf{L}} + \tilde{\mathbf{U}}) \Phi^{(n)} + \mathbf{Q} \right]$$

or

$$\Phi_i^{(n+1)} = \frac{1}{A_{ii}} \left[ Q_i - \sum_{\substack{j=1 \\ j \neq i}}^N A_{ij} \Phi_j^{(n)} \right]$$

Required number of iterations:  $n \simeq \frac{1}{2} p N$  to obtain  $\| \mathbf{R}^{(n)} \| \leq 10^{-p}$

## (B) Gauss-Seidel (GS)

$$\mathbf{N} = \mathbf{D}\mathbf{E} - \tilde{\mathbf{L}}, \quad \mathbf{P} = \tilde{\mathbf{U}}$$

GS uses in each iteration cycle  $(n+1)$  the values  $\Phi_j^{(n+1)}$ , which have already been updated within this cycle  $(n+1)$ .

$$\Phi_i^{(n+1)} = \frac{1}{A_{ii}} \left( Q_i - \sum_{j=1}^{i-1} A_{ij} \Phi_j^{(n+1)} - \sum_{j=i+1}^N A_{ij} \Phi_j^{(n)} \right)$$

Required number of iterations:  $n \simeq \frac{1}{4} p N$  to obtain  $\| \mathbf{R}^{(n)} \| \leq 10^{-p}$   
(= two times faster than Jakobi)

## (C) Successive Overrelaxation (SOR)

SOR uses a **weighted** average value between  $\Phi_i^{(n)}$  and  $[\Phi_i^{(n+1)}]_{GS}$ :

$$\Phi_i^{(n+1)} = \lambda [\Phi_i^{(n+1)}]_{GS} + (1 - \lambda) \Phi_i^{(n)}$$

$$\Phi_i^{(n+1)} = \lambda \left[ \frac{1}{A_{ii}} \left( Q_i - \sum_{j=1}^{i-1} A_{ij} \Phi_j^{(n+1)} - \sum_{j=i+1}^N A_{ij} \Phi_j^{(n)} \right) \right] + (1 - \lambda) \Phi_i^{(n)}$$

with relaxation parameter  $\lambda : 0 < \lambda < 2$  (convergence restriction).

If:  $\lambda = 1 \rightarrow$  Gauss-Seidel (GS)

SOR converges significantly faster than GS using a well selected  $\lambda$ .

Required number of iterations:  $n \simeq \frac{1}{3} p \sqrt{N}$  to obtain  $\| \mathbf{R}^{(n)} \| \leq 10^{-p}$

Problem: no analytical method for estimating an appropriate value of  $\lambda$  is available.

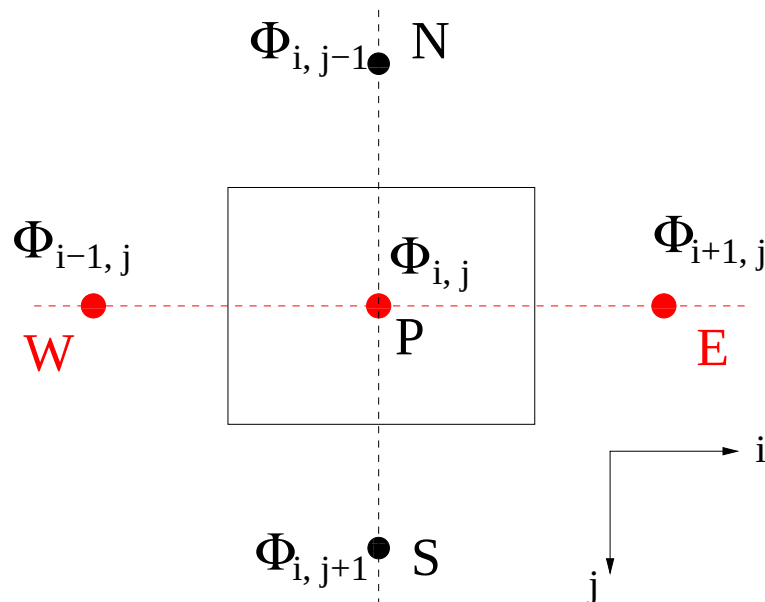
SOR and GS are **explicit** methods.

Combination with **implicit** methods are possible:

- Line Successive Overrelaxation (LSOR)
- Alternating Direction Implicit + SOR (ADISOR)

## Line Successive Overrelaxation (LSOR)

Implicit solution along one coordinate direction  
e.g. on  $j = \text{const.} \hat{=} W\text{-}E\text{-line}$ :



$$A_W \Phi_{i-1,j}^{(*)} + A_P \Phi_{i,j}^{(*)} + A_E \Phi_{i+1,j}^{(*)} = Q_{i,j} - A_N \Phi_{i,j-1}^{(n)} - A_S \Phi_{i,j+1}^{(n)}$$

yields a tridiagonal system for  $\Phi_{i,j}^{(*)}$  on each line  $j = \text{const.}$ :

solution  $\Phi_{i,j}^{(*)} \rightarrow \text{SOR}$

$$\Phi_{i,j}^{(n+1)} = \Phi_{i,j}^{(n)} + \lambda \left( \Phi_{i,j}^{(*)} - \Phi_{i,j}^{(n)} \right)$$

# Alternating Direction Implicit + SOR (ADISOR)

Implicit solution along a coordinate line with a change of the coordinate direction after every step, i.e.

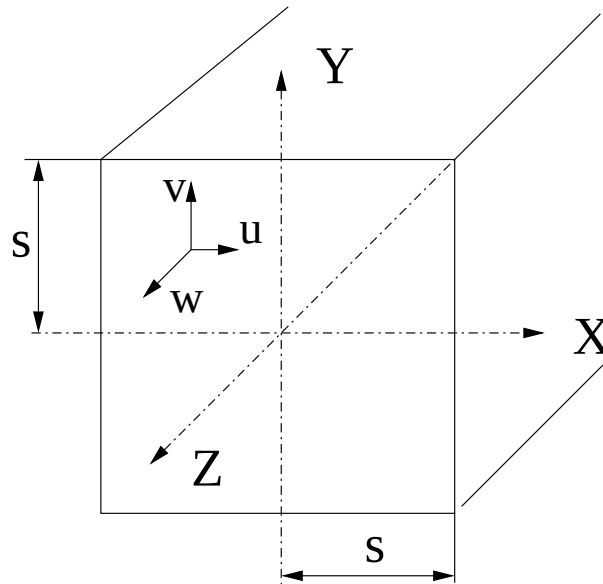
1<sup>st</sup> step: on  $j = \text{const.} \hat{=} W\text{-}E\text{-line}$ :

$$A_W \Phi_{i-1,j}^{(*)} + A_P \Phi_{i,j}^{(*)} + A_E \Phi_{i+1,j}^{(*)} = Q_{i,j} - A_N \Phi_{i,j-1}^{(n)} - A_S \Phi_{i,j+1}^{(n)}$$
$$\Phi_{i,j}^{(**)} = \Phi_{i,j}^{(n)} + \lambda \left( \Phi_{i,j}^{(*)} - \Phi_{i,j}^{(n)} \right)$$

2<sup>nd</sup> step: on  $i = \text{const.} \hat{=} N\text{-}S\text{-line}$ :

$$A_N \Phi_{i,j-1}^{(***)} + A_P \Phi_{i,j}^{(***)} + A_S \Phi_{i,j+1}^{(***)} = Q_{i,j} - A_W \Phi_{i-1,j}^{(**)} - A_E \Phi_{i+1,j}^{(**)}$$
$$\Phi_{i,j}^{(n+1)} = \Phi_{i,j}^{(**)} + \lambda \left( \Phi_{i,j}^{(***)} - \Phi_{i,j}^{(**)} \right)$$

Example: Flow in a square channel



Assumptions: incompressible, laminar, fully developed flow

$$\rightarrow u = 0, v = 0, w = w(X, Y), \frac{\partial p}{\partial Z} = \text{const.}$$

Navier-Stokes in  $z$ -direction:

$$0 = -\frac{\partial p}{\partial Z} + \mu \left( \frac{\partial^2 w}{\partial X^2} + \frac{\partial^2 w}{\partial Y^2} \right)$$

Non-dimensionalized:

$$x = \frac{X}{s}, \quad y = \frac{Y}{s}, \quad \Phi = w \frac{\mu}{s^2 \left( -\frac{\partial p}{\partial Z} \right)}, \quad \text{note } \frac{\partial p}{\partial Z} < 0$$

$$\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} + 1 = 0 \quad (\text{III.14})$$

Boundary conditions:

$$\begin{aligned} \Phi(-1, y) &= 0, \quad \Phi(1, y) = 0 \\ \Phi(x, -1) &= 0, \quad \Phi(x, 1) = 0 \end{aligned}$$

Discretization of (III.14) with 2<sup>nd</sup>-order Central Difference scheme:

$$\frac{\Phi_{i-1,j} - 2\Phi_{i,j} + \Phi_{i+1,j}}{\Delta x^2} + \frac{\Phi_{i,j-1} - 2\Phi_{i,j} + \Phi_{i,j+1}}{\Delta y^2} + 1 = 0 \quad (\text{III.15})$$

Jakobi:

$$\Phi_{i,j}^{(n+1)} = \frac{1}{\frac{2}{\Delta x^2} + \frac{2}{\Delta y^2}} \left( 1 + \frac{\Phi_{i-1,j}^{(n)} + \Phi_{i+1,j}^{(n)}}{\Delta x^2} + \frac{\Phi_{i,j-1}^{(n)} + \Phi_{i,j+1}^{(n)}}{\Delta y^2} \right)$$

SOR (  $\hat{=}$  Gauss-Seidel if  $\lambda = 1$ ):

$$\begin{aligned} \Phi_{i,j}^{(*)} &= \frac{1}{\frac{2}{\Delta x^2} + \frac{2}{\Delta y^2}} \left( 1 + \frac{\Phi_{i-1,j}^{(n+1)} + \Phi_{i+1,j}^{(n)}}{\Delta x^2} + \frac{\Phi_{i,j-1}^{(n+1)} + \Phi_{i,j+1}^{(n)}}{\Delta y^2} \right) \\ \Phi_{i,j}^{(n+1)} &= \lambda \Phi_{i,j}^{(*)} + (1 - \lambda) \Phi_{i,j}^{(n)} \end{aligned}$$

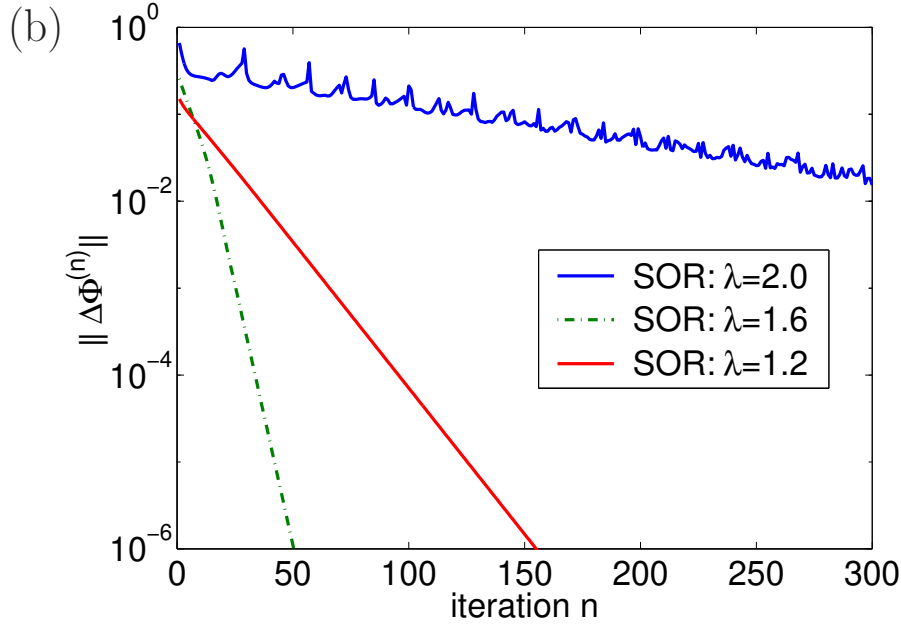
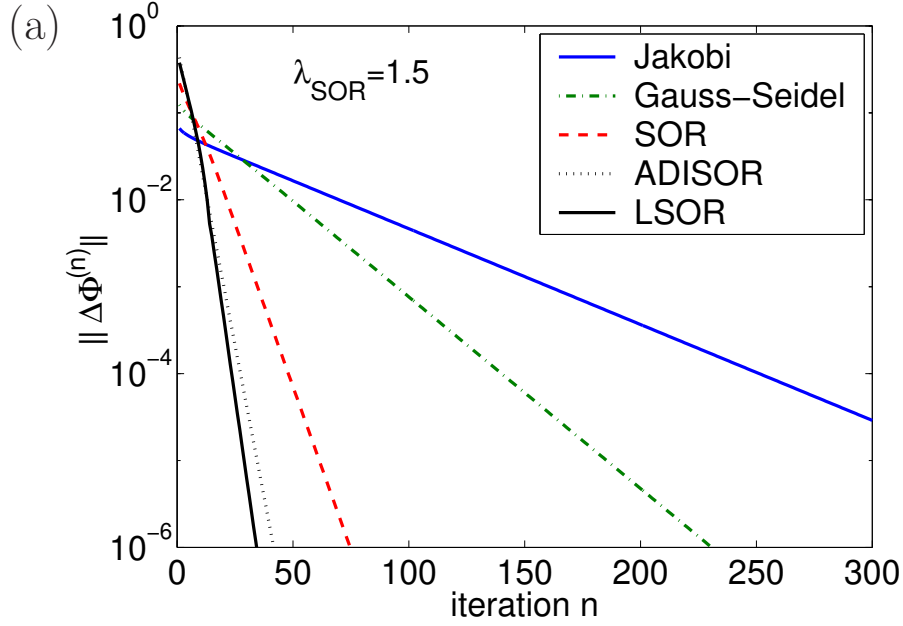
LSOR:

$$\begin{aligned} -\frac{1}{\Delta x^2} \Phi_{i-1,j}^{(*)} + \left( \frac{2}{\Delta x^2} + \frac{2}{\Delta y^2} \right) \Phi_{i,j}^{(*)} - \frac{1}{\Delta x^2} \Phi_{i+1,j}^{(*)} &= 1 + \frac{\Phi_{i,j-1}^{(n+1)} + \Phi_{i,j+1}^{(n)}}{\Delta y^2} \\ \Phi_{i,j}^{(n+1)} &= \lambda \Phi_{i,j}^{(*)} + (1 - \lambda) \Phi_{i,j}^{(n)} \end{aligned}$$

ADISOR

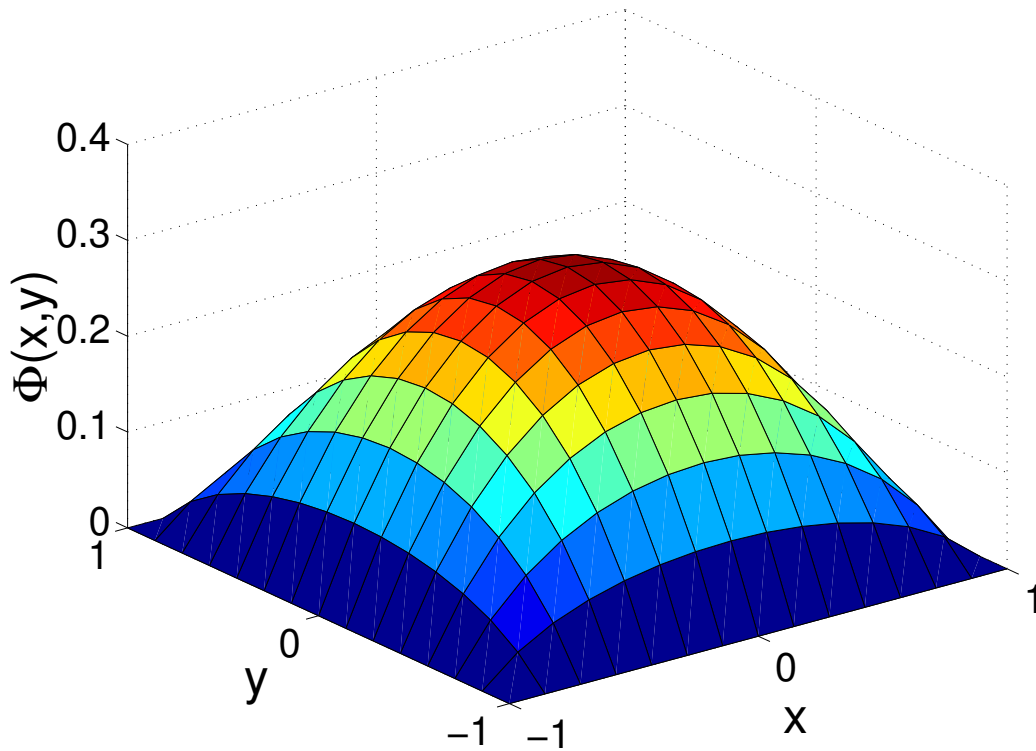
$$\begin{aligned} -\frac{1}{\Delta x^2} \Phi_{i-1,j}^{(*)} + \left( \frac{2}{\Delta x^2} + \frac{2}{\Delta y^2} \right) \Phi_{i,j}^{(*)} - \frac{1}{\Delta x^2} \Phi_{i+1,j}^{(*)} &= 1 + \frac{\Phi_{i,j-1}^{(**)} + \Phi_{i,j+1}^{(n)}}{\Delta y^2} \\ \Phi_{i,j}^{(**)} &= \lambda \Phi_{i,j}^{(*)} + (1 - \lambda) \Phi_{i,j}^{(n)} \\ -\frac{1}{\Delta y^2} \Phi_{i,j-1}^{(***)} + \left( \frac{2}{\Delta x^2} + \frac{2}{\Delta y^2} \right) \Phi_{i,j}^{(***)} - \frac{1}{\Delta y^2} \Phi_{i,j+1}^{(***)} &= 1 + \frac{\Phi_{i-1,j}^{(n+1)} + \Phi_{i+1,j}^{(**)}}{\Delta x^2} \\ \Phi_{i,j}^{(n+1)} &= \lambda \Phi_{i,j}^{(***)} + (1 - \lambda) \Phi_{i,j}^{(**)} \end{aligned}$$

$$\Delta\Phi_{RMS} : \quad \|\Delta\Phi^{(n)}\| = \left[ \sum_i \sum_j \left( \Phi_{i,j}^{(n+1)} - \Phi_{i,j}^{(n)} \right)^2 \right]^{\frac{1}{2}}$$



Convergence of the iterative solution of the elliptic equation (III.15),  
(Discretization:  $13 \times 13$  internal grid points) using

- (a) Jakobi, GS, SOR, ADISOR, LSOR ( $\lambda_{SOR} = 1.5$  in the latter three cases),
- (b) SOR for different values of the overrelaxation parameter  $\lambda$



Solution of the non-dimensional velocity  $\Phi(x, y)$ .

### Notes on the **Relaxation methods** Jakobi, GS and SOR:

- SOR and GS converge well for **purely** or **strongly elliptic** systems: Poisson-, potential-, heat conduction equation.  
Bad convergence in the case of significant convective transport: Navier-Stokes-, enthalpy transport equation, etc.  
→ Splitting methods (ADI), Multigrid-methods.
- Jakobi and GS are too slow for large systems, i.e., large number of grid points  $N$ . The methods eliminate the high-frequency error components very fast ( $\text{RHS} \sim \text{diffusion}$ ), but the low-frequency error contents very slow.
- SOR is by a factor  $\sqrt{N}$  faster than GS, but the optimum overrelaxation parameter  $\lambda$  cannot be determined analytically.



## (D) Conjugate Gradient Method (CG)

Solution of  $\mathbf{A} \Phi = \mathbf{Q}$  (III.14)

is computed by a minimization of the function

$$\mathbf{f}(\Phi) = \frac{1}{2} \Phi^T \mathbf{A} \Phi - \Phi^T \mathbf{Q} + \mathbf{C}$$

The vector  $\Phi$  at the minimum of  $\mathbf{f}$ , where the condition

$$\nabla \mathbf{f} = 0 \rightarrow \mathbf{f} = \mathbf{f}_{min}$$

is met, satisfies the linear system of equations (III.14) due to:

$$\nabla \mathbf{f} = \mathbf{A} \Phi - \mathbf{Q} \quad \text{(III.15)}$$

Solution strategy: Starting from an assumed initial value  $\Phi_0$ , approach step-by-step the minimum of  $\mathbf{f}$

$$\Phi_0, \Phi_1, \dots, \Phi_n, \Phi_{n+1}, \dots \rightarrow \mathbf{f}(\Phi) = \mathbf{f}_{min}$$

without a deterioration of the minimization of every previous step, i.e.

$$\mathbf{f}(\Phi_{n+1}) \stackrel{!}{\leq} \mathbf{f}(\Phi_n)$$

In effect, one is moving along the so-called *conjugate* directions

Constraints for the use of CG:

- (1) Matrix  $\mathbf{A}$  is symmetric:  $\mathbf{A}^T = \mathbf{A}$
- (2) Matrix  $\mathbf{A}$  is positive definite:  $\forall \mathbf{x} : \mathbf{x}^T \mathbf{A} \mathbf{x} \geq 0$

## Local minimization along a direction $\mathbf{u}$ :

Proceed at each step  $n$  into direction  $\mathbf{u}_n$

$$\Phi_{n+1} = \Phi_n + \mathbf{u}_n$$

on surface  $\mathbf{f}$  until the local minimum  $\min_{\mathbf{u}_n}(\mathbf{f})$  is reached,  
where:

$$\mathbf{u}_n \nabla \mathbf{f} = 0. \quad (*)$$

## Determination of the conjugate directions

Goal: determine the direction of continuation  $\mathbf{u}_{n+1}$  without a deterioration of the earlier minimization (along  $\mathbf{u}_n$ ):

Introducing an arbitrary  $\delta(\Phi)$  into (III.15) yields

$$\delta(\nabla \mathbf{f}) = \delta(\mathbf{A}\Phi - \mathbf{Q}) = \mathbf{A} \delta(\Phi)$$

Variation  $\delta(\Phi)$  must not deteriorate the local minimization  $(*)$

$$\begin{aligned} \underbrace{\mathbf{u}_n [\nabla \mathbf{f}]}_{=0} + \delta(\nabla \mathbf{f}) &= 0 \\ \Rightarrow \mathbf{u}_n \delta(\nabla \mathbf{f}) &= \mathbf{u}_n \mathbf{A} \underbrace{\delta(\Phi)}_{:= \mathbf{u}_{n+1}} = 0 \end{aligned}$$

Therein, the variation  $\delta(\Phi)$  represents the direction of continuation  $\mathbf{u}_{n+1}$ .

We can write:

$$\begin{aligned} \mathbf{u}_n \mathbf{A} \mathbf{u}_{n+1} &= 0, \text{ and} \\ \mathbf{u}_{n+1} \mathbf{A} \mathbf{u}_n &= 0 \quad (\mathbf{A} \text{ is symmetric}) \end{aligned}$$

The vectors  $\mathbf{u}_n$  and  $\mathbf{u}_{n+1}$  are the so called *conjugate* vectors with respect to matrix  $\mathbf{A}$ .

In the following we demonstrate the concept of CG with the help of a simple example:

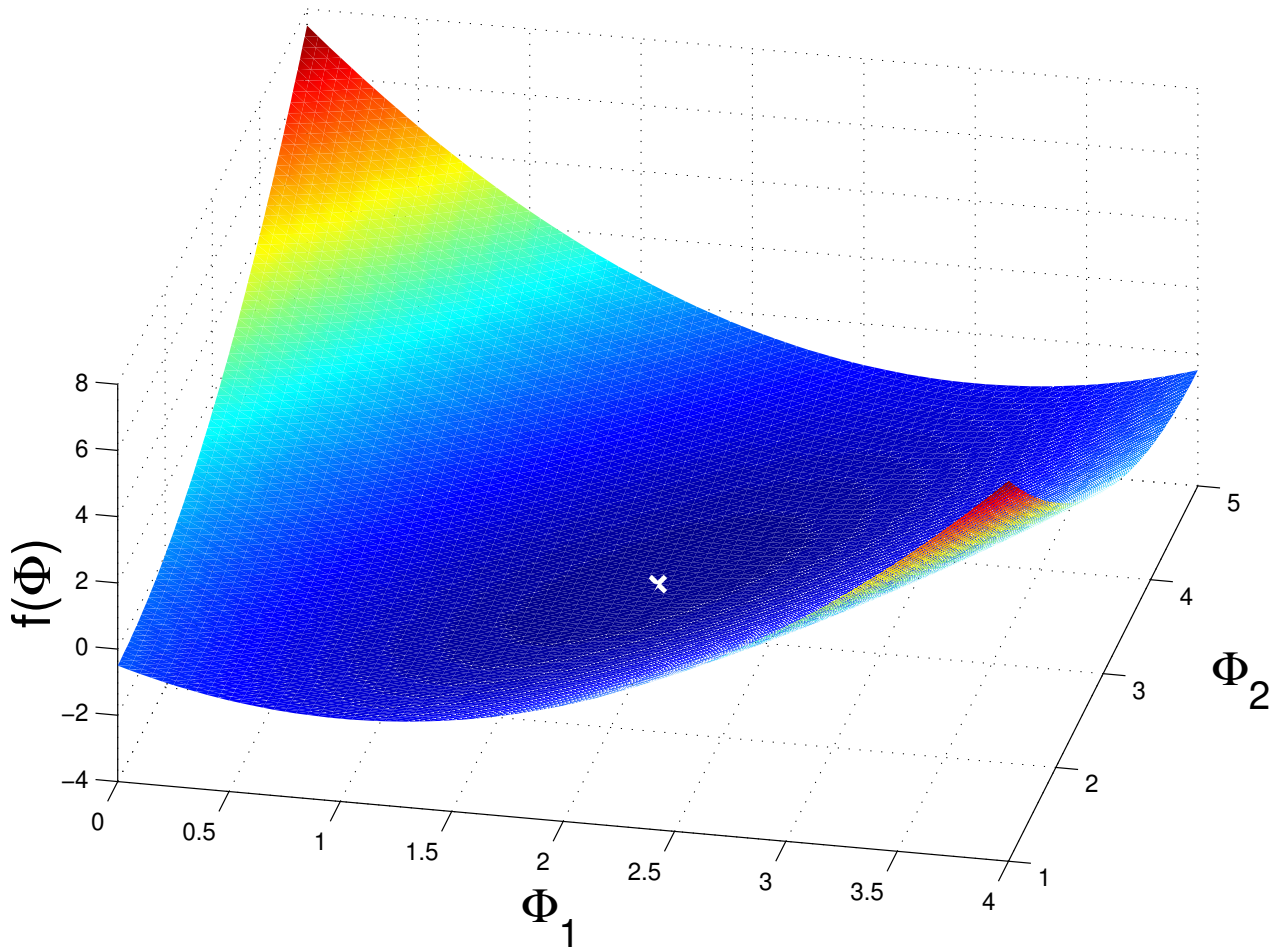
The solution of

$$\mathbf{A} \boldsymbol{\Phi} = \mathbf{Q}$$

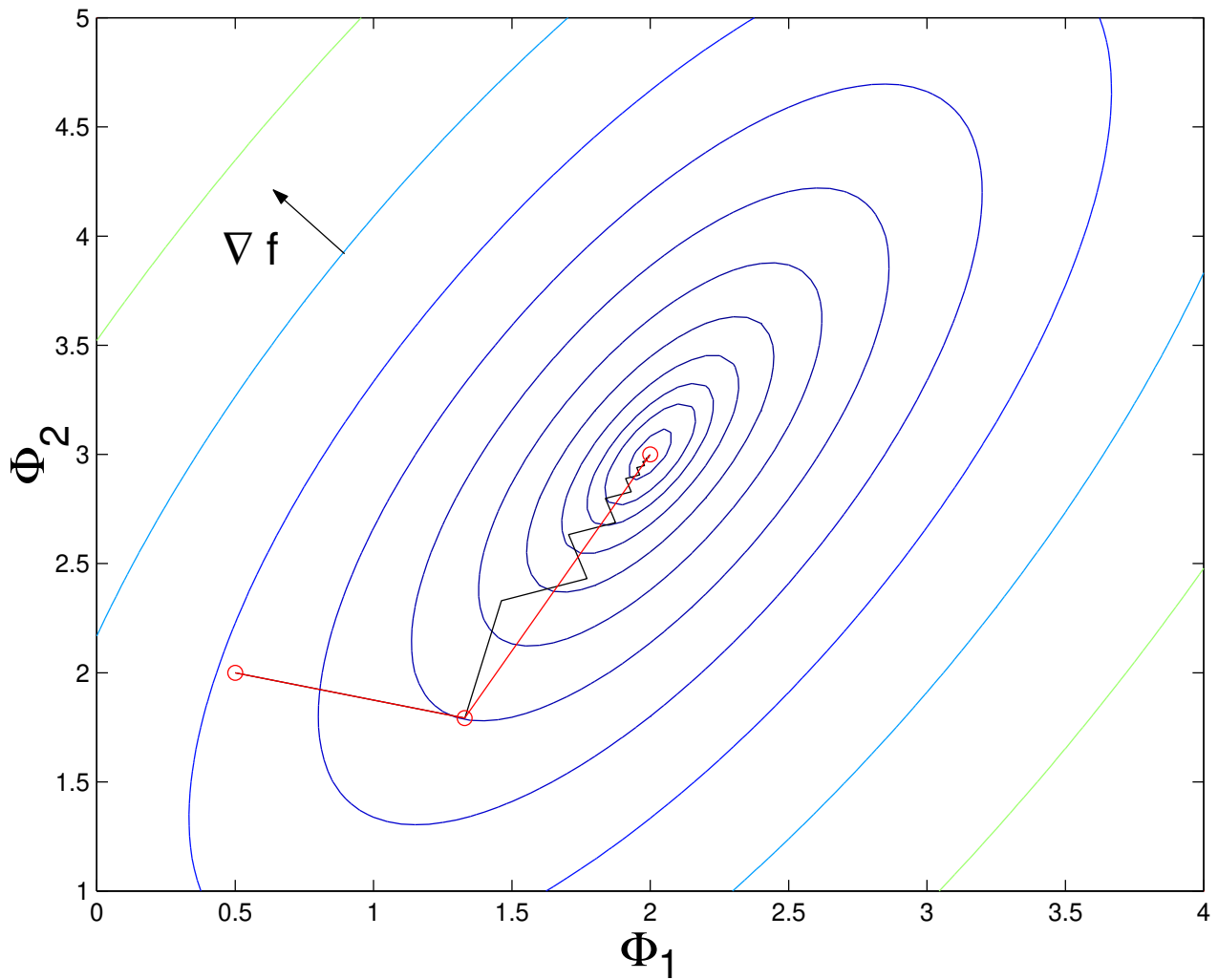
is sought, where

$$\mathbf{A} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}, \quad \mathbf{Q} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

The corresponding surface  $\mathbf{f}(\boldsymbol{\Phi})$  is displayed in the diagram below. The solution of the linear systems of equations above is exactly the minimum of this area (marked with the white cross).



The diagram below displays the iso-contours of the surface  $\mathbf{f}(\Phi)$  and the solution process of CG. The method approaches the minimum of  $\mathbf{f}(\Phi)$ , i.e. the sought solution, along two conjugated directions (red lines).



The also shown black zig-zag-line, represents a minimum search of  $\mathbf{f}$  using the “*Steepest Descent*”-method. Thereby, one always proceeds into the direction of the biggest negative gradient of  $\mathbf{f}$  at every individual step, i.e. into the  $-\nabla\mathbf{f}$ -direction. Evidently, this method requires much more iterations than CG. The “*Steepest Descent*”-method is therefore principally too slow.

### Convergence of CG:

For  $N$ -dimensional  $\Phi \in \mathcal{R}^N$ , exactly  $N$  conjugate directions exist:

$$\mathbf{P}^{(i)} \mathbf{A} \mathbf{P}^{(j)} = 0$$

$$\text{with } i, j = 0, 1, \dots, N-1, \quad \text{and} \quad i \neq j$$

$\Rightarrow$  theoretically (if numerical round off error  $\varepsilon_{rndoff} = 0$ ), the solution will be **always** achieved after  $N$  steps.

## (E) Biconjugate Gradient Method (BCG)

BCG is an extension of CG to linear systems with a **unsymmetric** coefficient matrix  $\mathbf{A}$

The original system is thereby extended to an equivalent system with a symmetric coefficient matrix:

$$\begin{pmatrix} 0 & \mathbf{A} \\ \mathbf{A}^T & 0 \end{pmatrix} \cdot \begin{pmatrix} \Psi \\ \Phi \end{pmatrix} = \begin{pmatrix} \mathbf{Q} \\ 0 \end{pmatrix}$$

### Acceleration of the convergence of BCG

Via *pre-conditioning* of the linear system of equations (??)

$$\left( \tilde{\mathbf{A}}^{-1} \cdot \mathbf{A} \right) \Phi = \tilde{\mathbf{A}}^{-1} \cdot \mathbf{Q} \quad (\text{III.16})$$

- where
- $\tilde{\mathbf{A}}$  is close to  $\mathbf{A}$ , i.e.,  $\tilde{\mathbf{A}}^{-1} \cdot \mathbf{A} \approx \mathbf{E}$
  - $\tilde{\mathbf{A}}$  should be easily inverted

Applying a pre-conditioning the eigenvalues in the modified system (III.16) can be bundled

$$\left( \frac{\| \Lambda_{max} \|}{\| \Lambda_{min} \|} \right)_{(III.16)} < \left( \frac{\| \Lambda_{max} \|}{\| \Lambda_{min} \|} \right)_{(??)}$$

such that the convergence will be accelerated.

→ *Preconditioned biconjugate gradient* method (PBCG)

## (F) Multigrid Method (MG)

- Advantages:  $\oplus$  **Fast** method: for the solution vector  $\Phi \in R^N$ ,  $\approx O[N]$  operations are required  
 (cf.: direct solver requires  $\approx O[N \log_2(N)]$ )  
 $\oplus$  very wide field of application ( $>$  direct solver),  
 e.g., non-constant coefficients, unstructured mesh, also applicable to non-linear equations
- Disadvantage:  $\ominus$  There is a big variation of Multigrid methods,  
 i.e., “*The Multigrid-method*” does not exist.  
 The most appropriate MG-method often depends on the individual application and must be additionally adapted.

### (G.1) From the 2-grid- to Multigrid-method

Elliptic model-problem:

$$\mathcal{L} \Phi = Q \quad (\text{III.17})$$

where  $\mathcal{L}$  represents the linear elliptic differential operator.

The equation (III.17) discretized with a constant mesh width  $h$  reads

$$\mathcal{L}_h \Phi_h = Q_h. \quad (\text{III.18})$$

Be  $\tilde{\Phi}_h$  an approximation of the exact solution of the system (III.18)  $\Phi_h$ , and we may write

$$\Phi_h = \tilde{\Phi}_h + \underbrace{\mathcal{E}_h}_{\text{correction}}$$



The residual or the error reads

$$\mathbf{D}_h = \mathcal{L}_h \tilde{\Phi}_h - \mathbf{Q} \quad (\text{III.19})$$

Due to the linearity of the operator  $\mathcal{L}_h$  one gets

$$\mathcal{L}_h \mathcal{E}_h = -\mathbf{D}_h \quad (\text{III.20})$$

Equation (III.20) on a coarser mesh with the mesh width  $H$ ,  
e.g.  $H = 2h$ , reads analogously

$$\mathcal{L}_H \mathcal{E}_H = -\mathbf{D}_H \quad (\text{III.21})$$

$\Rightarrow$  Coarse - grid correction scheme:  $\tilde{\Phi}_h \xrightarrow{Cgc} \tilde{\Phi}_h^{new}$

(1) Compute  $\mathbf{D}_h$  on a fine mesh using equation (III.19)

$$\mathbf{D}_h = \mathcal{L}_h \tilde{\Phi}_h - \mathbf{Q}$$

(2) Restriction  $\mathbf{D}_h \rightarrow \mathbf{D}_H$

$$\mathbf{D}_H = \mathcal{R}\mathbf{D}_h$$

(3) Compute solution  $\mathcal{E}_H$  of equation (III.21)

(4) Prolongation  $\mathcal{E}_H \rightarrow \mathcal{E}_h$  ( $\hat{=}$  interpolation)

$$\mathcal{E}_h = \mathcal{P}\mathcal{E}_H$$

$\rightarrow$  new approximated solution on the fine mesh is obtained as

$$\tilde{\Phi}_h^{new} = \tilde{\Phi}_h + \mathcal{E}_h$$

**Spectral splitting** of the error  $\mathcal{E}_h$  into low-frequency “smooth” component  $\mathcal{E}_h^{LF}$  and high-frequency “coarse” component  $\mathcal{E}_h^{HF}$

$$\mathcal{E}_h = \mathcal{E}_h^{LF} + \mathcal{E}_h^{HF}$$

## Comparison:

$$\boxed{\begin{array}{l} \textit{Relaxation-methods} \\ \text{(Jakobi, GS, ADI,...)} \end{array}} \Leftrightarrow \boxed{\textit{Coarse-grid correction}}$$

*Relaxation:* for mesh width  $h \rightarrow 0$  (refined mesh):

The reduction of the amplitudes of the “smooth” component  $\mathcal{E}_h^{LF}$  proceeds very slowly with every iteration.

The reduction of the amplitudes of the “coarse” component  $\mathcal{E}_h^{HF}$  proceeds fast with every iteration.

The overall convergence is very slow, **but**, very fast and efficient “smoothing” of the  $\mathcal{E}_h^{HF}$ -components is provided

*Coarse-grid correction:*

the corresponding “coarse” mesh width  $H = 2h$  can resolve only errors down to a wave-length

$\lambda_{min} = 2H$ , high-frequency contents ( $\mathcal{E}_h^{HF}$ ), are unresolved, and hence reduced to zero immediately!

→ fast convergence of the low-frequency contents  $\mathcal{E}_h^{LF}$  achievable, high-frequency contents (with  $\lambda < 2H$ ) no “present” on coarse-grid



**Basic idea of the 2-grid- and Multigrid-method:**

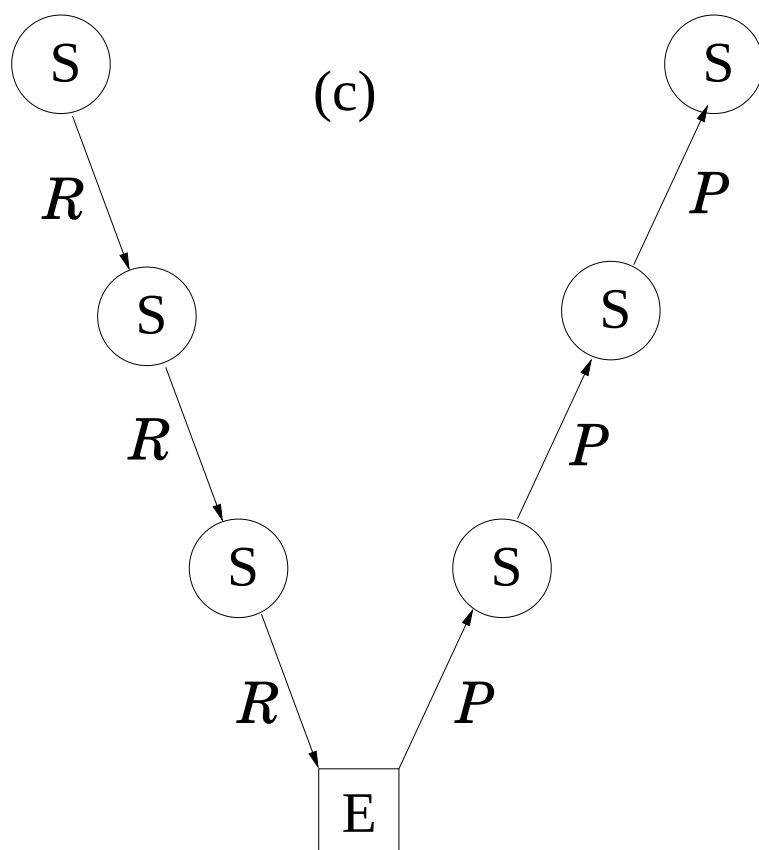
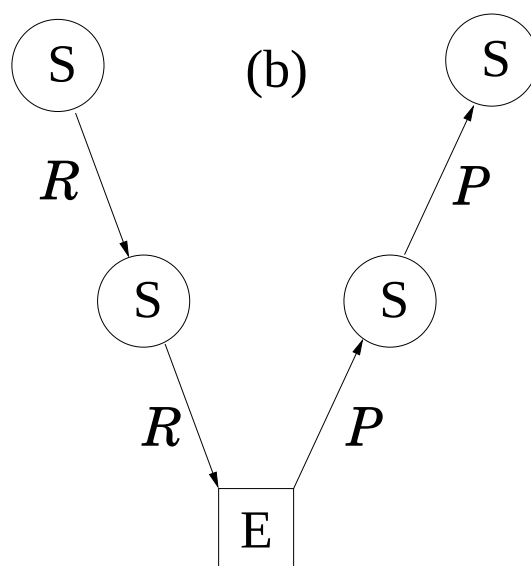
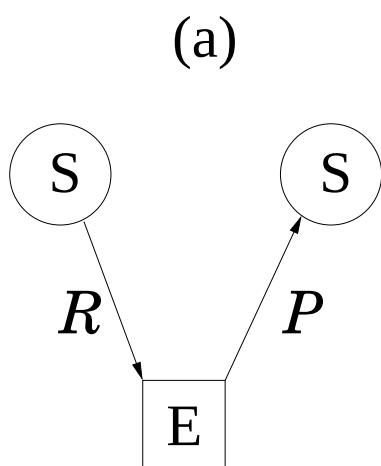
= Combination of the advantages of *Relaxation*- and *Coarse-grid*-method

## 2-grid-iteration ( $\hat{=}$ basic cycle)

- (1) *Pre-smoothing*: Apply  $n_1 \geq 0$  times a relaxation method to  $\tilde{\Phi}_h$  in order to achieve the “pre-smoothed” solution  $\bar{\Phi}_h$
- (2) *Coarse-grid correction*: As shown above,  
compute:  $\bar{\Phi}_h \rightarrow \bar{\Phi}_h^{new}$
- (3) *Post-smoothing*: Apply  $n_2 \geq 0$  times a relaxation method to  $\bar{\Phi}_h^{new}$  to achieve the “post-smoothed” solution  $\tilde{\Phi}_h^{new}$

**Important:** *Successive Overrelaxation* (SOR) is no good candidate for a relaxation method, because overrelaxation counteracts the smoothing effect. Gauss-Seidel (GS) is a widely used relaxation method in Multigrid context. GS is a simple algorithm and has good smoothing qualities.

$$\text{GS:} \quad \Phi_i = - \left( \sum_{\substack{j=1 \\ j \neq i}}^N \mathcal{L}_{ij} \Phi_j - \mathbf{Q}_i \right) \frac{1}{\mathcal{L}_{ii}} \quad i = 1, \dots, N$$



2-grid  $\rightarrow$  Multigrid

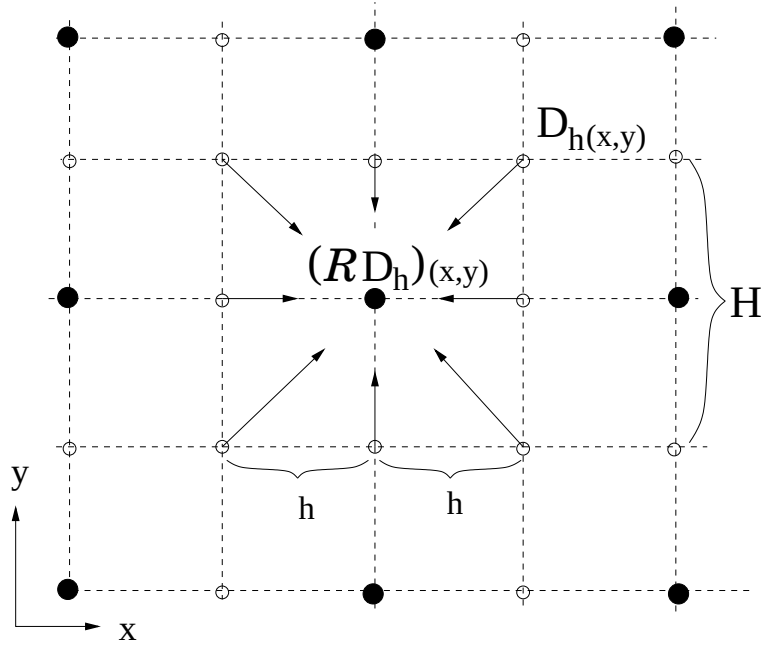
(a) 2-grid

(b) 3-grid

(c) 4-grid

## Restriction $\mathbf{D}_H = \mathcal{R}\mathbf{D}_h$

e.g.:  $H = 2h$



“straight injection”

$$\mathcal{R} = [1] : \quad (\mathcal{R}D_h)_{(x,y)} = D_h(x, y)$$

“full weighting”:

$$\mathcal{R} = \begin{bmatrix} \frac{1}{16} & \frac{1}{8} & \frac{1}{16} \\ \frac{1}{8} & \frac{1}{4} & \frac{1}{8} \\ \frac{1}{16} & \frac{1}{8} & \frac{1}{16} \end{bmatrix}$$

$$(\mathcal{R}D_h)_{(x,y)} = \frac{1}{4}D_h(x, y) + \frac{1}{8}D_h(x+h, y) + \frac{1}{16}D_h(x+h, y+h) + \dots$$

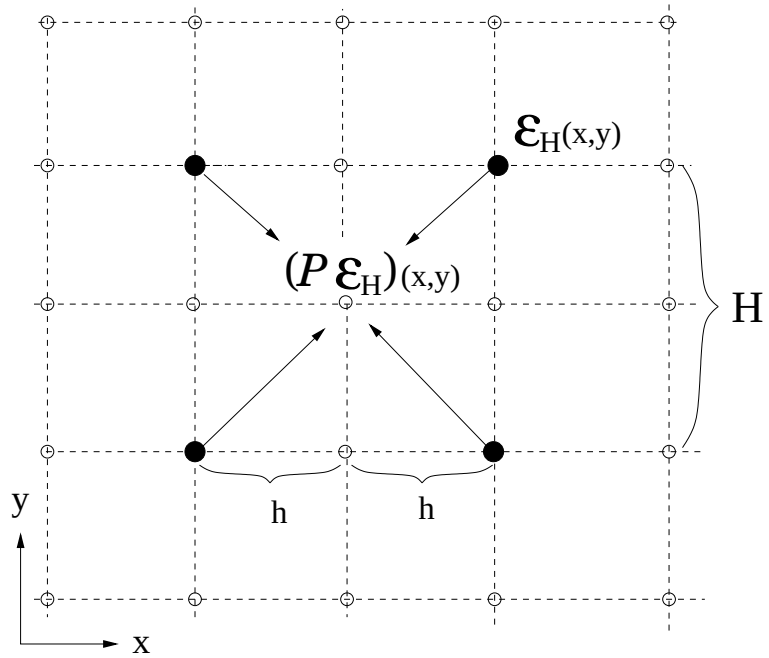
“half weighting”:

$$\mathcal{R} = \begin{bmatrix} 0 & \frac{1}{8} & 0 \\ \frac{1}{8} & \frac{1}{2} & \frac{1}{8} \\ 0 & \frac{1}{8} & 0 \end{bmatrix}$$

$$(\mathcal{R}D_h)_{(x,y)} = \frac{1}{2}D_h(x, y) + \frac{1}{8}D_h(x+h, y) + \frac{1}{8}D_h(x, y+h) + \dots$$

## Prolongation $\mathcal{E}_h = \mathcal{P}\mathcal{E}_H$

e.g.:  $H = 2h$



e.g. using bilinear interpolation:

$$\mathcal{P} = \begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix}$$

## Symbolic representation of the **differential operator** $\mathcal{L}_h$

e.g.:  $\mathcal{L}_h = \nabla_h^2$

$$\mathcal{L}_h = \frac{1}{h^2} \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Basic Multigrid-algorithm:  
 Reiteration of the

“V”-cycle:    fine  $\rightarrow$  coarse  $\rightarrow$  fine

until convergence.

Improvement in efficiency of the MG-method by the use of  
*Full-Multigrid*-method (FMG)

## (G.2) Full-Multigrid-Method

- (1) Computation of the right side of (III.18) on all  $m = 1 + \log_2(n)$  grid-levels:

$$\begin{array}{lll}
 \text{level} & (1) : & \mathbf{Q}_{2h} = \mathcal{R}\mathbf{Q}_h \\
 & (2) : & \mathbf{Q}_{4h} = \mathcal{R}\mathbf{Q}_{2h} \\
 & \vdots & \vdots \\
 & (m) : & \mathbf{Q}_{nh} = \mathcal{R}\mathbf{Q}_{nh/2}
 \end{array}$$

- (2) Compute the solution  $\Phi_{nh}$  on the lowest (coarsest) mesh-level  $m$

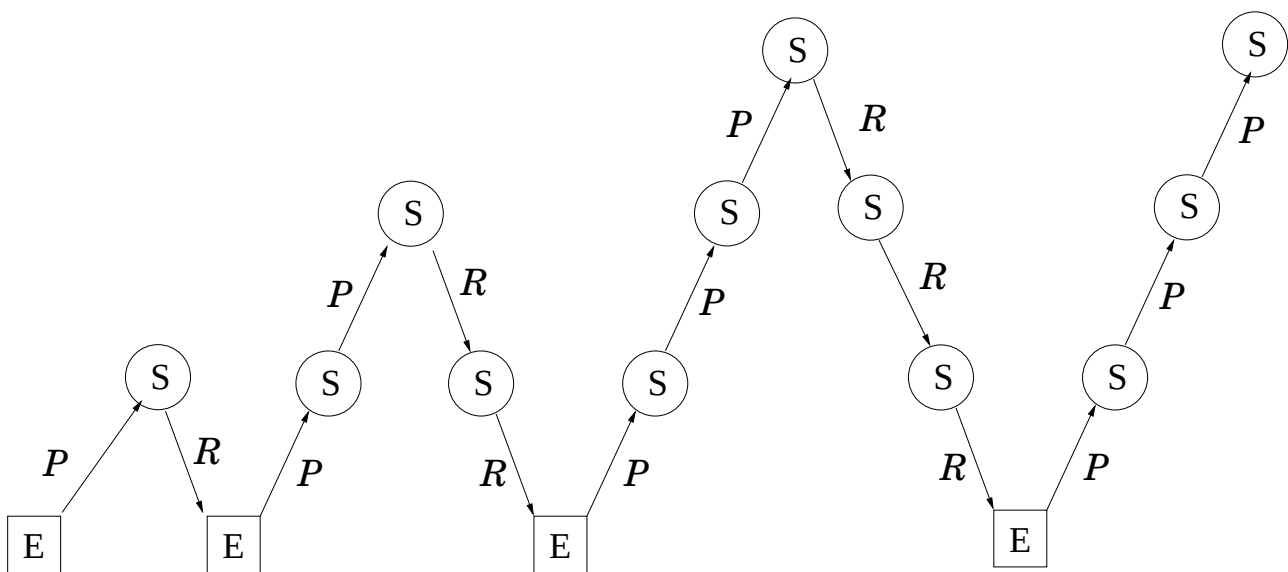
- (3) Prolongation  $(m) \rightarrow (m-1) : \quad \Phi_{nh/2} = \mathcal{P}\Phi_{nh}$

- (4) Beginning from the level  $(m-1)$ , start a V-cycle, and interpolate the solution at the end of the “V” to the next finer grid-level using a prolongation

$$\Phi_{kh} = \mathcal{P}\Phi_{2kh}$$

until the top (finest) level (1) is reached.





Full-Multigrid-Algorithm with 4-grids

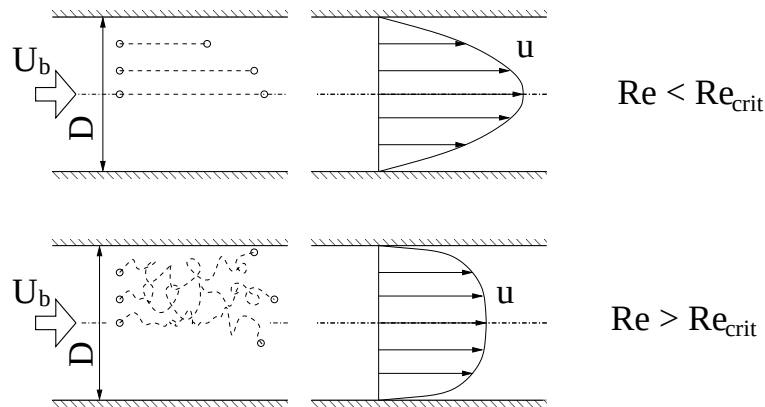
# IV Turbulent flow

## IV.1 Origin of Turbulence

### Stability of viscous flow (transition laminar $\rightarrow$ turbulent)

Hydrodynamic Instability = necessary condition for  
transition to turbulence

essentially dependent on flow configuration and Reynolds number  
e.g. pipe flow



dimensional analysis:  $u = f(D, U_b, \varrho, \mu) \rightarrow \frac{u}{U_b} = f\left(\frac{\varrho U_b D}{\mu}\right)$

$\Rightarrow$  key parameter is the **Reynolds-number**

$$Re = \frac{U_b D}{\nu} \quad \text{with} \quad U_b = \frac{Q}{D^2 \pi}, \quad \nu = \frac{\mu}{\varrho}$$

$Re > Re_{crit} \rightarrow$  unstable, **transition** into turbulent regime **possible**  
(depends on the existence of disturbances upstream)

$Re < Re_{crit} \rightarrow$  flow remains **always** laminar

## IV.2 Fluidmechanical instabilities

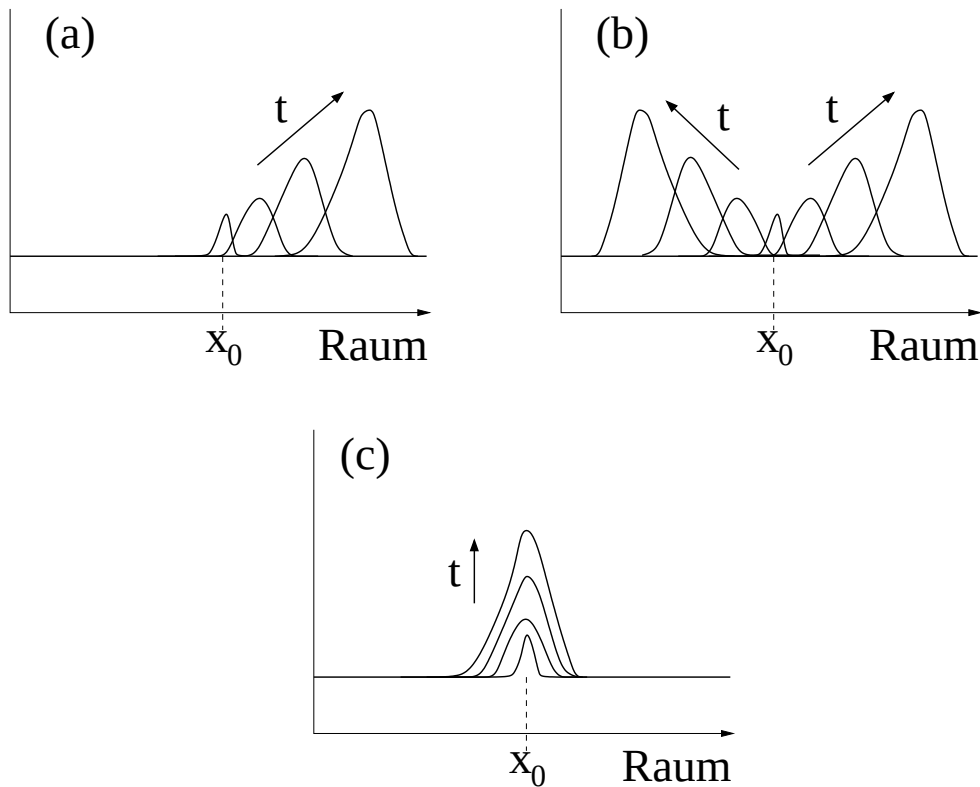
Definition of instability in time-space-system:

*Amplitude of any initially infinitesimally small perturbation grows unlimited in time and/or space*

↪ We distinguish two types of flow instabilities

- *Absolute instability*: amplitude of the perturbation grows in time, but stays at its local origin
- *Convective instability*: amplitude of the perturbation grows in time, but moves convectively away from its local origin (upstream and/or downstream)

Growth of perturbation in time-space system



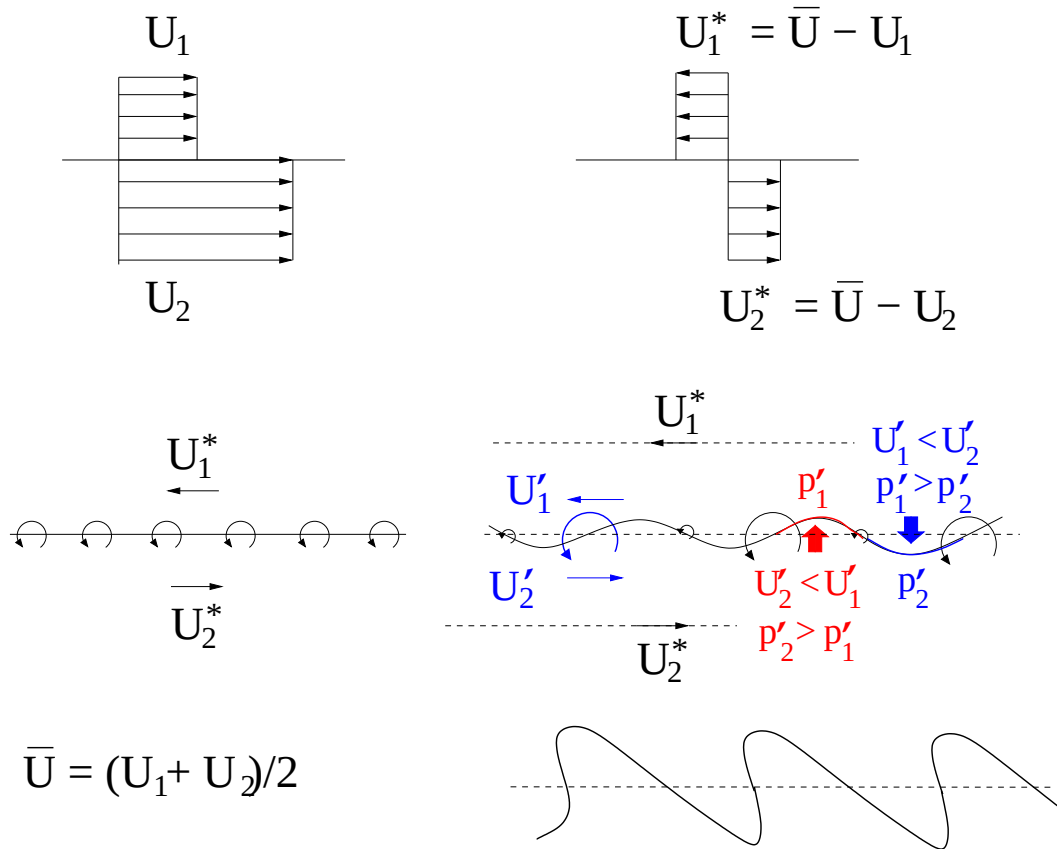
(a) and (b): convective Instability; (c) absolute instability

## IV.2.1 Inviscid stability

Viscous forces play no role in triggering the instability

Example: Kelvin-Helmholtz Instability

= one of the most important hydrodynamic instabilities



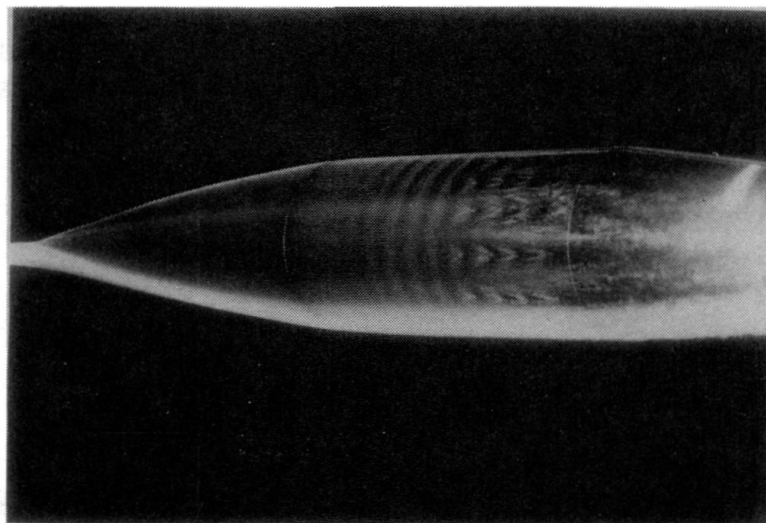
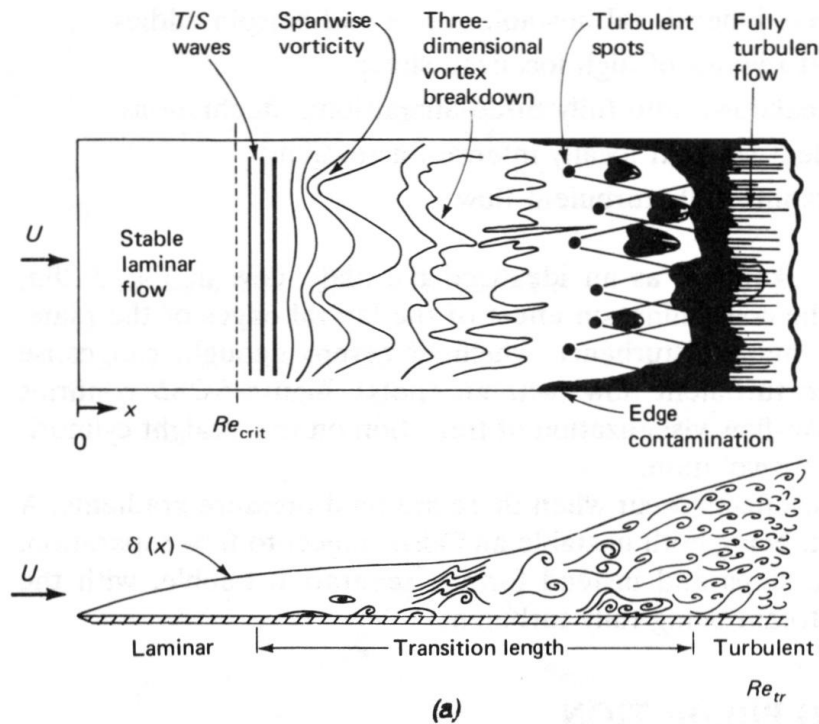
Kelvin-Helmholtz instability in nature



## IV.2.2 Viscous instability

Viscosity plays important role in instability mechanism

⇒ e.g.: Transition to turbulence in wall boundary layer



a) idealized sketch, b) measurements by Kegelman and Müller (AIAA J. 1986)

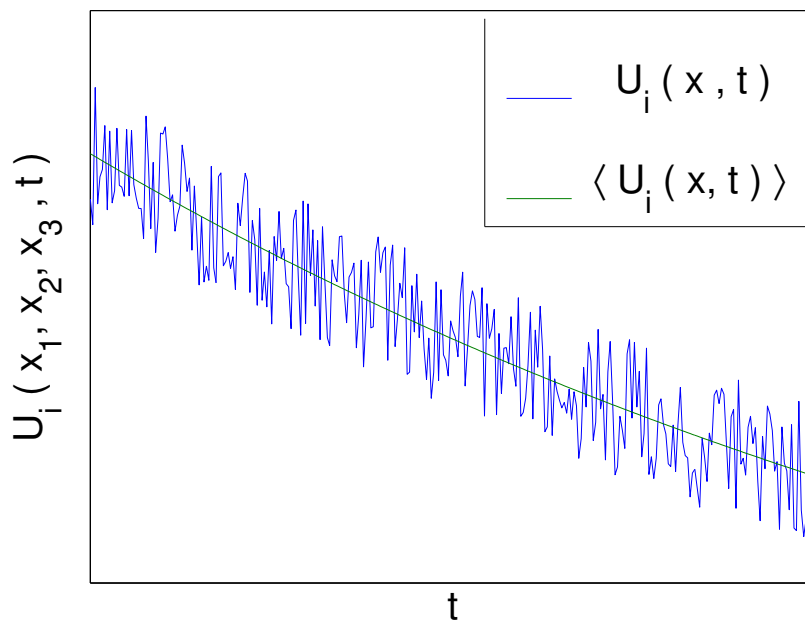
## IV.3 Computation of incompressible turbulent flow

Supercritical flow with  $Re > Re_{crit}$  is met in most technically relevant applications

We recognize: turbulent flow is **inherently**

**spatially three-dimensional**  
and **instationary** !  $\Rightarrow u_i = u_i(x_1, x_2, x_3, t)$

all relevant flow quantities flow quantities  $(u_i, p)$  fluctuate in time and space.



Example of instantaneous local velocity  $u_i$  fluctuating in time

## ⇒ Basic concepts of mathematical description

- Fully resolved description in space and time:

*solves directly* the unsteady discretized representation of the Navier-Stokes equations, as, e.g., generally specified in (II.1), with *highest possible accuracy*, so as to

↪ capture convective/diffusive transport on *all physically relevant scales*

⇒ **D**irect **N**umerical **S**imulation (**DNS**)

- Statistical description:

*solves numerically a statistically averaged representation* of the Navier-Stokes equations,  $\langle(\text{I.6})\rangle$  &  $\langle(\text{I.7})\rangle$ , such as

$$\left\langle \frac{\partial u_j}{\partial x_j} \right\rangle = 0$$
$$\left\langle \frac{\partial u_i}{\partial t} + \frac{\partial u_i u_j}{\partial x_j} \right\rangle = \left\langle -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \frac{\mu}{\rho} \frac{\partial^2 u_i}{\partial x_j \partial x_j} \right\rangle, \quad i = 1, 2, 3$$

for the statistically averaged flow quantities

$\langle u_i \rangle, \langle p \rangle$  (first order moments).

⇒ **R**eynolds **A**veraged **N**avier-**S**tokes (**RANS**)

- Spatially averaged description:

*solves numerically a spatially averaged (= spatially filtered) representation* Navier-Stokes equations,  $\overline{(\text{I.6})}$  &  $\overline{(\text{I.7})}$ , such as

$$\begin{aligned} \overline{\frac{\partial u_j}{\partial x_j}} &= 0 \\ \overline{\frac{\partial u_i}{\partial t} + \frac{\partial u_i u_j}{\partial x_j}} &= \overline{-\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \frac{\mu}{\rho} \frac{\partial^2 u_i}{\partial x_j \partial x_j}}, \quad i = 1, 2, 3 \end{aligned}$$

for the spatially averaged (filtered) flow quantities  $(\bar{u}_i, \bar{p})$ .

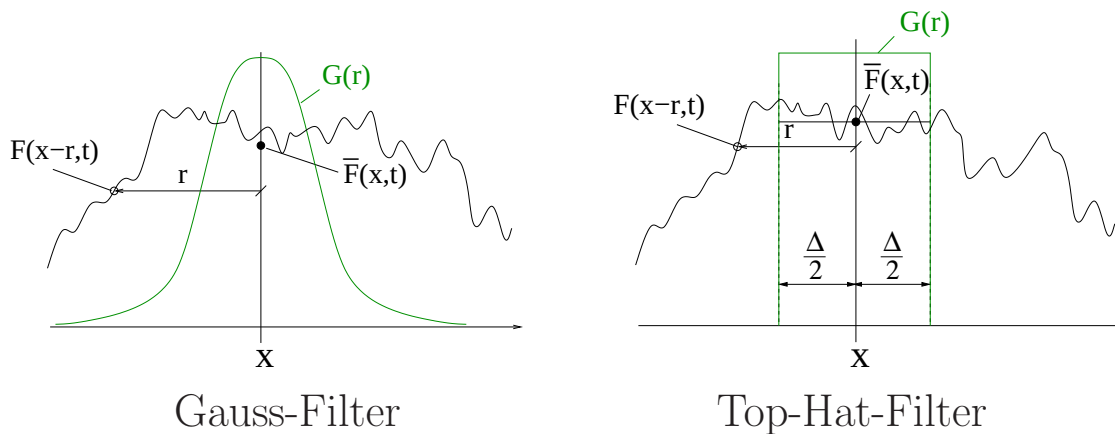
$\Rightarrow$  **Large-Eddy Simulation (LES)**

(= **underresolved** direct description)

Spatial filtering procedure:

$$\bar{F}(\mathbf{x}, t) = \int_{\mathcal{D}} F(\mathbf{x} - \mathbf{r}, t) G(\mathbf{x}, \mathbf{r}) d\mathbf{r}$$

$G(\mathbf{x}, \mathbf{r}) \dots$  filter-function





## IV.4 Statistical averages and correlations

Velocity represents a fluctuating quantity (random variable):

$$\mathbf{U} = U_i(\mathbf{x}, t), \quad \text{with the spatial coordinate-indices } i = 1, 2, 3$$

$N$  realizations of the turbulent flow field (e.g.  $N$  experiments)

$$\hat{=} \text{Ensemble } U_i^{(n)} : \quad n = 1, \dots, N$$

**Ensemble average:**

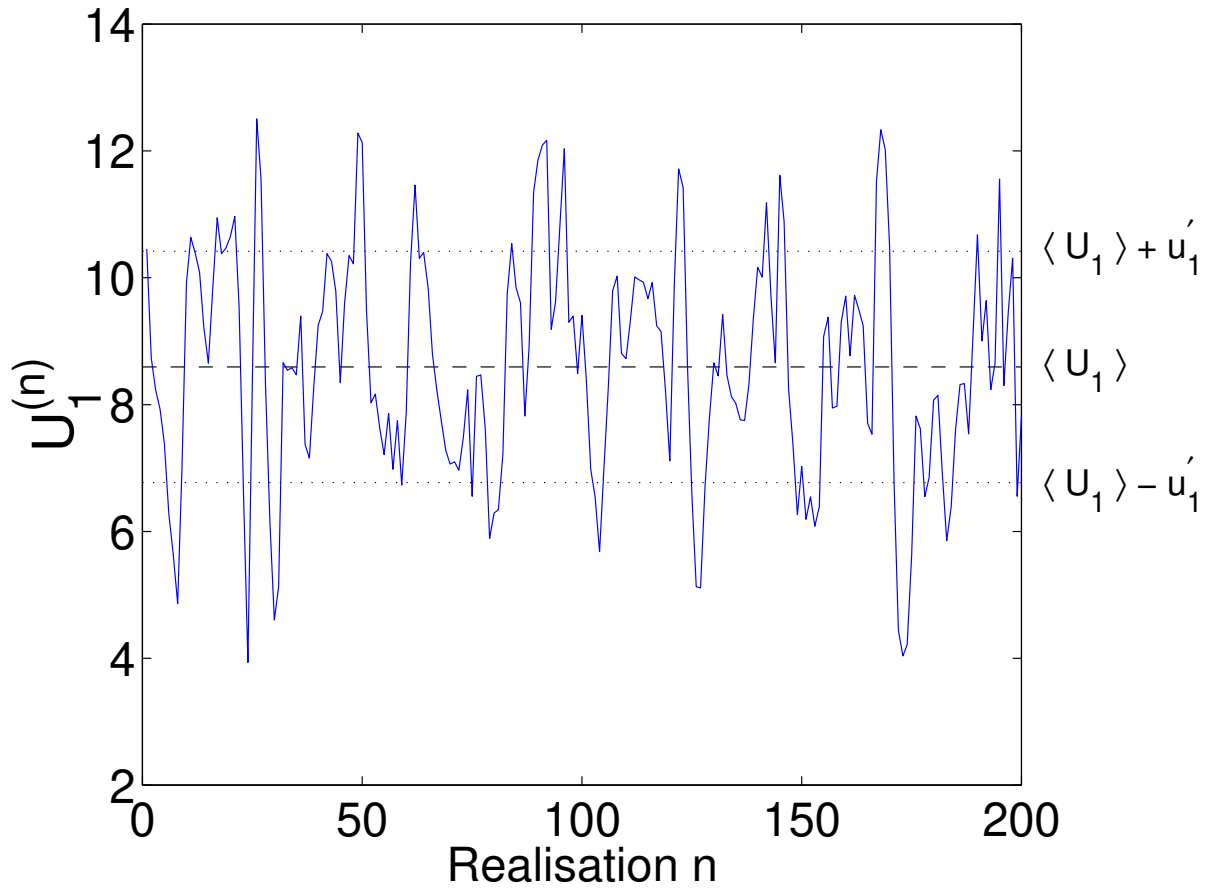
$$\langle U_i \rangle = \frac{1}{N} \sum_{n=1}^N U_i^{(n)}$$

$$\text{Decomposition:} \quad U_i = \langle U_i \rangle + u_i$$

$$\text{where:} \quad \langle u_i \rangle = 0$$

**Standard deviation**

$$u'_i = \left\langle (U_i - \langle U_i \rangle)^2 \right\rangle^{\frac{1}{2}} = \langle u_i^2 \rangle^{\frac{1}{2}}$$



Ensemble of the velocity component  $U_1$  with  $N = 200$

**Correlation**  $R_{ij}$ : e.g. between two variables  $U_i$  and  $U_j$

$$R_{ij} = \left\langle (U_i - \langle U_i \rangle) (U_j - \langle U_j \rangle) \right\rangle = \langle u_i u_j \rangle$$

and nondimensionalized with corresponding standard deviation  $u'_k$ :

$$\varrho_{ij} = \frac{\langle u_i u_j \rangle}{u'_i u'_j}, \quad (-1 \leq \varrho_{ij} \leq +1)$$

If variables  $U_i$  and  $U_j$  are statistically independent,  
 $\rightarrow U_i$  and  $U_j$  **“decorrelate”**:

$$\begin{aligned} \Rightarrow \quad & \langle U_i U_j \rangle = \langle U_i \rangle \langle U_j \rangle \\ & R_{ij} = \langle u_i u_j \rangle = 0 ; \quad \varrho_{ij} = 0 \end{aligned}$$

## IV.4.1 Statistically stationary flows

Ensemble average is **time-independent** in the case of sufficiently big  $N$

$$\langle U_i \rangle = \langle U_i \rangle_{(\mathbf{x})} \neq f(t)$$

Definition of the temporal average:

$$\bar{U}_i(\mathbf{x}) = \frac{1}{T} \int_0^T U_i(\mathbf{x}, t) dt$$

For large  $T$ , i.e.  $T \rightarrow \infty$  we get:

$$\bar{U}_i = \langle U_i \rangle$$

If the time interval for averaging  $T$  is sufficiently large, in a **statistically stationary** flow:

**temporal average = ensemble average.**

Accordingly, all correlations read:

$$R_{ij} = \overline{u_i u_j}, \quad \varrho_{ij} = \frac{\overline{u_i u_j}}{\overline{u'_i u'_j}}; \quad u'_i = \left( \overline{u_i^2} \right)^{\frac{1}{2}}$$

**!** **Difference** to statistically non-stationary flow:

$$\begin{array}{ccc} \text{temporal average} & \neq & \text{ensemble average} \\ \bar{u} & \neq & \langle u \rangle \end{array}$$

**Reynolds Averaged Navier-Stokes (RANS)** methods solve for the

- temporal averages in the case of statistically stationary flow
- ensemble averages in the case of statistically non-stationary flow

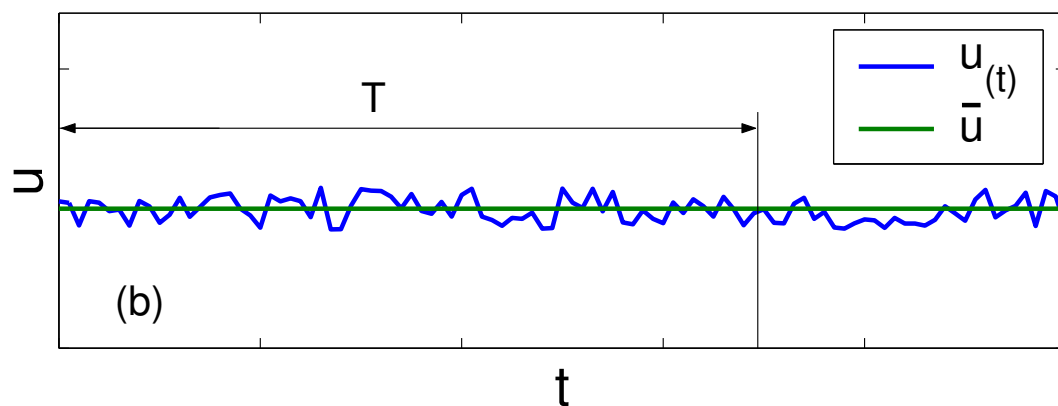
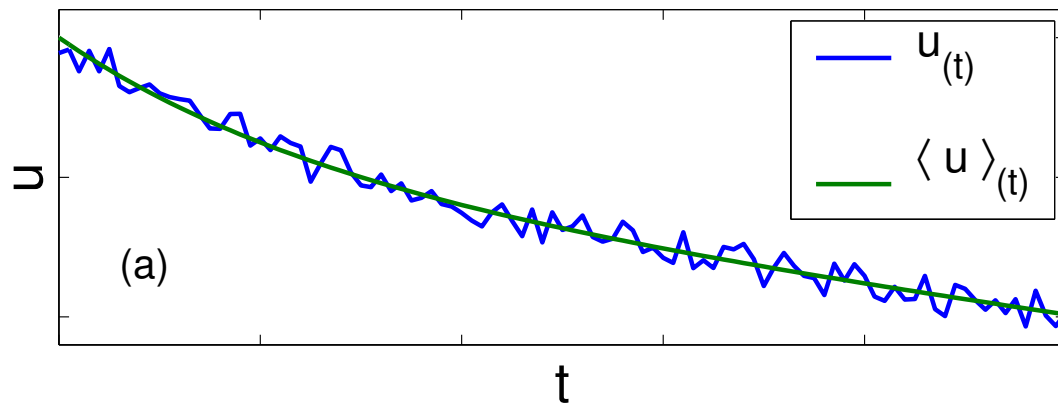
The higher-order terms occurring in the RANS equations

$$R_{ij} = \overline{u_i u_j}, \text{ and } \langle u_i u_i \rangle$$

must be **modelled** (“One-Point closure”):

$$\Rightarrow k - \varepsilon, \text{RSM, etc.}$$

Averaging over ensemble, respectively in time:

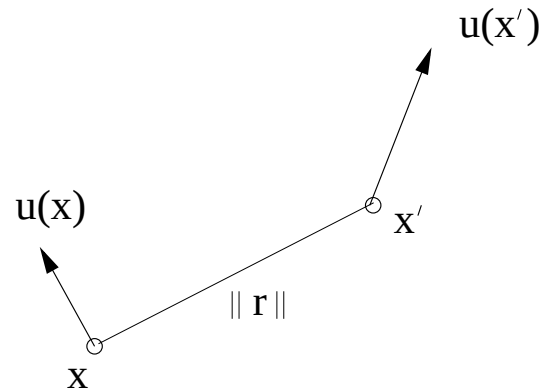


(a) statistically instationary, (b) statistically stationary

## IV.5 Length scales of the turbulence

Correlation tensor between  
two points  $\mathbf{x}$  and  $\mathbf{x}'$ :

$$R_{ij}(\mathbf{x}, \mathbf{x}', t) = \left\langle u_i(\mathbf{x}, t) u_j(\mathbf{x}', t) \right\rangle$$



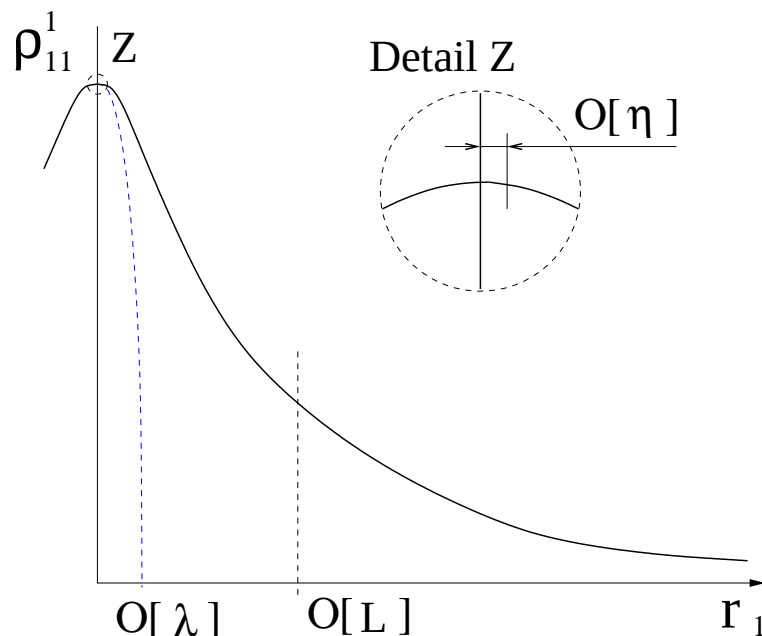
Isotropic turbulence:

$$R_{ij}(\mathbf{x}, \mathbf{x}', t) = R_{ij}(r, t), \quad \dots \quad r = \|\mathbf{x} - \mathbf{x}'\|$$

$$\begin{array}{lll} r \rightarrow \infty : & R_{ij} \rightarrow 0 & \dots u_i, u_j \text{ decorrelates} \\ r \rightarrow 0 : & R_{ij} \rightarrow \langle u_i u_j \rangle_{(\mathbf{x}, t)} & \dots \text{single-point-correlation} \end{array}$$

e.g. typical spatial variation of the correlation coefficient

$$\varrho_{11}^1 = \varrho_{11}(r_1, r_2 = 0, r_3 = 0)$$



## (A) Integral length scale

$$L_{11}^1 = \int_0^\infty \varrho_{11}(r_1, r_2 = 0, r_3 = 0) \, dr_1$$

Isotropy:  $L_{11}^1 \simeq L_{ij}^k = L$

## (B) Energy transfer in spectral space: Kolmogorov scale

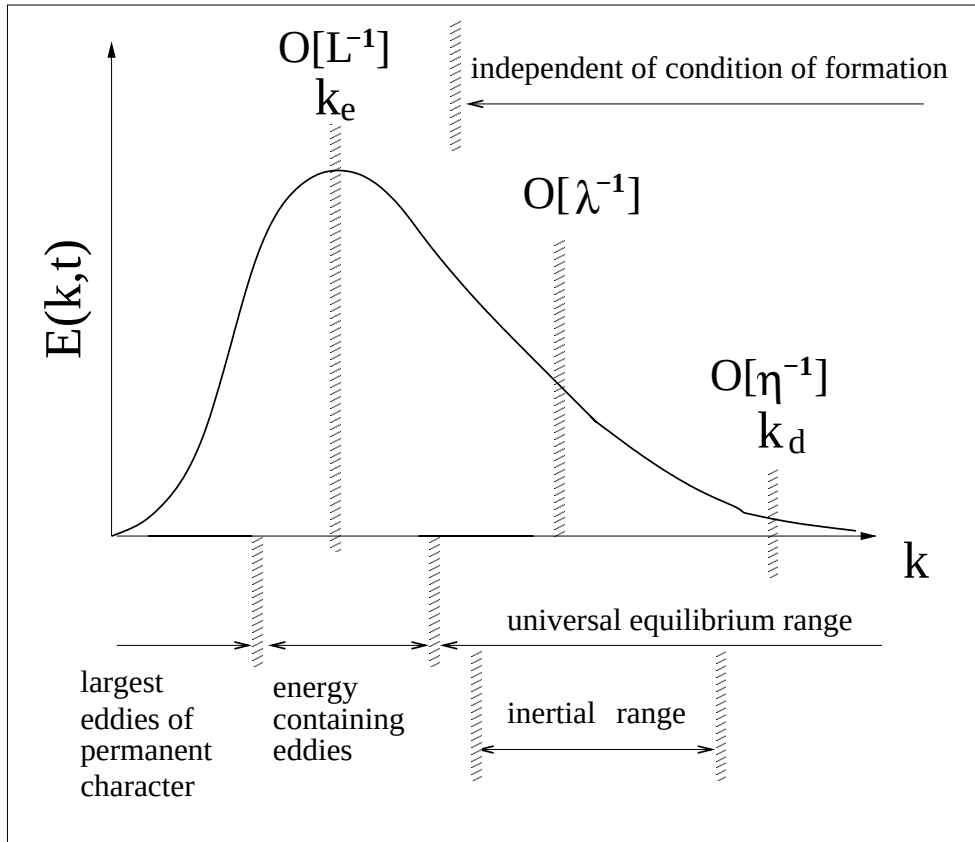
Turbulent energy spectrum  $E$  represents the distribution of the turbulent energy in respect in wave-number space ( $\hat{=}$  “spectral energy-density”):

$$\frac{1}{2} \langle u_i u_i \rangle = \int_0^\infty E(\mathbf{k}, t) d\mathbf{k}$$

Isotropic turbulence:  $E(\mathbf{k}, t) = E(k, t)$  with  $k = \|\mathbf{k}\|$

$$\frac{1}{2} \langle u_i u_i \rangle = \int_0^\infty E(k, t) dk$$

Turbulent energy-spectrum (isotropic):



The turbulence gets more and more isotropic with increasing wave-number,  $\rightarrow$  it becomes independent of its origin.

The evolution of  $E(k, t)$  is given by

$$\frac{\partial E(k, t)}{\partial t} = \underbrace{T(k, t)}_{\text{inert.transfer}} - \underbrace{2\nu k^2 E(k, t)}_{\text{dissipation } \varepsilon(k, t)} \quad (\text{IV.1})$$

$T(k, t) \dots$  non-linear transfer of turbulent energy  
from large to small eddies,  
i.e. from low  $k$  to high  $k$   
(= energy cascade)

$\varepsilon(k, t) \dots$  viscous dissipation of  
the turbulent energy in wave-number space  
comes primarily into effect at high  $k$

$$\int_0^\infty (\text{IV.1}) dk \quad \text{yields}$$

$$\frac{1}{2} \frac{\partial \langle u_i u_i \rangle}{\partial t} = \underbrace{\int_0^\infty T(k, t) dk}_{= 0} - \underbrace{2\nu \int_0^\infty k^2 E(k, t) dk}_{= \bar{\varepsilon}} \quad (\text{IV.2})$$

so

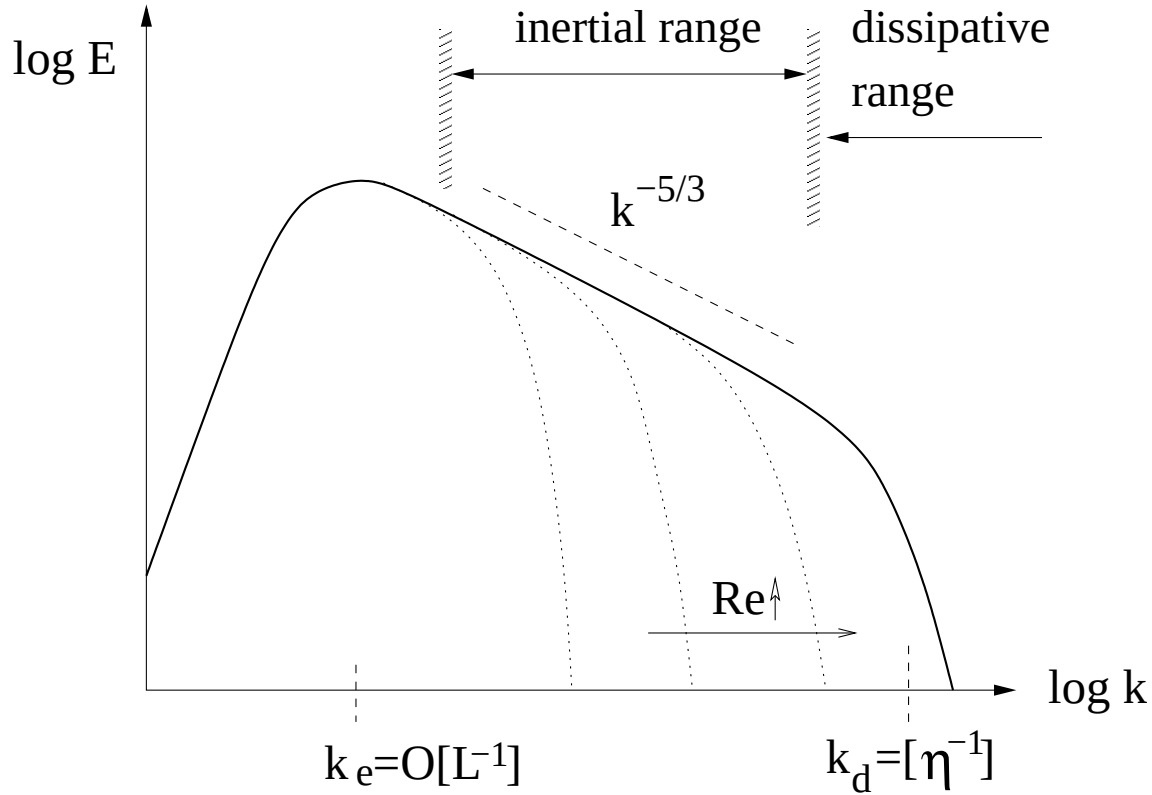
|                                                                   |   |                              |
|-------------------------------------------------------------------|---|------------------------------|
| total turbulent<br>energy loss                                    | = | total viscous<br>dissipation |
| $\frac{1}{2} \frac{\partial \langle u_i u_i \rangle}{\partial t}$ | = | $-\bar{\varepsilon}$         |



## Kolmogorov-theory about the small scales

Pre-condition: - sufficient high Re-number

$\Rightarrow$  broad “inertial range”:  $k_e \lll k_d$



Integration of (IV.1) from  $k$  to  $\infty$  with  $k_e \ll k \ll k_d$

$$\frac{d}{dt} \int_k^\infty E(k', t) dk' = \underbrace{\int_k^\infty T(k', t) dk'}_{= \mathcal{T}} - 2\nu \int_k^\infty k'^2 E(k', t) dk' \quad (\text{IV.3})$$

Due to the broad “inertial range”:

- major part of the turbulent energy is in the range of:  $k' < k$
- major part of the viscous dissipation is in the range of:  $k' > k$

This leads to the following assumptions for equation (IV.3):

- “*inertial range*” is in developed balance  
(full developed turbulence)

$$\frac{d}{dt} \int_k^\infty E(k', t) dk' \approx 0$$

- dissipation is mainly effective at high wave-numbers

$$2\nu \int_k^\infty k'^2 E(k', t) dk' \approx 2\nu \int_0^\infty k'^2 E(k', t) dk' = \bar{\varepsilon}$$

So, (IV.3) is reduced to

$$\mathcal{T} = \bar{\varepsilon}$$

The turbulence is **only** dependent on the spectral transfer  $\mathcal{T}$  in the developed “*inertial range*”, which is equal to the total dissipation  $\bar{\varepsilon}$ . Thus, written this range, the turbulence is determined by a **single** key-parameter  $\bar{\varepsilon}$ .

A dimensional analyzing gives:

$$E(k, t) = C \bar{\varepsilon}^{\frac{2}{3}} k^{-\frac{5}{3}} \quad (C = \text{const.})$$

The “*dissipative range*” is next to the “*inertial range*” on the high wave number side. In the “*dissipative range*”, the viscosity  $\nu$ , becomes relevant as the second parameter.

$\Rightarrow$  viscous (or dissipative) length scale = *Kolmogorov length scale*:

$$\eta = \left( \frac{\nu^3}{\bar{\varepsilon}} \right)^{\frac{1}{4}}$$

$\hat{=}$  **smallest length scale of the turbulence**

Estimate for the ratio  $\frac{\eta}{L}$  :

---

Dimension of  $\bar{\varepsilon}$  reads according to equation (IV.2):

$$\bar{\varepsilon} \simeq O \left[ \underbrace{\frac{u'_i u'_i}{t}}_{\simeq O[\frac{L}{u'_i}]} \right] \simeq O \left[ \frac{u'^3_i}{L} \right]$$

$$\frac{\eta}{L} = O \left[ \left( \frac{\nu^3}{u'^3_i L^3} \right)^{\frac{1}{4}} \right] = O \left[ Re_L^{-\frac{3}{4}} \right] \ll 1$$

Generally:

$$Re_L \approx 0.01 Re$$

# V Fully resolved description: Direct Numerical Simulation (DNS)

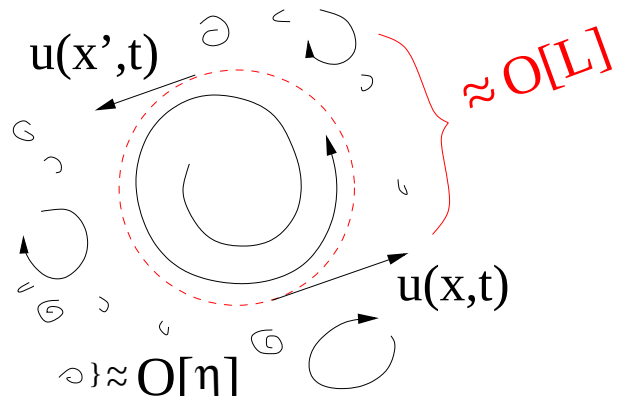
$\hat{=}$  **the most accurate** solution method of the Navier-Stokes equations

- DNS captures numerically the motion on **all** physically relevant length scales.
- **no** averaging (temporal or spatial)
- **no** approximation for unclosed terms
- basically the simplest concept for flow simulation
  - **no** modelling required!
- discretization = only approximation

Resolution requirements for homogeneous turbulence:

Length of the largest turbulent eddies:  $\approx O[L]$

Length of the smallest turbulent eddies:  $\approx O[\eta]$



$\Rightarrow$  Domain-length:  $\sim n \cdot L$  (with, e.g.  $n = 3 \div 6$ )

$\Rightarrow$  Minimum number of the points/dimension in space:  $N_i = \frac{L}{\eta}$

$\rightarrow N_{min} \geq \left(\frac{L}{\eta}\right)^3 = Re_L^{\frac{9}{4}}$  (3-Dim)

Resolution requirement for inhomogeneous turbulent flow on the basis of DNS of **turbulent pipe flow** (Leonard 1994):

$$\text{Re-number: } Re = \frac{\bar{U} D}{\nu}$$

$$\text{Kolmogorov-length: } \frac{\eta}{D} \sim Re^{-\frac{3}{4}}$$

$$\text{pipe length: } L = 10 D$$

$$\text{mesh width: } \Delta = 3\eta$$

| Reynolds-<br>number: | Kolmogorov-length<br>$\frac{\eta}{D}$ | required number of grid points<br>$N = \frac{10\pi}{4} \left(\frac{D}{\Delta}\right)^3$ |
|----------------------|---------------------------------------|-----------------------------------------------------------------------------------------|
|----------------------|---------------------------------------|-----------------------------------------------------------------------------------------|

---

|                 |         |                      |
|-----------------|---------|----------------------|
| $5 \times 10^3$ | 0.00450 | $3.1 \times 10^6$    |
| $1 \times 10^4$ | 0.00280 | $1.3 \times 10^7$    |
| $5 \times 10^4$ | 0.00093 | $3.6 \times 10^8$    |
| $1 \times 10^5$ | 0.00058 | $1.5 \times 10^9$    |
| $5 \times 10^5$ | 0.00019 | $4.2 \times 10^{10}$ |

⇒ **Very high** number of grid points (  $N_{\text{DNS}} \gg N_{\text{RANS}}!$  )

limits DNS to very small Re-numbers

(e.g.  $\text{Re} = 3000 \div 10000$ )

however, technically relevant Re-numbers often  $> O[10^4]!$

Conclusion:

highest resolution requirements

+ 3-dimensionality (symmetries can not be exploited)

⇒

**DNS is too expensive for the  
most technical applications!**

Why is DNS still important?

- DNS supplies the most detailed information (complete statistic!)
- better understanding of the mechanism:  
production, transfer, dissipation of turbulent energy
- DNS-data useful for modelling and model-evaluation  
(RANS, LES)

## V.1 Numerical integration in DNS

### Temporal discretization

Accuracy (and stability) requires small time steps  $\Delta t$ .

→ explicit methods ( $2^{nd} \div 4^{th}$ -order accurate) often applied  
(Adams-Bashforth, Runge-Kutta, Leapfrog, etc.)

Nevertheless, instability due to high spatial gradients (e.g. near the wall):

→ Implicit schemes (e.g. Crank-Nicholson) in corresponding direction

### Spatial discretization

DNS must resolve numerically the whole spectrum in length scales  $0 \leq k \leq k_d$ :

→ very accurate discretization required  
(minimal dissipation- and dispersion error)

Typical DNS-schemes:

- high-order ( $\geq O[\Delta x^4]$ ) FD-schemes
- Padé-schemes (Compact schemes)  $\geq O[\Delta x^5]$
- spectral methods

Strong dissipative schemes (e.g. low-order FD, Upwind, TVD) are often not applicable. The high numerical dissipation of these schemes can “dampen” the high-frequency turbulent motion (→ laminarization).

On the other hand, highly accurate, low-dissipative schemes tend to instability.

**Important:** schemes should be **conservative**.

## V.2 Initial- and boundary conditions for DNS

High sensibility to the setting of boundary conditions

- DNS requests **unsteady** boundary conditions (at “inflow”)
- Errors and deficits of the boundary conditions may “contaminate” the entire flow field
- Periodic boundary conditions in homogeneous directions are best suited
- No-slip conditions at solid walls ( $\mathbf{u}_w = 0$ ), very high local resolution near to the wall is required!
- Inflow-conditions often from another DNS (“Inflow-generator”)
- Outflow-conditions often as convective boundary conditions:

$$\frac{\partial \Phi}{\partial t} - U \frac{\partial \Phi}{\partial n} = 0$$

$$U = \frac{1}{A_{out}} \int_{A_{out}} u_n dA, \quad \vec{n} \perp dA$$

= very robust, virtually no backflow

Required total computation time of DNS:

$$t_{total} \sim \text{several “Flow-through-times” } t_f$$

$$t_f = \frac{L}{U_m}$$

$L \dots$  Length of the computational domain in the main-flow direction

$U_m \dots$  average velocity in the main-flow direction



**Important:** at the end of the computation, where  $t = t_{total}$

- DNS must have “forgotten” its initial conditions
- the simulated turbulent flow field must be statistically stationary (for statically stationary flows)

$\Rightarrow t_{total}$  very long (compared to RANS)

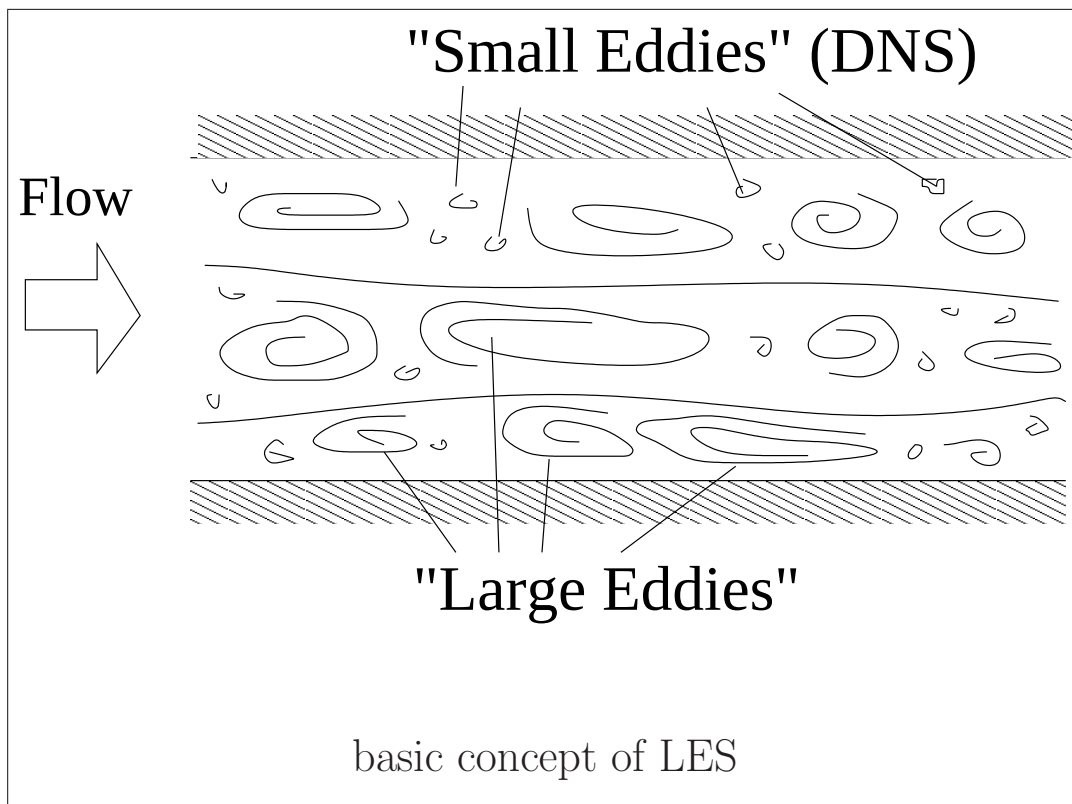
$\Rightarrow$  The long computation time increase the computational costs of DNS!

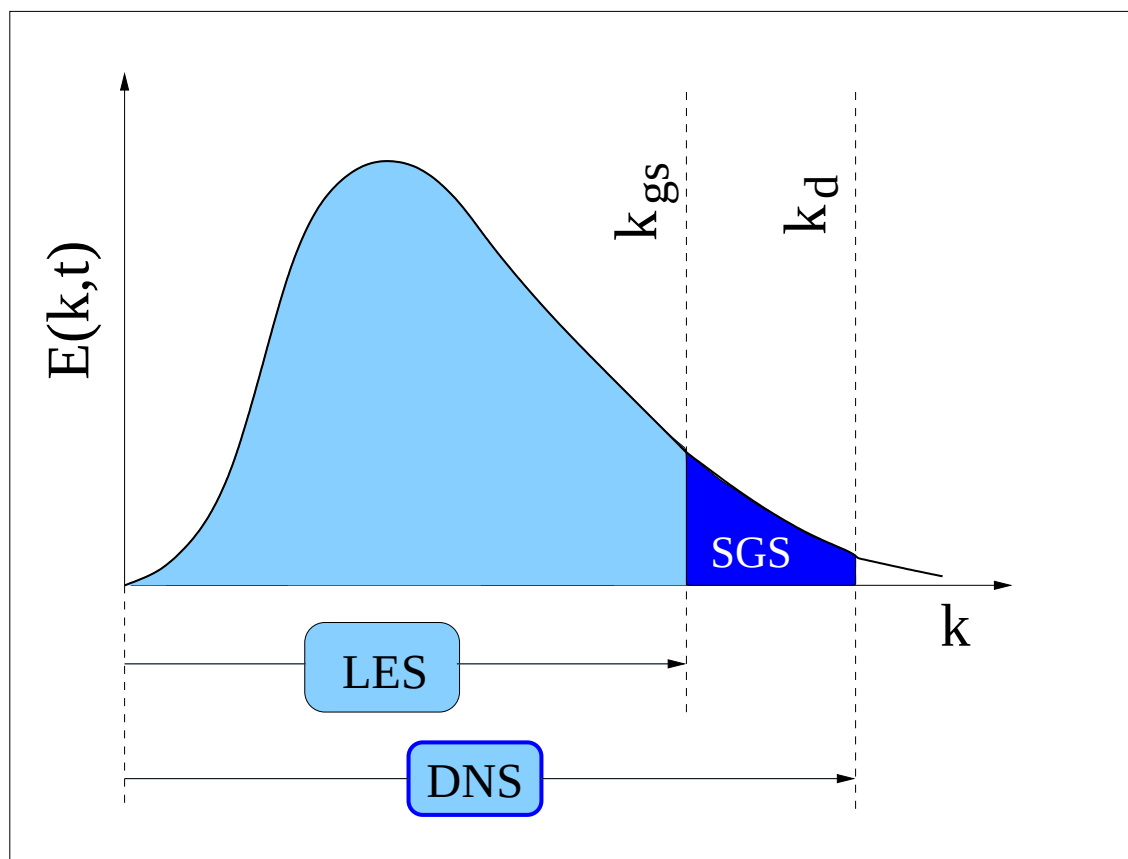
## VI Underresolved direct description: Large Eddy Simulation (LES)

Direct Numerical Simulation computationally too expensive for high Re-numbers

Compromise  $\Rightarrow$  Large Eddy Simulation (LES):

- Numerical resolution only of the large eddies (structures)  
→ requires less grid points than DNS
- Modelling of the small eddies (in general, small eddies are more isotropic and so simpler to model)
- Computation time is also long, but acceptable compared to DNS.





Conceptual comparison between LES and DNS in the energy spectrum

## VI.1 LES equations - filtering

LES resolves “spatially filtered” transport equations:

→ **filtered** velocity- and scalar-fields

Spatial filtering of an arbitrary variable  $F(\mathbf{x}, t)$ :

$$\overline{F}(\mathbf{x}, t) = \int_{\mathcal{D}} F(\mathbf{x} - \mathbf{r}, t) G(\mathbf{x}, \mathbf{r}) d\mathbf{r} \quad (\text{VI.4})$$

$G(\mathbf{x}, \mathbf{r})$  ... filter-function

$$\text{with} \quad \int_{\mathcal{D}} G(\mathbf{x}, \mathbf{r}) d\mathbf{r} = 1$$

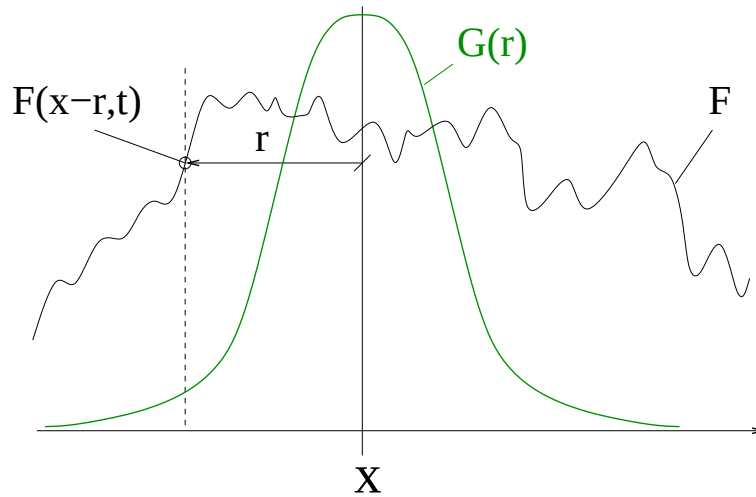
Filters are often *homogeneous*:  $G(\mathbf{x}, \mathbf{r}) = G(\mathbf{r}) \neq f(\mathbf{x})$

and *symmetric*:  $G(\mathbf{r}) = G(-\mathbf{r})$

### Filtering in one dimension

$$\overline{F}(x, t) = \int_{-\infty}^{\infty} F(x - r, t) G(r) dr$$

Schematic  
representation  
of 1-dimensional  
filtering



### Filter

general

$$G(r)$$

$$\begin{aligned}\hat{G}^*(k) &= \int_{-\infty}^{\infty} G(r) e^{-ikr} dr \\ &= 2\pi \mathcal{F}\{G(r)\} = 2\pi \hat{G}(k)\end{aligned}$$

Box

$$\begin{cases} \frac{1}{\Delta} & \text{if } |r| \leq \frac{\Delta}{2} \\ 0 & \text{else} \end{cases}$$

$$\frac{\sin\left(\frac{1}{2} k \Delta\right)}{\frac{1}{2} k \Delta}$$

Gaussian

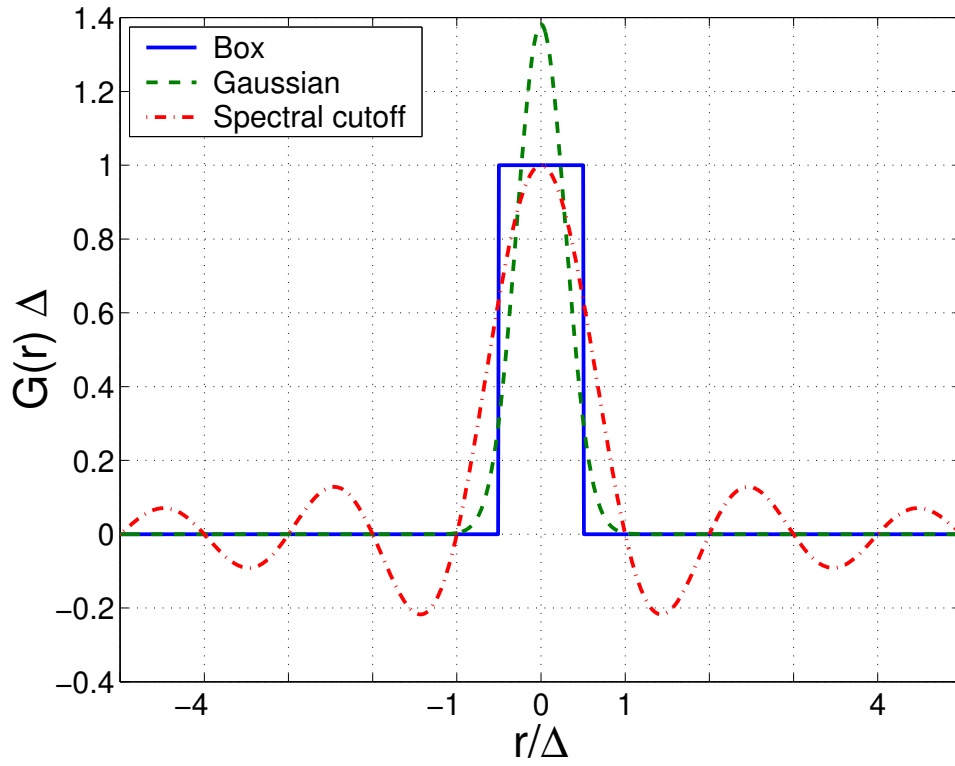
$$\left(\frac{6}{\pi \Delta^2}\right)^{\frac{1}{2}} \exp\left(-\frac{6r^2}{\Delta^2}\right)$$

$$\exp\left(-\frac{k^2 \Delta^2}{24}\right)$$

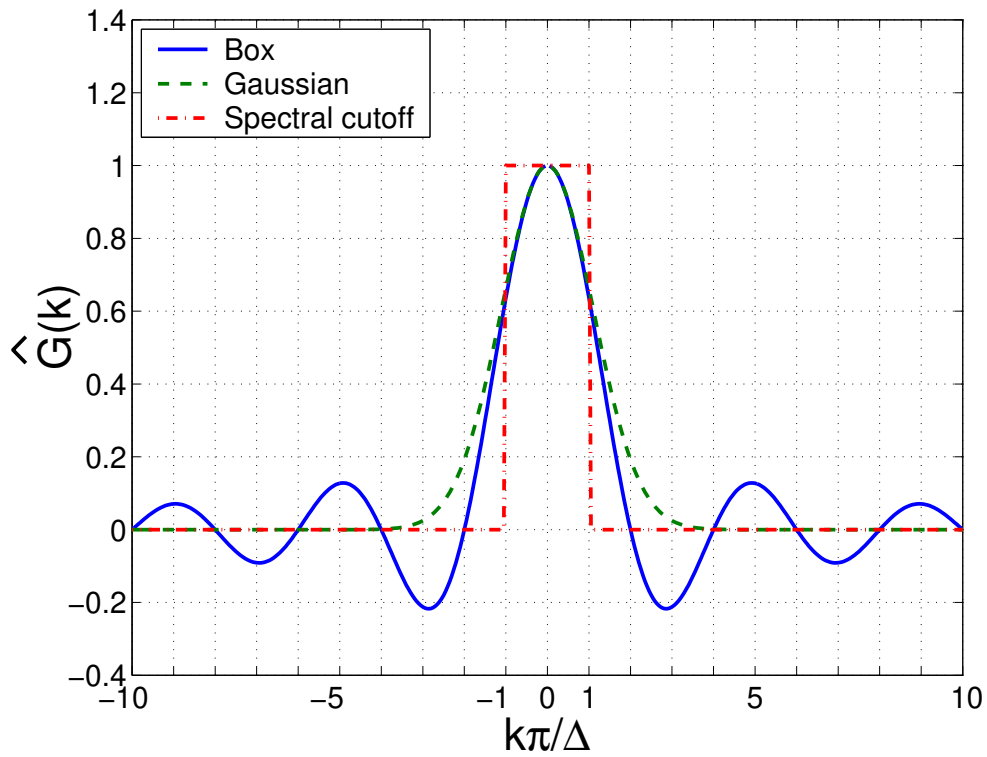
Sharp spectral  
(cutoff)

$$\frac{\sin\left(\frac{\pi r}{\Delta}\right)}{\pi r}$$

$$\begin{cases} 1 & \text{if } |k| \leq \frac{\pi}{\Delta} \\ 0 & \text{else} \end{cases}$$



Filter-function in the physical space  $G(r)$

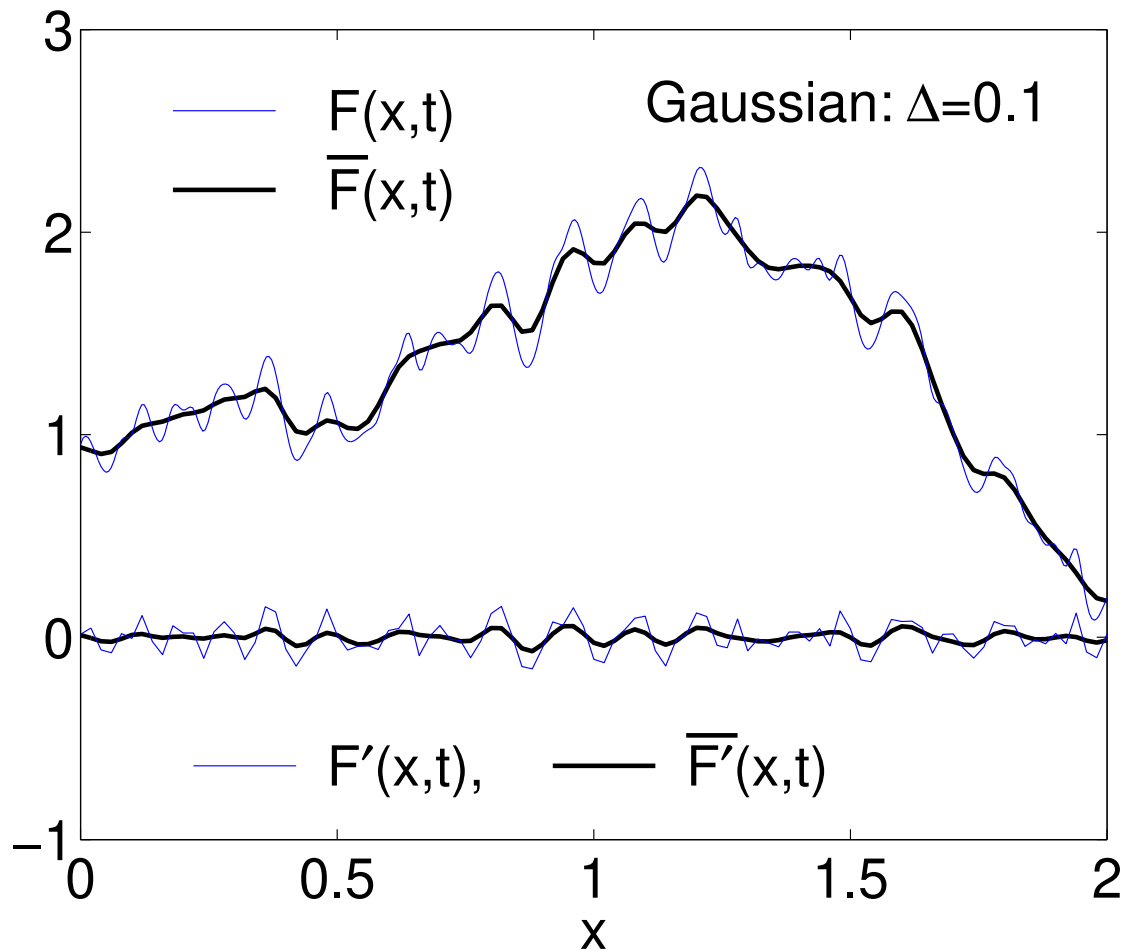


Spectral transfer function  $\hat{G}(k)$

Splitting:  $F = \overline{F} + F'$

$F'$  . . . . . Subgrid-scale (SGS) fluctuations

Note:  $\overline{F'} \neq 0$



Application of the Gaussian filter with filter width  $\Delta = 0.1$  to  $F(x)$

filter width      mesh width  
 $\Delta$        $\sim$        $\Delta x_i$

## Filtered energy spectrum

1-dimensional energy spectrum in  $r$ -direction:

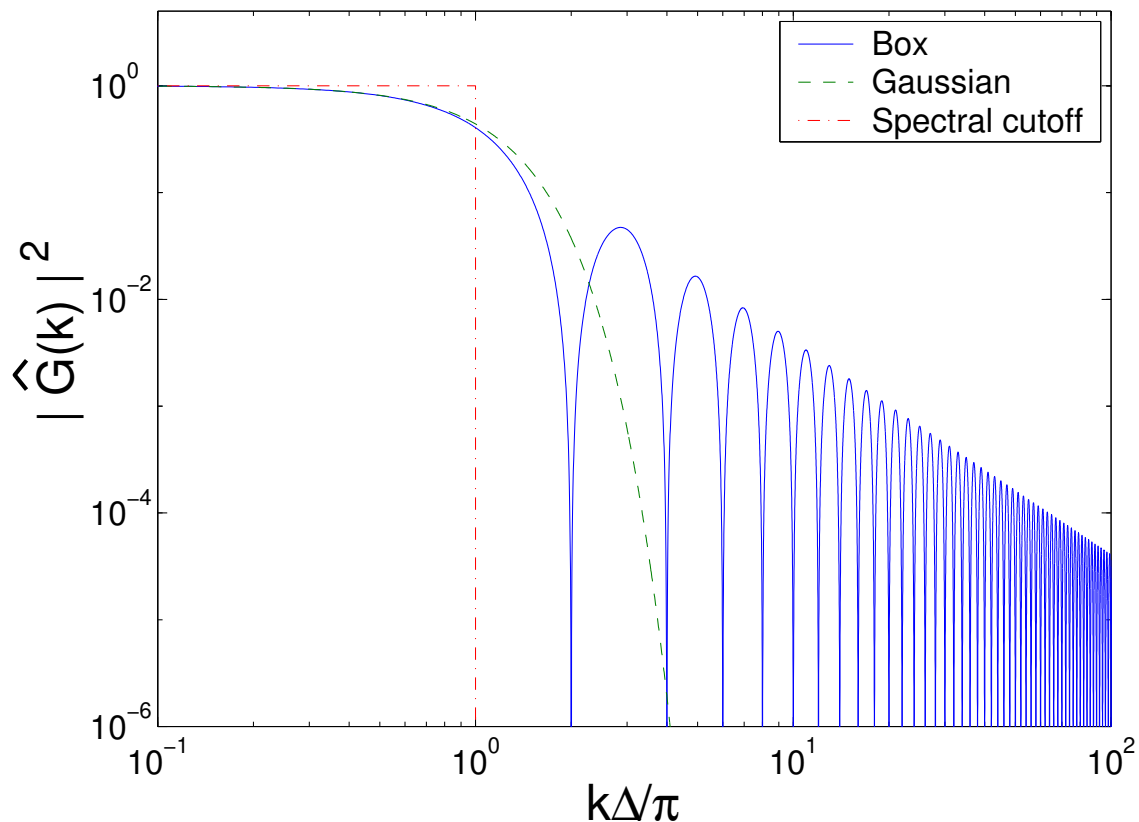
general definition : 
$$E_{ij} = \frac{1}{\pi} \int_{-\infty}^{\infty} R_{ij} e^{-ikr_1} dr_1$$

$i = 1, j = 1$  : 
$$R_{11}(r) = \langle u_1(x+r) u_1(x) \rangle$$

$$E_{11}(k) = \frac{1}{\pi} \int_{-\infty}^{\infty} R_{11} e^{-ikr} dr$$

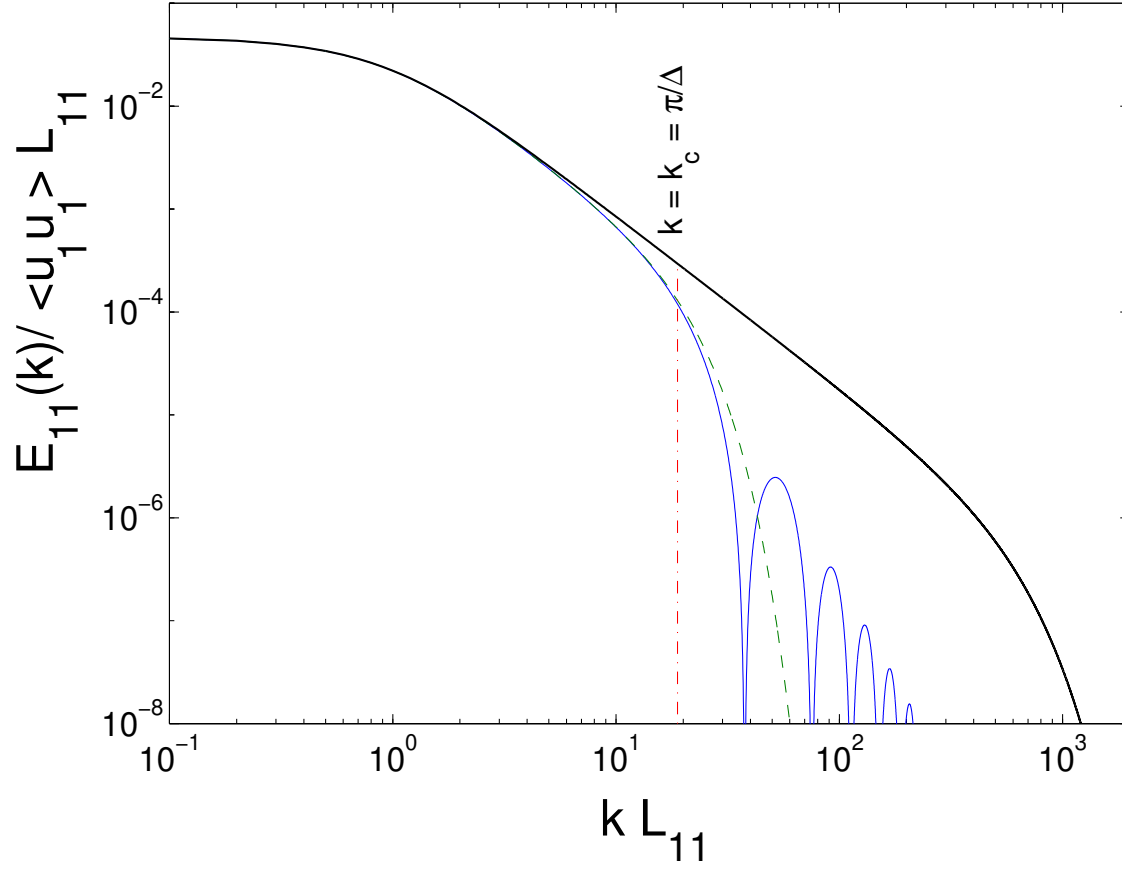
$$L_{11} = \frac{1}{\langle u_1^2 \rangle} \int_0^{\infty} R_{11}(r) dr$$

$$\overline{E}_{11}(k) = |\hat{G}(k)|^2 E_{11}(k) \quad \text{isotropic turbulence}$$



Spectral damping coefficient  $|\hat{G}(k)|^2$  for three types of filters





1-dimensional energy spectrum at  $Re_\lambda = 500$  (black thick line)  
 and the corresponding filtered spectra  
 (filter width at all three filters:  $\Delta = L_{11}/6$ )

## Filtered transport equations

The filter operation (VI.4), carried out by using a *homogeneous* and *symmetric* filter function  $G$

$$\bar{F}(\mathbf{x}, t) = \int_{\mathcal{D}} F(\mathbf{x} - \mathbf{r}, t) G(\mathbf{r}) d\mathbf{r}, \quad (\text{VI.5})$$

yields a filtering, which “*commutes*” with the temporal and spatial derivations, i.e

$$\overline{\left(\frac{\partial F}{\partial t}\right)} = \frac{\partial \bar{F}}{\partial t}, \quad \overline{\left(\frac{\partial F}{\partial x_j}\right)} = \frac{\partial \bar{F}}{\partial x_j}.$$

The commuting filter operation (VI.5), applied to the incompressible transport equations for **mass**, **momentum** and an arbitrary **scalar**  $\Phi$ , yields: ( $\rho, \nu_m, D_m = \text{const.}$ ):

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0 \quad (\text{VI.6})$$

$$\frac{\partial \bar{u}_i}{\partial t} + \frac{\partial \overline{u_i u_j}}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + \nu_m \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} \quad (\text{VI.7})$$

$$\frac{\partial \bar{\Phi}}{\partial t} + \frac{\partial \overline{u_j \Phi}}{\partial x_j} = D_m \frac{\partial^2 \bar{\Phi}}{\partial x_j \partial x_j} \quad (\text{VI.8})$$

Introduction of the *Residual (or subgrid-scale) stress and flux*

$$\tau_{ij}^R = \overline{u_i u_j} - \bar{u}_i \bar{u}_j \quad (\text{VI.9})$$

$$\sigma_j^R = \overline{u_j \Phi} - \bar{u}_j \bar{\Phi} \quad (\text{VI.10})$$

substituted into (VI.7)-(VI.8) yields

$$\frac{\partial \bar{u}_i}{\partial t} + \frac{\partial \bar{u}_i \bar{u}_j}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + \nu_m \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} - \frac{\partial}{\partial x_j} \underbrace{(\overline{u_i u_j} - \bar{u}_i \bar{u}_j)}_{\tau_{ij}^R}$$

$$\frac{\partial \bar{\Phi}}{\partial t} + \frac{\partial \bar{u}_j \bar{\Phi}}{\partial x_j} = D_m \frac{\partial^2 \bar{\Phi}}{\partial x_j \partial x_j} - \frac{\partial}{\partial x_j} \underbrace{(\overline{u_j \Phi} - \bar{u}_j \bar{\Phi})}_{\sigma_j^R}$$

The effect of the turbulent motion, which associated with the small scale range

$$\eta < l^R < O[\Delta],$$

which is not resolved by LES, is represented by the *Residual stress*-terms  $\tau_{ij}^R$  and  $\sigma_j^R$ , respectively.

Generally: Three contributions (please note: generally  $\bar{\bar{u}} \neq \bar{u}$ ):

$$\tau_{ij}^R = \underbrace{\overline{\bar{u}_i \bar{u}_j} - \bar{u}_i \bar{u}_j}_{(I)} + \underbrace{\overline{\bar{u}_i u'_j} + \overline{u'_i \bar{u}_j}}_{(II)} + \underbrace{\overline{u'_i u'_j}}_{(III)}$$

(I) Interaction between the eddies resolved by the LES

$$= \text{"Leonard term"} : \quad resolved \leftrightarrow resolved$$

(II) Interaction between the resolved and unresolved eddies

$$\hat{=} \text{Energy transfer} : \quad resolved \quad \overset{\leftarrow}{\rightarrow} \quad unresolved$$

(III) Interaction between the unresolved eddies, which may produce large eddies again

$$= \text{"backscatter"} : \quad unresolved \leftrightarrow unresolved$$

$$\Downarrow$$

$$resolved$$

$\tau_{ij}^R$  and  $\sigma_j^R$  must be modelled using so-called

### **Subgrid-Scale-models** ( "*SGS models*")

Most of the SGS-models model  $\tau_{ij}^R$  and  $\sigma_j^R$  as **one** term, i.e. **without** the splitting above (I)+(II)+(III) .

## Explicit $\leftrightarrow$ implicit filtration

**Explicitly** filtered LES-equations:

- filter operation (VI.5) with a defined filter function  $G$  applied explicitly to the transport equation.
- discretization of the explicit filtered equations

**Implicitly** filtered LES-equations:

- no explicitly defined filter function  $G$
- numerical discretization determines implicitly the filter  
 $\rightarrow$  discretization scheme  $\hat{=}$  filter (implicit)
- mesh width  $\Delta x$ ,  $[\Delta x^n]$ -order of the scheme determine the filter operation significantly,

e.g.: 1-dimensional  $2^{nd}$ -order central difference  
approximation for the first derivative

$$\frac{\partial u}{\partial x} \simeq \frac{u(x + \Delta x) - u(x - \Delta x)}{2\Delta x}$$

is equivalent to a Box-filter with filter width  $2\Delta x$

$$\frac{u(x + \Delta x) - u(x - \Delta x)}{2\Delta x} = \frac{d}{dx} \underbrace{\left[ \frac{1}{2\Delta x} \int_{x-\Delta x}^{x+\Delta x} u(r) dr \right]}_{\bar{u}(x)}$$

Generally (3-dimensional case), the filter function  $G$  **can not be determined analytically** for the actually used difference scheme.

## **VI.2 Subgrid-Scale models (SGS)**

### **(A) Implicit LES (ILES)**

No SGS modelling explicitly applied to capture the unresolved interactions.

Numerical discretization scheme does this job,  
i.e., scheme provides sufficient numerical dissipation

### **(B) Constant Coefficient models**

- Gradient Model (GR)
- Scale Similarity Model (SS)
- Smagorinsky (SM)
- Mixed model (SS+SM)

### **(C) Dynamic models**

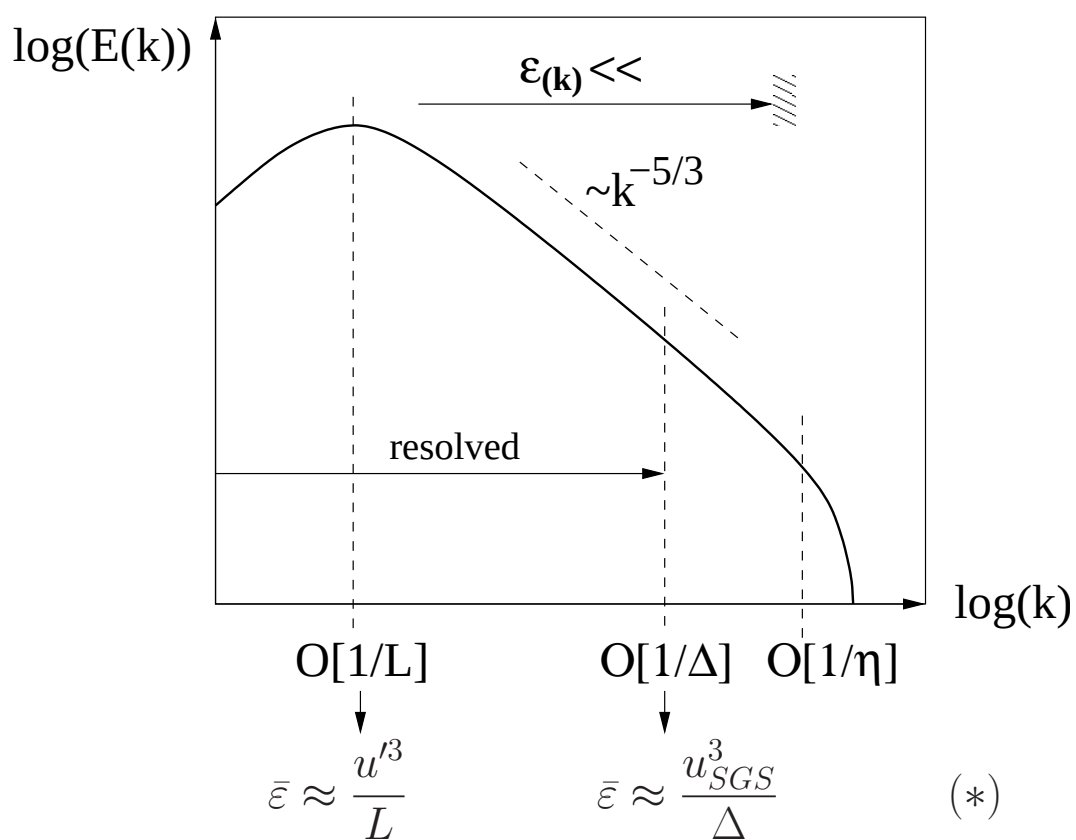
- Germano
- Lilly

## ad (B): Smagorinsky Model

Boussinesq Ansatz:

$$\tau_{ij}^R - \frac{1}{3} \tau_{kk}^R \delta_{ij} = -\nu_T \left( \frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) = -2 \nu_T \bar{S}_{ij}$$

Estimate for the eddy-viscosity  $\nu_T$ :



$$\begin{aligned}
\bar{\varepsilon} = \text{dissipation:} \quad \bar{\varepsilon} &\approx \nu_T \frac{u_{SGS}^2}{\Delta^2} \\
\text{yields with } (*): \quad \nu_T &\approx u_{SGS} \Delta \\
\text{due to } (*): \quad u_{SGS} &= u' \left( \frac{\Delta}{L} \right)^{\frac{1}{3}} \\
\nu_T &\approx u' \Delta^{\frac{4}{3}} L^{-\frac{1}{3}}
\end{aligned}$$

$$\text{estimate: } u' \approx L \left( \bar{S}_{ij} \bar{S}_{ij} \right)^{\frac{1}{2}} = L |\bar{S}_{ij}|$$

$$\nu_T \approx \underbrace{\Delta^{\frac{4}{3}} L^{\frac{2}{3}}}_{\hookrightarrow \Delta^2} |\bar{S}_{ij}|$$

$$\Rightarrow \boxed{\nu_T = (C_S \Delta)^2 |\bar{S}_{ij}|}$$

so, the Smagorinsky-model reads:

$$\tau_{ij}^R - \frac{1}{3} \tau_{kk}^R \delta_{ij} = -2 (C_S \Delta)^2 |\bar{S}_{ij}| \bar{S}_{ij} \quad (\text{VI.11})$$

Smagorinsky constant  $C_S = \text{const.}$ :

Assumption of the Kolmogorov-spectrum:

$$E(k) \sim C k^{(-5/3)} \quad \rightarrow \quad C_S \approx 0.17$$

proposed values in literature



|                      |              |
|----------------------|--------------|
| Isotropic turbulence | $C_S = 0.23$ |
| Channel flow         | $C_S = 0.1$  |
| Rotating flow        | $C_S = 0.08$ |

**Main problem:** = constancy of  $C_S$   
 Smagorinsky-model excludes laminar flow  
 (e.g. near to the wall)!

laminar flow requires:

$$C_S \stackrel{!}{=} 0 \quad \text{so} \quad \nu_T = 0$$

Possible improvements:

- *van-Driest*-damping near to the wall

$$C_S \Delta \rightarrow C_S \Delta \left[ 1 - \exp \left( -\frac{y^+}{A^+} \right) \right]$$

$$\text{with } y^+ = \sqrt{\frac{\tau_w}{\rho}} \frac{y}{\nu}, \quad A^+ \approx 25$$

- dynamic modelling

## ad (C): Dynamic model

Second filtering of the already filtered equations

$$\frac{\partial \bar{u}_i}{\partial t} + \frac{\partial \overline{u_i u_j}}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + \nu_m \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} \quad (\text{VI.12})$$

using a coarse *test-filter* with the filter width

$$\tilde{\Delta} > \Delta \quad (\text{e.g. } \tilde{\Delta} = 2\Delta).$$

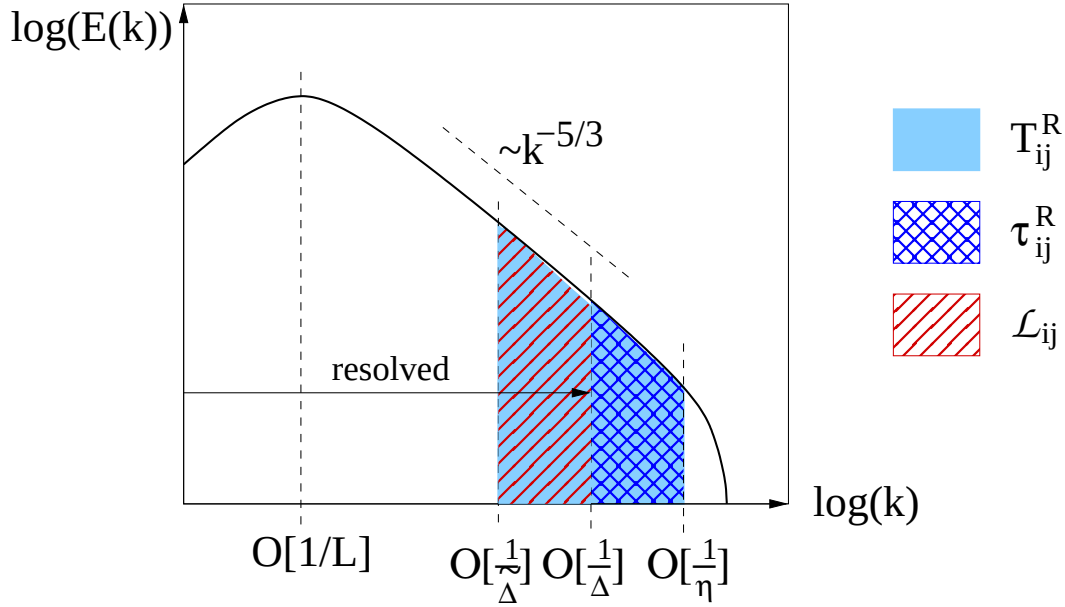
This yields

$$\frac{\partial \tilde{u}_i}{\partial t} + \frac{\partial \widetilde{\overline{u_i u_j}}}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \tilde{p}}{\partial x_i} + \nu_m \frac{\partial^2 \tilde{u}_i}{\partial x_j \partial x_j} \quad (\text{VI.13})$$

$$\begin{aligned} \text{where: } \tilde{u}_i(\mathbf{x}, t) &= \int_{\mathcal{D}} \bar{u}_i(\mathbf{x} - \mathbf{r}, t) \tilde{G}(\mathbf{r}; \tilde{\Delta}) d\mathbf{r} \\ &= \int_{\mathcal{D}} u_i(\mathbf{x} - \mathbf{r}, t) \underbrace{\tilde{G}(\mathbf{r}; \underbrace{\tilde{\Delta}}_{= f(\Delta, \tilde{\Delta})})}_{= f(\Delta, \tilde{\Delta})} d\mathbf{r} \end{aligned}$$

Using the definition of the *Residual stress* (VI.9), the test filtered equations (VI.13) are rewritten as

$$\frac{\partial \tilde{u}_i}{\partial t} + \frac{\partial \tilde{u}_i \tilde{u}_j}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \tilde{p}}{\partial x_i} + \nu_m \frac{\partial^2 \tilde{u}_i}{\partial x_j \partial x_j} - \frac{\partial}{\partial x_j} \left( \widetilde{\overline{u_i u_j}} - \tilde{u}_i \tilde{u}_j \right)$$



## Two levels of resolution:

Gridfilter level:

$$\frac{\partial \bar{u}_i}{\partial t} + \frac{\partial \bar{u}_i \bar{u}_j}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + \nu_m \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} - \frac{\partial}{\partial x_j} (\overline{u_i u_j} - \bar{u}_i \bar{u}_j)$$

Testfilter level:

$$\frac{\partial \tilde{u}_i}{\partial t} + \frac{\partial \tilde{u}_i \tilde{u}_j}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \tilde{p}}{\partial x_i} + \nu_m \frac{\partial^2 \tilde{u}_i}{\partial x_j \partial x_j} - \frac{\partial}{\partial x_j} (\widetilde{u_i u_j} - \tilde{u}_i \tilde{u}_j)$$

$$\rightarrow \text{unknown residual stresses } \tau_{ij}^R = \overline{u_i u_j} - \bar{u}_i \bar{u}_j \quad (\text{VI.14})$$

$$T_{ij}^R = \widetilde{u_i u_j} - \tilde{u}_i \tilde{u}_j \quad (\text{VI.15})$$

Test-filtering of (VI.14) and subtraction from (VI.15)

$\Rightarrow$  Germano's identity:

$$\begin{aligned} \mathcal{L}_{ij} &= T_{ij}^R - \tilde{\tau}_{ij}^R = \widetilde{u_i u_j} - \tilde{u}_i \tilde{u}_j - (\widetilde{u_i u_j} - \tilde{u}_i \tilde{u}_j) \\ &= \widetilde{u_i u_j} - \tilde{u}_i \tilde{u}_j \end{aligned}$$

The *Leonard stress*-term

$$\mathcal{L}_{ij} = \mathcal{L}_{ij}(\bar{u}_i, \bar{u}_j)$$

only depends on quantities, which are resolved by LES  
(*resolved* and *grid-scale*).

so  $\mathcal{L}_{ij}$  is **known!**

$\Downarrow \quad \Downarrow \quad \Downarrow$

Dynamic determination of the Smagorinsky-constant  $C_S$   
(*Dynamic modelling*)

## Dynamic computation of Smagorinsky's $C_S$

Smagorinsky model (VI.11) for *residual stress* with  $c_S = C_S^2$  on

$$\text{gridfilter level:} \quad \tau_{ij}^r = \tau_{ij}^R - \frac{1}{3} \tau_{kk}^R \delta_{ij} = -2 c_S \Delta^2 |\bar{S}_{ij}| \bar{S}_{ij}, \quad \text{and}$$

$$\text{the testfilter level:} \quad T_{ij}^r = T_{ij}^R - \frac{1}{3} T_{kk}^R \delta_{ij} = -2 c_S \tilde{\Delta}^2 |\tilde{S}_{ij}| \tilde{S}_{ij}$$

*Germano's identity:*

$$\begin{aligned} \mathcal{L}_{ij}^r &= \mathcal{L}_{ij} - \frac{1}{3} \mathcal{L}_{kk} \delta_{ij} = T_{ij}^r - \tilde{\tau}_{ij}^r \\ &= -2\Delta^2 \underbrace{\left\{ \left( \frac{\tilde{\Delta}}{\Delta} \right)^2 c_S |\tilde{S}_{ij}| \tilde{S}_{ij} - c_S \widetilde{|\bar{S}_{ij}| \bar{S}_{ij}} \right\}}_{= c_S M_{ij}} \quad (\text{VI.16}) \end{aligned}$$

Solution of (VI.16) for the parameter  $c_S$  applying a contraction with  $\bar{S}_{ij}$  yields

$$\mathcal{L}_{ij} \bar{S}_{ij} = -2 c_S \Delta^2 M_{kl} \bar{S}_{kl}$$

$$c_S \Delta^2 = -\frac{1}{2} \frac{\mathcal{L}_{ij} \bar{S}_{ij}}{M_{kl} \bar{S}_{kl}}$$

or, alternatively, applying a Least-squares Ansatz

$$[\mathcal{L}_{ij}^r + 2c_s \Delta^2 M_{ij}]^2 \rightarrow \min.$$

$$\text{gives} \quad c_s \Delta^2 = -\frac{1}{2} \frac{M_{ij} \mathcal{L}_{ij}}{M_{kl} M_{kl}} \quad (\text{VI.17})$$

Annotations to the dynamic model:

- Dynamic model also captures laminar flow:

$$c_S \rightarrow 0$$

- If:  $c_S < 0 \rightarrow$  backscatter

$$\text{energy transfer:} \quad \begin{array}{ccc} \textit{unresolved} & \rightarrow & \textit{resolved} \\ (\textit{subgrid-scale}) & & (\textit{grid-scale}) \end{array}$$

- For stability reasons, an averaging in the homogeneous directions is applied, e.g. if the solution is homogeneous in  $x$  and  $z$ , (VI.17) is evaluated as

$$c_s \Delta^2 = -\frac{1}{2} \frac{\langle M_{ij} \mathcal{L}_{ij} \rangle_{x,z}}{\langle M_{kl} M_{kl} \rangle_{x,z}}$$

- For statistically stationary flows, averaging in time is additionally possible.